

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\115-C.pvdb

Date of Analysis : Sep 9, 2026 9:24am

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User
 FileName : 115-C Page 2 of 184
 Vessel Design Summary: Step: 28 9:24am Sep 9,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1670.000 mm ID
 Vessel Design Length, Tangent to Tangent 9845.97 mm
 Specified Datum Line Distance 0.00 mm
 Shell Side Design Temperature 200 °C
 Channel Side Design Temperature 82 °C
 Shell Side Design Pressure 4550.000 KPa.
 Channel Side Design Pressure 1000.000 KPa.
 Shop Shell Side Test Pressure 6558.139 KPa.
 Field Shell Side Test Pressure 5915.000 KPa.
 Shop Channel Side Test Pressure 1648.255 KPa.
 Field Channel Side Test Pressure 1300.000 KPa.
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-266 2	...	...	K03506	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 2	...	...	K03506	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Blind Flg	1018.212	103.42	1095.000	3.0000	N/A	No
LHS Ch Flg	1016.173	103.42	1781.000	3.0000	N/A	No
LHS Ch Shell	1016.399	103.42	1781.000	3.0000	N/A	No
Main Shell	4566.398	0.00	4977.000	3.0000	N/A	No
Rhs Dish End	4566.398	0.00	4929.000	3.0000	No	No

Liquid Level: 1858.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
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Vessel Design Summary:

Step: 28 9:24am

Sep 9, 2026

Body Flg	123.2	120.0	128.0	120.0	114.8	...	1.00	1.00
Body Flg	331.2	208.0	168.0	160.0	152.8	...	1.00	1.00
Cylinder	1931.2	1600.0	14.0	14.0	9.2	...	1.00	1.00
Cylinder	9795.2	7600.0	34.0	34.0	31.3	...	1.00	1.00
Ellipse	9846.0	50.8	40.0	33.0	30.7	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	120	90.1192	103.419	No Calc
20	30	160	152.756	103.419	No Calc
30	Tsht	14	8.30599	103.419	643
Tsht	40	14	No Calc	103.419	No Calc
40	80	34	No Calc	...	1452
80	90	33	No Calc	...	1838

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1599.97	8957.51	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	80	7790	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3-Tube Inlet	WRC-297	0.892	Shell	Passed
N4-Tube Outlet	WRC-297	0.892	Shell	Passed
N1-Shell Inlet	WRC-297	0.949	Shell	Passed
N2-Shell outle	WRC-297	0.957	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	350.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1430.000 mm
Wear Pad Width	650.000 mm
Wear Pad Thickness	14.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1520.000 mm

Baseplate Length	1540.000 mm
Baseplate Thickness	28.000 mm
Baseplate Width	375.000 mm
Number of Ribs (including outside ribs)	4
Rib Thickness	10.000 mm
Web Thickness	10.000 mm
Height of Center Web	605.000 mm
Number of Bolts in Baseplate	6



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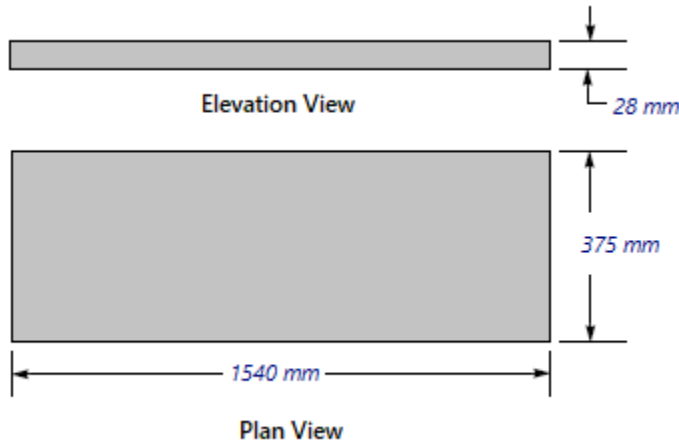
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Vessel Design Summary:

Step: 28

9:24am

Sep 9, 2026

Baseplate Sketch**Summary of Maximum Saddle Loads, Operating Case, Un-Factored:**

Maximum Vertical Saddle Load	57909.93	Kgf
Maximum Transverse Saddle Shear Load	8082.66	Kgf
Maximum Longitudinal Saddle Shear Load	4063.53	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	51128.82	Kgf
Maximum Transverse Saddle Shear Load	222.69	Kgf
Maximum Longitudinal Saddle Shear Load	175.05	Kgf

Weights:

Fabricated - Bare w/o removable internals	50477.0	kg.
Shop Test - Fabricated + water (full)	69732.0	kg.
Shipping - Fab. + rem. intls.+ shipping app.	50477.0	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	50477.0	kg.
Empty - Fab. + intls. + details + wghts.	50852.5	kg.
Operating - Empty + operating liquid (no ca)	70106.5	kg.
Field Test - Empty weight + test medium (full)	70107.5	kg.

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Nozzle Summary:

Step: 26 9:24am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3-Tube Inlet	1781	OK	...	OK	9.68	OK	Passed
N4-Tube Outlet	1781	OK	...	OK	9.78	OK	Passed
N1-Shell Inlet	4977	...	...	OK	11.33	OK	Passed
N2-Shell outlet	4853	...	...	OK	11.33	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 4853 Nozzle : N2-Shell outlet [Shellside]
 Minimum MAWP Nozzles : 1781 Nozzle : N3-Tube Inlet [Tubeside]

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N3-Tube Inlet	1031.175	0.000	1161.401
30	N4-Tube Outlet	1031.175	180.000	1161.401
40	N1-Shell Inlet	5995.175	0.000	756.800
40	N2-Shell outlet	5995.175	180.000	756.800

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 25 9:24am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1-Shell Inlet	16.000 in	Actual	WNF	16.000	17.000	...	...	359.10	600
N2-Shell outlet	16.000 in	Actual	WNF	16.000	17.000	...	...	359.10	600
N3-Tube Inlet	24.000 in	40	None	24.000	17.450	...	...	481.62	...
N4-Tube Outlet	24.000 in	40	None	24.000	17.450	...	...	481.62	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1-Shell In	SA-266 2	32.000	10.000	...	...	...
N2-Shell ou	SA-266 2	32.000	10.000	...	...	...
N3-Tube Inl	SA-266 2	14.000	10.000	...	...	...
N4-Tube Out	SA-266 2	14.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N1-Shell Inlet	5995.175	0.0	300.00	0.00	Main Shell
N2-Shell outlet	5995.175	180.0	300.00	0.00	Main Shell
N3-Tube Inlet	1031.175	0.0	400.00	0.00	LHS Ch Shell
N4-Tube Outlet	1031.175	180.0	400.00	0.00	LHS Ch Shell

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Bill of Material:

Step: 24 9:24am Sep 9,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 128.0mm THK X 1880.0mm OD	SA-266 2
1	WELD NECK FLANGE: 168.0mm THK X 1880.0mm OD	SA-266 2
1	CYLINDER: 14.0mm THK X 1670.0mm ID X 1600.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 40.0mm THK X 1670.0mm ID X 50.8mm	SA-516 70
1	INSULATION: 90mm X 80mm THK	Hot Insulation
1	INSULATION: 143mm X 80mm THK	Hot Insulation
1	INSULATION: 1600mm X 80mm THK	Hot Insulation
4	BAFFLES: 1642.0mm DIA X 8.0mm THK	SA-516 70
12	TIE RODS: 15mm DIA X 6999mm	SA-193 B7M
1	TUBESHEET: 1738.0mm X 264.0mm THK	SA-266 2
1996	TUBES: 7000.0mm X 25.4mm DIA X 2.8mm THK	SA-179
2	SADDLE: 350.0mm X 120 DEG	SA-516 70
1	GASKET: 1740.0mm OD X 1702.0mm ID	...
64	BODY FLANGE BOLTS: 31.8mm DIA	SA-193 B7
128	NUTS FOR BODY FLANGE BOLTS: 31.8mm DIA	...
1	CYLINDRICAL SEGMENT 34.0mm THK X 3048.0mm X 7600.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 34.0mm THK X 2198.4mm X 7600.0mm	SA-516 70
1	NAMEPLATE	...

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MDMT Summary:

Step: 27 9:24am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Main Shell [8]	B	7	5		0.956	34.000	1.00	Yes
Rhs Dish End [10]	B	7	5		0.966	33.000	1.00	Yes
Rhs Dish End [7]	B	11	0		0.801	40.000	1.00	Yes
LH saddle flo[24]	B	-35	-37		0.956	14.000	1.00	No
RHS saddle fi[24]	B	-35	-37		0.956	14.000	1.00	No
N1-Shell Inlet[1]	B	7	5		0.956	34.000	1.00	No
Nozzle Flg [5]	A	-18	-48					
N2-Shell outle[1]	B	7	5		0.959	34.000	1.00	No
Nozzle Flg [5]	A	-18	-48					
Tubesheet: SS[13]	B	6	1		0.899	66.000	1.00	No
Warmest MDMT:		11	5					
LHS Blind Flg[11]	B	4	-4		0.854	30.000	1.00	No
LHS Ch Flg [11]	B	-18	-48	-29	0.478	14.000	1.00	No
LHS Ch Shell [8]	B	-18	-40	-29	0.608	14.000	1.00	No
N3-Tube Inlet [1]	B	-2	-104	-29	0.318	24.000	1.00	No
N4-Tube Outlet[1]	B	-2	-104	-29	0.323	24.000	1.00	No
Tubesheet: CS[14]	B	23	-104		0.299	66.000	1.00	No
Bolting [21]		-48						
Warmest MDMT:		23	-4					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	5.0	-12.0	Fail
Channel/Tube	-4.0	-12.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:



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MDMT Summary:

Step: 27 9:24am Sep 9, 2026

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 9:24am Sep 9, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 9:24am Sep 9,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	4550	KPa.
Channel Side Design Pressure	999.99	KPa.
Shell Side Design Temperature	200.0	°C
Channel Side Design Temperature	82.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Sour	
Service Type, Channel Side	Sour	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



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Input Echo:

Step: 1 9:24am Sep 9,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	16000 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Input Echo:

Step: 1 9:24am Sep 9, 2026

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes

Do not use Nozzle MDMT Interpretation VIII-1 01-37 No

Use Table G instead of exact equation for "A" Yes

Shell Head Joints are Tapered Yes

Compute "K" in corroded condition Yes

Use Code Case 2286 No

Use the MAWP to compute the MDMT Yes

For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes

For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10
Element To Node	20
Element Type	Flange
Description	LHS Blind Flg
Distance "FROM" to "TO"	120 mm
Flange Outside Diameter	1880 mm
Element Thickness	120 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	128 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.08 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1858 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	10
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm



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Input Echo:

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Height/Length of Insulation	90	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element#	2/5
Element From Node	20
Element To Node	30
Element Type	Flange
Description	LHS Ch Flg
Distance "FROM" to "TO"	208 mm
Flange Inside Diameter	1670 mm
Element Thickness	160 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	168 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1650 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	143 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm ³

Element#	3/5
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	LHS Ch Shell
Distance "FROM" to "TO"	1600 mm
Inside Diameter	1670 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm



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Yield Stress, Operating	242.6	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1670	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1600	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N3-Tube Inlet	
Dist. from "FROM" Node / Offset dist	700	mm
Nozzle Diameter	24	in.
Nozzle Schedule	40	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	97.544	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-266 2	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N4-Tube Outlet	
Dist. from "FROM" Node / Offset dist	700	mm
Nozzle Diameter	24	in.
Nozzle Schedule	40	
Nozzle Class	None	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	139.72	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-266 2	

Element From Node	30	
Detail Type	Weight	
Detail ID	pass partition	
Dist. from "FROM" Node / Offset dist	600	mm
Miscellaneous Weight	200	Kgf
Offset from Element Centerline	0	mm

Element#	4/5
Element From Node	40
Element To Node	80
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	7600 mm



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Inside Diameter	1670	mm
Element Thickness	34	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	200	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	LH saddle float	
Dist. from "FROM" Node / Offset dist	1520	mm
Width of Saddle	350	mm
Height of Saddle at Bottom	1430	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	650	mm
Thickness of Wear Plate	14	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Saddle	
Detail ID	RHS saddle fixe	
Dist. from "FROM" Node / Offset dist	6080	mm
Width of Saddle	350	mm
Height of Saddle at Bottom	1430	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	650	mm
Thickness of Wear Plate	14	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1670	mm
Liquid Density	0.0009984	kg./cm³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N1-Shell Inlet	
Dist. from "FROM" Node / Offset dist	3800	mm
Nozzle Diameter	16	in.
Nozzle Schedule	60	
Nozzle Class	600	



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Input Echo:

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Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	334.2	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N2-Shell outlet	
Dist. from "FROM" Node / Offset dist	3800	mm
Nozzle Diameter	16	in.
Nozzle Schedule	60	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	334.2	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element From Node	40	
Detail Type	Weight	
Detail ID	Fire prrof	
Dist. from "FROM" Node / Offset dist	1520	mm
Miscellaneous Weight	450	Kgf
Offset from Element Centerline	1420	mm

Element From Node	40	
Detail Type	Weight	
Detail ID	Fire prrof 1	
Dist. from "FROM" Node / Offset dist	6180	mm
Miscellaneous Weight	450	Kgf
Offset from Element Centerline	1420	mm

Element#	5/5	
Element From Node	80	
Element To Node	90	
Element Type	Elliptical	
Description	Rhs Dish End	
Distance "FROM" to "TO"	50.8	mm
Inside Diameter	1670	mm
Element Thickness	33	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	40	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	200	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1670	mm
Liquid Density	0.0009984	kg./cm³



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XY Coordinate Calculations:

Step: 2 9:24am

Sep 9,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Blind Flg		123.175	...	120	...
LHS Ch Flg		331.175	...	208	...
LHS Ch Shell		1931.17	...	1600	...
Main Shell		9795.17	...	7600	...
Rhs Dish End		9845.97	...	50.8	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: Ch Cover

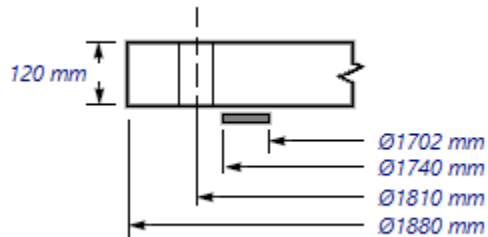
Flng: 5 9:24am Sep 9,2026

Flange Input Data Values

Description: Ch Cover :

LHS Blind Flg

Flange Type		TEMA Channel Cover	
Design Pressure	P	1018.21	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1880.000	mm
Flange Thickness	t	120.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	5686.283	mm
Youngs Modulus for Blind Flange Matl.	Sy	197846.66	N./mm ²
Allowed Channel Cover Deflection	dfAllow	2.0875	mm
Diameter of Bolt Circle	C	1810.000	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads		UNC Thread Series	
Number of Bolts		64	
Flange Face Outside Diameter	Fod	1751.000	mm
Flange Face Inside Diameter	Fid	1670.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1740.000	mm
Gasket Inside Diameter	Gi	1702.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	1686.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²





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Flg Calc [Int P]: Ch Cover

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Gasket Contact Width,	$N = (G_o - G_i)/2$	19.000 mm
Basic Gasket Width,	$b_o = N/2$	9.500 mm
Effective Gasket Width,	$b = C_b \cdot \sqrt{b_o}$	7.767 mm
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	1724.466 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1724^2 \cdot 1018$$

$$= 242512.469 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 7.77 \cdot \pi \cdot 1724 \cdot 3 \cdot 1018$$

$$+ 2 \cdot 1686 \cdot 5 \cdot 3 \cdot 1018$$

$$= 31466.230 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(242512 + 31466 + 0, 0)$$

$$= 273978.688 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 69 \cdot 7.77 \cdot \pi \cdot 1724 + 68.9 \cdot 5 \cdot 1686$$

$$= 355111.125 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4M_e/G) / S_b, W_g / S_a)$$

$$= \max((273979 + 0 + 4 \cdot 0 / 1724) / 172, 355111 / 172)$$

$$= 202.475 \text{ cm}^2$$

Bolting Information for UNC Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	202.475	367.483	
Radial Distance between Bolts and Edge:	31.750	35.000	
Circ. Spacing between the Bolts:	71.450	88.812	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 367 \cdot 172 / (69 \cdot \pi (1740 + 1702))$$

$$= 8.477 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 367$$

$$= 644509.50 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o$$

$$= 273978.69 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1810 - 1724) / 2$$

$$= 42.7669 \text{ mm}$$

Flange Stresses



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Flg Calc [Int P]: Ch Cover

Flng: 5 9:24am

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Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 273979 \cdot 42.8 \cdot 1 / (117^2 \cdot 1724) + 0.3 \cdot 1 \cdot 1018 \cdot 1724^2 / 117^2$$

$$= 75.61 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 644510 \cdot 42.8 \cdot 1 / (117^2 \cdot 1724)$$

$$= 21.76 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 273979 / 367$$

$$= 73.12 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (355111 / 367)$$

$$= 94.77 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Tangential Flange	75.61	138.00	21.76	138.00
Bolting	73.12	172.00	94.77	172.00

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 1.9717 mm

Allowed deflection at center of Cover 2.0875 mm

Reqd. Blind Flange Thickness at Center 89.605 mm

Reqd. Blind Flange Thickness at Gasket 49.457 mm

Min. Required Flange Thickness per TEMA 114.796 mm

Estimated M.A.W.P. (Operating) 1114 KPa.

Estimated Finished Weight of Flange at given Thk. 2581.8 kg.

Estimated Unfinished Weight of Forging at given Thk 2581.8 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.85, Temperature Reduction per Fig. UCS 66.1 = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 4 °C

Min Metal Temp. at Required thickness (UCS 66.1) -4 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Tangential Flange	71.88	138.00	20.68	138.00
Bolting	73.12	172.00	94.77	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 114.148 mm

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Flg Calc [Int P]: FLANGE

Flng: 6 9:24am Sep 9,2026

Flange Input Data Values

Description: FLANGE :

LHS Ch Flg

Flange Type	Integral Weld Neck, Integral type		
Design Pressure	P	1016.17	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1670.000	mm
Flange Outside Diameter	A	1880.000	mm
Flange Thickness	t	160.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	21.0000	mm
Length of Hub	h	48.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1810.000	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads	UNC Thread Series		
Number of Bolts		64	
Flange Face Outside Diameter	Fod	1751.000	mm
Flange Face Inside Diameter	Fid	1670.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1740.000	mm
Gasket Inside Diameter	Gi	1702.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	1686.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



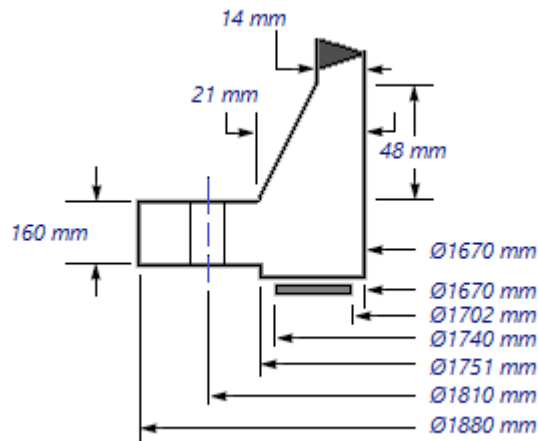
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Flg Calc [Int P]: FLANGE

Flng: 6 9:24am Sep 9, 2026



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1016 \cdot (1670/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 1016) + C_a$$

$$= 9.1985 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 11) / (838 + 0.6 \cdot 11)$$

$$= 1797.164 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (835 + 0.6 \cdot 14)$$

$$= 2290.554 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	157.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1676.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	18.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	11.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	49.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	19.000 mm
Basic Gasket Width, $bo = N/2$	9.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{bo}$	7.767 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1724.466 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1724^2 \cdot 1016$$

$$= 242026.891 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{part} \cdot m_{part} \cdot P$$

$$= 2 \cdot 7.77 \cdot \pi \cdot 1724 \cdot 3 \cdot 1016$$

$$+ 2 \cdot 1686 \cdot 5 \cdot 3 \cdot 1016$$

$$= 31403.223 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1676^2 \cdot 1016$$

$$= 228613.703 \text{ Kg}$$



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Flg Calc [Int P]: FLANGE

Flng: 6 9:24am Sep 9, 2026

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 242027 - 228614$$

$$= 13413.192 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(242027 + 31403 + 0, 0)$$

$$= 273430.125 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 7.77 \cdot \pi \cdot 1724 + 68.9 \cdot 5 \cdot 1686$$

$$= 355111.094 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((273430 + 0 + 4 \cdot 0 / 1724) / 172, 355111 / 172)$$

$$= 202.475 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 31.8 + 6 \cdot 157 / (3 + 0.5)$$

$$= 332.643 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1810 \cdot \sin(\pi / 64)$$

$$= 88.812 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ (Bs / (2a + t)) }, 1)$$

$$= \max(\sqrt{ (88.8 / (2 \cdot 31.8 + 157)) }, 1)$$

$$= 1.0000$$

Bolting Information for UNC Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	202.475	367.483	
Radial Distance between Hub and Bolts:	44.450	49.000	
Radial Distance between Bolts and Edge:	31.750	35.000	
Circ. Spacing between the Bolts:	71.450	88.812	332.643

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 367 \cdot 172 / (68.9 \cdot \pi (1740 + 1702))$$

$$= 8.477 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 367$$

$$= 644509.50 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 273430 - 242027$$

$$= 31403.24 \text{ Kgf}$$



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Flg Calc [Int P]: FLANGE

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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1810 - 1724) / 2$$

$$= 42.7669 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((1810 - 1676) / 2 + 42.8)$$

$$= 54.8834 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1810 - 1676 - 18) / 2$$

$$= 58.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	228614.	58.0000	1.0000	13260.
Face Pressure, Mt	13413.	54.8834	1.0000	736.
Gasket Load, Mg	31403.	42.7669	1.0000	1343.
Gasket Seating, Matm	644510.	42.7669	1.0000	27564.

Total Moment for Operation, Mo 15339. Kg-m.
Total Moment for Gasket seating, Mg 27564. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 135.779 mm
Hub Ratio, h/h0 = HL / H0 0.354
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.636

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.866
Factor V per 4.16.5 0.327
Factor f per 4.16.5 1.210

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1880 / 1676$$

$$= 1.122$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.12^2 \cdot \log_{10}(1.12))) / (1.12^2 - 1)) / (1.12 - 1)$$

$$= 16.907$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.12^2 (1 + 8.55246 \cdot \log_{10}(1.12)) - 1) / ((1.0472 + 1.9448 \cdot 1.12^2) \cdot (1.12 - 1))$$

$$= 1.869$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.12^2 (1 + 8.55246 \cdot \log_{10}(1.12)) - 1) / (1.36136 (1.12^2 - 1) \cdot (1.12 - 1))$$

$$= 18.579$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.12^2 + 1) / (1.12^2 - 1)$$

$$= 8.744$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$



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Flg Calc [Int P]: FLANGE

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$$= 18.6 \cdot 11^2 \cdot 136 / 0.33$$

$$= 934545.250 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.87 / 136$$

$$= 0.006 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$

$$= (157 \cdot 0.0064 + 1) / 1.87 + 157^3 / 934545$$

$$= 5.212$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g1^2)$$

$$= (1.21 \cdot 15339 / 1676) / (5.21 \cdot 18^2)$$

$$= 64.32 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g1^2)$$

$$= (1.21 \cdot 27564 / 1676) / (5.21 \cdot 18^2)$$

$$= 115.59 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.34 \cdot 15339 / 1676) / (5.21 \cdot 157^2)$$

$$= 1.63 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.34 \cdot 27564 / 1676) / (5.21 \cdot 157^2)$$

$$= 2.93 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (16.9 \cdot 15339 / (157^2 \cdot 1676)) - 8.74 \cdot 1.63$$

$$= 47.30 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (16.9 \cdot 27564 / (157^2 \cdot 1676)) - 8.74 \cdot 2.93$$

$$= 84.99 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (64.3 + \max(1.63, 47.3)) / 2$$

$$= 55.81 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (116 + \max(2.93, 85)) / 2$$

$$= 100.29 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 273430 / 367$$

$$= 72.97 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (355111 / 367)$$

$$= 94.77 \text{ N./mm}^2$$



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Flg Calc [Int P]: FLANGE

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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	64.32	207.00	115.59	207.00
Radial Flange	1.63	138.00	2.93	138.00
Tangential Flange	47.30	138.00	84.99	138.00
Maximum Average	55.81	138.00	100.29	138.00
Bolting	72.97	172.00	94.77	172.00

Minimum Required Flange Thickness [Rigidity] 152.756 mm

Estimated M.A.W.P. (Operating) 1797 KPa.

Estimated Finished Weight of Flange at given Thk. 760.6 kg.

Estimated Unfinished Weight of Forging at given Thk 943.9 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	481.04	499.99	0.962	481.04	499.99	0.962
Tangential Flange	St	353.12	374.99	0.942	353.12	374.99	0.942
Tangential Flange	Sto	158.33	250.00	0.633	158.33	250.00	0.633
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	365.31	749.99	0.487	365.31	749.99	0.487
Combined	Sh + Sto	639.38	749.99	0.853	639.38	749.99	0.853
Radial Flange	Sr	12.18	499.99	0.024	12.18	499.99	0.024
Maximum Stress Ratio				1.000			1.000



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Flg Calc [Int P]: FLANGE

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Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.62/1.33)

= 1.2121

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.51/1.29) if $h/h0 < 0.35$, else max (0.25; 0.3/0.76)

= 0.3990

In this case, the hub ratio $h/h0$, 0.3535 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	714.6 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Root Area of a Single Bolt	Abs	5.7419 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.608 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$$

$$= \pi/4 (1740^2 - 1702^2) + (0.5 \cdot 1686 \cdot 5)$$

$$= 1069.4 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207 \cdot 1069 / (64 \cdot 5.74)$$

$$= 602.4 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(602; 508)$$

$$= 507.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(508; 290)$$

$$= 507.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(508; 715)$$

$$= 507.5 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 508 \geq 140 (1069 / (64 \cdot 5.74))$$

$$:: 507.5 \geq 407.4 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 508 \geq (97 \cdot 1069 + \pi/4 \cdot 1016 \cdot 1702^2) / (0.7 \cdot 5.74 \cdot 64)$$

$$:: 507.5 \geq 493.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 508 \leq 276 (1069 / (64 \cdot 5.74))$$

$$:: 507.5 \leq 803.2 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(thetagmax / thetafmax)$$

$$:: 508 \leq 715 (1/0.61)$$

$$:: 507.5 \leq 1174.9 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 508 \cdot 0.2 \cdot 5.74 \cdot 31.8$$

$$= 188.7 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



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Flg Calc [Int P]: FLANGE

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Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 27564 / 1 \cdot 9807 \cdot 0.33 / (4.64 \cdot 201334 \cdot 11^2 \cdot 136 \cdot 0.3)$
 $= 1.000$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 15339 / 1 \cdot 9807 \cdot 0.33 / (4.64 \cdot 197847 \cdot 11^2 \cdot 136 \cdot 0.3)$
 $= 0.566$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 27564 / 1 \cdot 9807 \cdot 0.33 / (5.21 \cdot 201334 \cdot 11^2 \cdot 136 \cdot 0.3)$
 $= 0.890$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 15339 / 1 \cdot 9807 \cdot 0.33 / (5.21 \cdot 197847 \cdot 11^2 \cdot 136 \cdot 0.3)$
 $= 0.504$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.48, Temperature Reduction per Fig. UCS 66.1 = 35 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	62.13	207.00	109.10	207.00
Radial Flange	2.11	138.00	3.70	138.00
Tangential Flange	41.17	138.00	72.29	138.00
Maximum Average	51.65	138.00	90.70	138.00
Bolting	72.97	172.00	94.77	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 140.564 mm

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Internal Pressure Calculations: Step: 5 9:24am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Blind Flg		1018.2	128	3	1880	138	1
LHS Ch Flg		1016.2	168	3	1670	138	2
LHS Ch Shell		1016.4	14	3	1670	138	3
Main Shell		4566.4	34	3	1670	138	4
Rhs Dish End		4566.4	40	3	1670	138	5

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Blind Flg		1000	1095	1282	120	114.796	1
LHS Ch Flg		1000	1781	2291	160	152.756	2
LHS Ch Shell		1000	1781	2291	14	9.19991	3
Main Shell		4550	4977	5485	34	31.2932	4
Rhs Dish End		4550	4929	5432	33	30.6914	5

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 4777 KPa.
 Shell Side MAPnc = 5059 KPa.
 Channel Side MAWP = 1095 KPa.
 Channel Side MAPnc = 1282 KPa.

Elements Suitable for Design Internal Pressure.

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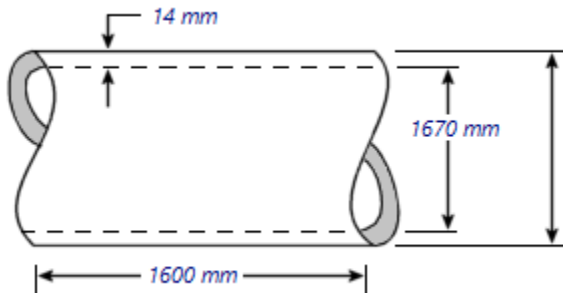
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Element: LHS Ch Shell

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Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	1016.399 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	1600.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1670.000 mm	Actual Stress	78.0 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	85350.8 cm ²	SG of Contents	1.000
Element Volume	3505262.8 cm ³	Weight of Contents	3503.1 Kg
Empty Weight	918.5 Kg	Operating Weight	4421.6 Kg
		Comp. All. Ope. Stress	118.5 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1016 \cdot 838) / (138 \cdot 1 - 0.6 \cdot 1016)$$

$$= 6.1999 + 3.0000 = 9.1999 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 16.399 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 11) / (838 + 0.6 \cdot 11)$$

$$= 1797 - 16.4 = 1781 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (835 + 0.6 \cdot 14)$$

$$= 2291 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1016 \cdot (838 + 0.6 \cdot 11)) / (1 \cdot 11)$$

$$= 78.045 \text{ N./mm}^2$$



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Element: LHS Ch Shell

Step: 6

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.831 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14$, $t_r = 6.68$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.61$, Temp. Reduction = 22 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -40 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1095.335 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.608	Temperature Difference	22 °C
UCS-66(c) red.	No	Computed Min. Temp.	-40 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: LHS Ch Shell

Cylindrical Shell From 30 to Tsht Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	1698.00	1599.97	154.36	0.9423	0.0007449	74.47

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 74.5) / (3 \cdot 154) = 643 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.306	1698.00	1599.97	320.02	0.9423	0.0002483	24.82

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 24.8) / (3 \cdot 320) = 103 \text{ KPa.}$$

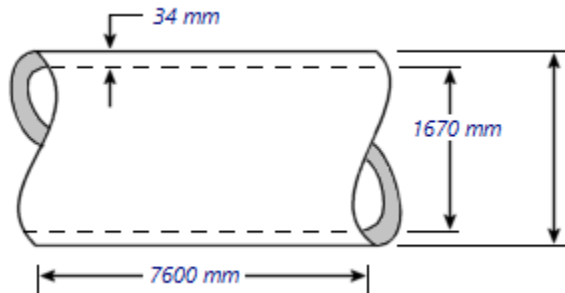
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	1698.00	8957.51	154.36	5.2753	0.0001199	11.98

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 12) / (3 \cdot 154) = 103 \text{ KPa.}$$



Internal Pressure Calculation Results:

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Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	4566.398 KPa.	Int. Design Temperature	200 °C
Material	SA-516 70	Ext. Design Temperature	200 °C
Length	7600.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1670.000 mm	Actual Stress	126.2 N./mm ²
Spec. Min. Thk t,e, or sa	34.0000 mm	Spec. Nominal Thk.	34.000 mm
Surface Area	414966.7 cm ²	SG of Contents	1.000
Element Volume	8547025.0 cm ³	Weight of Contents	8541.8 Kg
Empty Weight	10721.2 Kg	Operating Weight	19263.0 Kg
		Comp. All. Ope. Stress	126.6 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (4566 \cdot 838) / (138 \cdot 1 - 0.6 \cdot 4566) \\
 &= 28.2932 + 3.0000 = 31.2932 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 16.399 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 31) / (838 + 0.6 \cdot 31) \\
 &= 4994 - 16.4 = 4977 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 34) / (835 + 0.6 \cdot 34) \\
 &= 5485 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (4566 \cdot (838 + 0.6 \cdot 31)) / (1 \cdot 31) \\
 &= 126.187 \text{ N./mm}^2
 \end{aligned}$$



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Element: Main Shell

Step: 7

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f (1 - R_f / R_o)$) 1.995 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 34$, $t_r = 29.6$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.96$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B

7 °C

Min Metal Temp. at Required thickness (UCS 66.1)

5 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4777.354 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	34.000 mm	Basic MDMT	7 °C
Stress/Thk. Ratio	0.956	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	5 °C
		Joint Efficiency E^*	100 %

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Element: Rhs Dish End

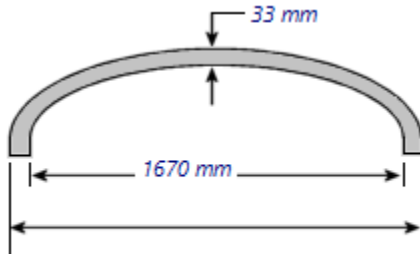
Step: 8

9:24am

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Rhs Dish End

Elliptical Head Design Information:			
Design Pressure P	4566.398 KPa.	Int. Design Temperature	200 °C
Material	SA-516 70	Ext. Design Temperature	200 °C
Length	50.800 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1670.000 mm	Actual Stress	127.4 N./mm ²
Spec. Min. Thk t,e, or sa	33.0000 mm	Spec. Nominal Thk.	40.000 mm
Surface Area	35989.9 cm ²	SG of Contents	1.000
Element Volume	721061.8 cm ³	Weight of Contents	720.6 Kg
Empty Weight	1169.6 Kg	Operating Weight	1890.2 Kg
		Comp. All. Ope. Stress	126.6 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 20.875 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.438 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (4566 \cdot 1676 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 4566)$$

$$= 27.6914 + 3.0000 = 30.6914 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 16.399 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 30) / (1 \cdot 1676 + 0.2 \cdot 30)$$

$$= 4946 - 16.4 = 4929 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 33) / (1 \cdot 1670 + 0.2 \cdot 33)$$

$$= 5432 \text{ KPa.}$$



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Element: Rhs Dish End

Step: 8

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Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (4566 \cdot (1 \cdot 1676 + 0.2 \cdot 30)) / (2 \cdot 1 \cdot 30)$$

$$= 127.413 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (4566 \cdot 838) / (138 \cdot 1 - 0.6 \cdot 4566) + 3$$

$$= 31.293 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 16.399 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 37) / (838 + 0.6 \cdot 37)$$

$$= 5935 - 16.4 = 5919 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (1676 / (2 \cdot 420)))^2 / 6$$

$$= 0.995252$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 9.872 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 33, tr = 29, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.97$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B
Min Metal Temp. at Required thickness (UCS 66.1)

7 °C

5 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 40, tr = 29.6, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.8$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B
Min Metal Temp. at Required thickness (UCS 66.1)

11 °C

0 °C

[Note:](#)

[Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.](#)

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4777.354 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	40.000 mm	Basic MDMT	11 °C
Stress/Thk. Ratio	0.801	Temperature Difference	11 °C
UCS-66(c) red.	No	Computed Min. Temp.	0 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 20.875 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.438 mm



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Shop Pressure Test Results: Step: 9 9:24am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6210.561 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	5915.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6558.139 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5255.090 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	6506.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	6506.500 KPa.
App 27-4	= M.A.W.P.	4777.354 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1423.935 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1648.255 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1204.868 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1095.335 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	100.3	234.0	0.429	1664.62
Main Shell	165.4	234.0	0.707	6574.51
Rhs Dish End	167.0	234.0	0.714	6574.51

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3-Tube Inlet	Nozzle	138.00	138.00	1.000
N4-Tube Outlet	Nozzle	138.00	138.00	1.000
N1-Shell Inlet	Nozzle	138.00	138.00	1.000
N2-Shell outlet	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Blind Flg	138.00	138.00	1.000
LHS Ch Flg	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Rhs Dish End	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
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Shop Pressure Test Results:

Step: 9

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Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3-Tube Inlet	28.41	225.00	0.126
N4-Tube Outlet	28.41	225.00	0.126
N1-Shell Inlet	75.96	225.00	0.338
N2-Shell outlet	75.96	225.00	0.338

Elements Suitable for Test Pressure.

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Field Pressure Test Results: Step: 10 9:24am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6210.561 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	5915.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6558.139 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5255.090 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	6506.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	6506.500 KPa.
App 27-4	= M.A.W.P.	4777.354 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1423.935 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1648.255 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1204.868 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1095.335 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 4550 • 1

= 5915.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 1000 • 1

= 1300.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	101.1	234.0	0.432	1316.37
Main Shell	163.9	234.0	0.700	5931.37
Rhs Dish End	165.5	234.0	0.707	5931.37

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3-Tube Inlet	Nozzle	138.00	138.00	1.000
N4-Tube Outlet	Nozzle	138.00	138.00	1.000
N1-Shell Inlet	Nozzle	138.00	138.00	1.000
N2-Shell outlet	Nozzle	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Blind Flg	138.00	138.00	1.000
LHS Ch Flg	138.00	138.00	1.000



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Field Pressure Test Results: Step: 10 9:24am Sep 9, 2026

LHS Ch Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Rhs Dish End	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3-Tube Inlet	22.47	225.00	0.100
N4-Tube Outlet	22.47	225.00	0.100
N1-Shell Inlet	68.53	225.00	0.305
N2-Shell outlet	68.53	225.00	0.305

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

Case: 1 9:24a Sep 9, 2026

Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		5915.00 KPa.
Shell Thickness	ts	34.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1670.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		1300.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1670.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1996
Tube Wall Thickness	et	2.7700 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	7000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6736.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	33.0000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.7700 mm
Groove Weld Leg	ag	2.7700 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	811.245 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	18959.3 cm ²
Perimeter of Tube Layout	Cp	5147.1 mm



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ASME TS Calc: Pressure Test

Case: 1 9:24a Sep 9,2026

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-266 2

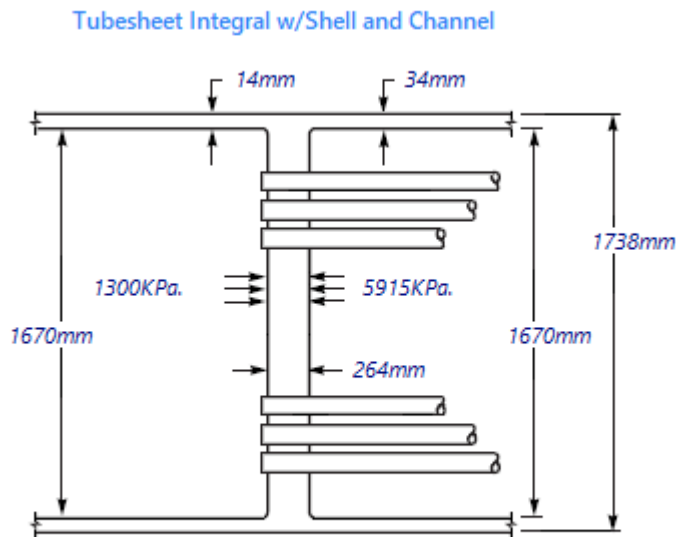
See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
Thickness of Tubesheet h 264.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
Tubesheet Outside Diameter A 1738.000 mm

Area of the Untubed Lanes AL 0.0 cm²

Junction Stress Reduction option Perform Plastic Calculation
Perform Differential Pressure Design No
Run Multiple Load Cases Yes
Channel Side Min. Design Pressure Ptd,min 103.4190 KPa.

Tubesheet Integral with Both
Tubesheet Extended as Flange No



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature 21.1 °C 0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.



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ASME TS Calc: Pressure Test

Case: 1 9:24a Sep 9,2026

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20132E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(1300 \cdot (25.4/2 - 0)) / (92.4 \cdot 1 + 0.4 \cdot 1300)$

= $0.1777 + 0.0000 = 0.1777$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	CS-1	at	21.11 °C
Elastic Modulus for Material			199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.7700	25.40	7547.24	9.17	50.0000	0.0130823	95.84

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 14665$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.2592	25.40	7547.24	20.17	50.0000	0.0027034	91.07

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 91.1) / (3 \cdot 20.2) = 6019$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.2592 mm
Allowable Internal Pressure at Corroded thickness	22078.03 KPa.
Required Internal Design Pressure	1300.00 KPa.
Allowable External Pressure at Corroded thickness	14664.90 KPa.
Required External Design Pressure	6018.42 KPa.
Required Thickness due to Shell Side pressure	1.2592 mm

Detailed Results for load Case D1 un-corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

= $2 \cdot ro + dt$
 = $2 \cdot 811 + 25.4$
 = 1647.890 mm

Basic ligament efficiency for shear [μ]:

= $(p - dt) / p$
 = $(33 - 25.4) / 33$
 = 0.230



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ASME TS Calc: Pressure Test

Case: 1 9:24a Sep 9, 2026

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 1670/1648$$

$$= 1.0134$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1670/1648$$

$$= 1.0134$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0/264$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (p); dt - 2tt)$$

$$= \max(25.4 - 2 \cdot 2.77 (202701232/201322320) \cdot (92.4/138) \cdot (0); 25.4 - 2 \cdot 2.77)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 33 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1648 \cdot 33) / (\pi \cdot 1648^2)) }$$

$$= 33.0000 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (33 - 25.4) / 33$$

$$= 0.2303$$

$h/p = 8.000000$; $\mu^* = 0.230303$; $E^* = 38647612 \text{ KPa}$.

$E^*/E = 0.191969$; $v^* = 0.418359$

Note: As h/p (8.000) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12(1 - v_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (1670 + 34) 34 }$$

$$= 0.0076 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2))$$

$$= 0.0076 \cdot 202701232 \cdot 34^3 / (6(1 - 0.3^2))$$

$$= 11020009472.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1670 + 34) \cdot 11.02E+09/264^3 \cdot (1 + 264 \cdot 0.0076 + (264 \cdot 0.0076)^2 / 2)$$

$$= 30503424.0 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 1670^2 / (4 \cdot 202701232 \cdot 34) \cdot (1 - 1670 \cdot 0.3 / (2(1670 + 34)))$$

$$= 0.86289E-01 \text{ mm/N./mm}^2$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$



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$$= 1.01 \cdot 11.02E+09 \cdot 0.0076 \cdot 0.08628948 (1 + 264 \cdot 0.0076)$$

$$= 21790.1309 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1670 + 14)14}$$

$$= 0.0118 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.012 \cdot 202701232 \cdot 14^3 / (6(1 - 0.3^2))$$

$$= 1206058880.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1670 + 14 / \cos(0)) \cdot 1206058880 / 264^3 \cdot (1 + 264 \cdot 0.012 \cdot (1 + 1) / 2 + (264 \cdot 0.012)^2 / 2 \cdot 1)$$

$$= 5967274.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1670^2 / (4 \cdot 202701232 \cdot 14) \cdot (1 - 1670 \cdot 0.3 / (2(1670 + 14)))$$

$$= 0.20913E+00 \text{ mm/N} \cdot \text{mm}^2$$

Channel coefficient [zeta α_c]:

$$zeta\alpha_c = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.01 \cdot 1206058880 [0.20913123 (1 + 0.012 \cdot 1 \cdot 264) - 0.000 (1/264 + 1 \cdot 0.012/2)]$$

$$= 12485.3799 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1738/1648$$

$$= 1.0547$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.42) / 38647612 \cdot (30503424 + 5967274 + 201322320 \cdot \ln(1.05))$$

$$= 0.7102$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot Pt]$$

$$= 1648^2 / 16 \cdot [(1.01 - 1) \cdot (1.01^2 + 1) \cdot 0 - (1.01 - 1) \cdot (1.01^2 + 1) \cdot 1300]$$

$$= -6000620.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps$$

$$= -6000620 + 12485 \cdot 1300 - 21790 \cdot 0$$

$$= 10230374.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:



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Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M \cdot) - Do^2/32 \cdot F (Ps - Pt)) / (1 + F)$$

$$= ((10230374) - 1648^2/32 \cdot 0.71 (0 - 1300)) / (1 + 0.71)$$

$$= 51793984.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu) \cdot (Ps - Pt)$$

$$= 51793984 + 1648^2/64 \cdot (3 + 0.42) \cdot (0 - 1300)$$

$$= -136760800.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp | ; | Mo |)$$

$$= \max(| 51793984 | ; | -136760800 |)$$

$$= 136760800.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 136760800 / (0.23 (264 - 8)^2)$$

$$= 54.3699 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot Sy$$

$$= 2 \cdot 0.95 \cdot 250$$

$$= 474.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 96372312 / (0.23 (80.7 - 8)^2)$$

$$= 474.6605 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 80.7 + 0 + 0$$

$$= 80.7316 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB; HreqS)$$

$$= \max(80.7 ; 10.9)$$

$$= 80.7316 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | Ps - Pt |$$

$$= | 0 - 1300 |$$

$$= 1300.000 \text{ KPa.}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h/Do \cdot 0.8 \cdot \beta T \cdot Sy$$

$$= 4 \cdot 0.23 \cdot 264/1648 \cdot 0.8 \cdot 0.95 \cdot 250$$

$$= 28.040 \text{ N./mm}^2$$

| Ps - Pt | is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot Sy)) \cdot (4 \cdot Ap/Cp) \cdot | Ps - Pt | + Cats + Catc$$

$$= 1 / (4 \cdot 0.23 \cdot 190) \cdot (4 \cdot 18959/5147) \cdot | 0 - 1300 | + 0$$

$$= 10.9441 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:



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Note: For a given Shell thickness of: 34.000mm
Min. Shell len. adjacent to the tubesheet is: 428.914 mm

Note: For a given Channel thickness of: 14.000 mm
Min. Channel len. adjacent to the tubesheet is: 275.229mm

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.77) 2.77$$

$$= 1.969 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.77 (25.4 - 2.77) 171$$

$$= 3433.791 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 171$$

$$= 3433.7905 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 171/171$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 171$$

$$= 2274.6389 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 3.372 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.6859 mm

Minimum Required Groove Weld Leg [agr]: 1.6859 mm

Tube-Tubesheet Jt allowable, 3434 is $\geq$ tube strength 3434 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 un-corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	54.4 $\leq$	475.0 N./mm ²	Ok
Tubesheet shear stress	7.9 $\leq$	190.0 N./mm ²	Ok

Thickness results for loadcase D1 un-corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	80.732	264.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.686	2.770	Ok
Tube-Tubesheet Groove Weld Leg :	1.686	2.770	Ok



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Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 811 + 25.4 \\ &= 1647.890 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (33 - 25.4) / 33 \\ &= 0.230 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 1670 / 1648 \\ &= 1.0134 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 1670 / 1648 \\ &= 1.0134 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned} &= l_{tx} / h \\ &= 0 / 264 \\ &= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned} &= \max(dt - 2tt(E_tT/E) \cdot (StT/S) \cdot (\rho); dt - 2tt) \\ &= \max(25.4 - 2 \cdot 2.77 (202701232/201322320) \cdot \\ &\quad (92.4/138) \cdot (0); 25.4 - 2 \cdot 2.77) \\ &= 25.4000 \text{ mm} \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\ &= 33 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1648 \cdot 33) / (\pi \cdot 1648^2)) } \\ &= 33.0000 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (33 - 25.4) / 33 \\ &= 0.2303 \end{aligned}$$

$h/p = 8.000000$; $\mu^* = 0.230303$; $E^* = 38647612 \text{ KPa}$.

$E^*/E = 0.191969$; $v^* = 0.418359$

Note: As h/p (8.000) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12 (1 - v_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$



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$$= ((12(1 - 0.3^2))^{1/4}) / ((1670 + 34)^{3/4})$$

$$= 0.0076 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0076 \cdot 202701232 \cdot 34^3 / (6(1 - 0.3^2))$$

$$= 11020009472.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1670 + 34) \cdot 11.02\text{E}+09 / 264^3 \cdot (1 + 264 \cdot 0.0076 + (264 \cdot 0.0076)^2 / 2)$$

$$= 30503424.0 \text{ KPa.}$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1670^2 / (4 \cdot 202701232 \cdot 34) \cdot (1 - 1670 \cdot 0.3 / (2(1670 + 34)))$$

$$= 0.86289\text{E}-01 \text{ mm/N./mm}^2$$

Parameter [ws]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.01 \cdot 11.02\text{E}+09 \cdot 0.0076 \cdot 0.08628948 (1 + 264 \cdot 0.0076)$$

$$= 21790.1309 \text{ mm}^2$$

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4}) / ((D_c + t_c)^{3/4})$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1670 + 14)^{3/4})$$

$$= 0.0118 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.012 \cdot 202701232 \cdot 14^3 / (6(1 - 0.3^2))$$

$$= 1206058880.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1670 + 14 / \cos(0)) \cdot 1206058880 / 264^3 \cdot (1 + 264 \cdot 0.012 \cdot (1 + 1) / 2 + (264 \cdot 0.012)^2 / 2 \cdot 1)$$

$$= 5967274.0 \text{ KPa.}$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1670^2 / (4 \cdot 202701232 \cdot 14) \cdot (1 - 1670 \cdot 0.3 / (2(1670 + 14)))$$

$$= 0.20913\text{E}+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [wc]:

$$= p_c \cdot k_c [\delta_c (CmTheta + \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.01 \cdot 1206058880 [0.20913123 (1 + 0.012 \cdot 1 \cdot 264) - 0.000 (1/264 + 1 \cdot 0.012/2)]$$

$$= 12485.3799 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1738/1648$$

$$= 1.0547$$



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Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.42) / 38647612 \cdot (30503424 + 5967274 + 201322320 \cdot \ln(1.05))$$

$$= 0.7102$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 1648^2 / 16 \cdot [(1.01 - 1) \cdot (1.01^2 + 1) \cdot 5915 - (1.01 - 1) \cdot (1.01^2 + 1) \cdot 1300]$$

$$= 21302202.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= 21302202 + 12485 \cdot 1300 - 21790 \cdot 5915$$

$$= -91355440.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^* - D_o^2 / 32 \cdot F(P_s - P_t)) / (1 + F)$$

$$= (-91355440 - 1648^2 / 32 \cdot 0.71 (5915 - 1300)) / (1 + 0.71)$$

$$= -216050880.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= -216050880 + 1648^2 / 64 \cdot (3 + 0.42) \cdot (5915 - 1300)$$

$$= 453318624.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|-216050880|; |453318624|)$$

$$= 453318624.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 453318624 / (0.23 (264 - 8)^2)$$

$$= 180.2189 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta_T \cdot S_y$$

$$= 2 \cdot 0.95 \cdot 250$$

$$= 474.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 402228864 / (0.23 (157 - 8)^2)$$

$$= 474.6359 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 157 + 0 + 0$$

$$= 156.5918 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(157; 38.9)$$

$$= 156.5918 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:



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Absolute pressure differential [| Ps - Pt |]:

$$= | P_s - P_t |$$

$$= | 5915 - 1300 |$$

$$= 4615.000 \text{ KPa.}$$

Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta T \cdot S_y$$

$$= 4 \cdot 0.23 \cdot 264 / 1648 \cdot 0.8 \cdot 0.95 \cdot 250$$

$$= 28.040 \text{ N./mm}^2$$

| Ps - Pt | is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{atc} + C_{atc}$$

$$= 1 / (4 \cdot 0.23 \cdot 190) \cdot (4 \cdot 18959 / 5147) \cdot | 5915 - 1300 | + 0$$

$$= 38.8515 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 34.000mm

Min. Shell len. adjacent to the tubesheet is: 428.914 mm

Note: For a given Channel thickness of: 14.000 mm

Min. Channel len. adjacent to the tubesheet is: 275.229mm

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.77) 2.77$$

$$= 1.969 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.77 (25.4 - 2.77) 171$$

$$= 3433.791 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 171$$

$$= 3433.7905 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 171 / 171$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 171$$

$$= 2274.6389 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 (\sqrt{ ((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) - 0.75 \cdot d_o })$$

$$= 2 (\sqrt{ ((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1 \cdot 1) - 0.75 \cdot 25.4 })$$

$$= 3.372 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.6859 mm

Minimum Required Groove Weld Leg [agr]: 1.6859 mm



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Tube-Tubesheet Jt allowable, 3434 is >= tube strength 3434 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	180.2 <=	475.0 N./mm ²	Ok
Tubesheet shear stress	28.0 <=	190.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	156.592	264.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.686	2.770	Ok
Tube-Tubesheet Groove Weld Leg :	1.686	2.770	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	80.732 ...	54	475	8 190	Ps+Pt D1	Ok
D2uc	182.941 ...	239	475	36 190	Ps+Pt D2	Ok
D3uc	156.592 ...	180	475	28 190	Ps+Pt D3	Ok
Max:	182.9414 ... mm	0.503	0.192	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Shell	Reqd. Thk. Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	370	...	370	Ok
D2uc	...	...	...	370	...	370	Ok
D3uc	...	...	...	370	...	370	Ok
Max	...	...	...	...	...	...	

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.



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ASME TS Calc: Pressure Test

Case: 1 9:24a Sep 9, 2026

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	182.941	264.000	Ok
Tube thickness :	1.259	2.770	Ok
Tube-tubesheet fillet weld leg :	1.686	2.770	Ok
Tube-tubesheet groove weld leg :	1.686	2.770	Ok

Given Shell Thickness: 34.0000 mm

Given Channel Thickness: 14.0000 mm

Min Shell length of thk, (34.000) adj. to tubesheet: 428.914 mm

Min Channel length of thk, (14.000) adj. to tubesheet: 275.229 mm

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	117	No Calc	No Calc
20	30	No Calc	...	157	No Calc	No Calc
30	Tsht	1599.97	1698	11	0.00074491	74.474
Tsht	40	No Calc	...	11	No Calc	No Calc
40	80	7790	1738	31	0.00063061	61.0472
80	90	No Calc	1736	30	0.0024002	95.7353

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	120	90.1192	103.419	No Calc
20	30	160	152.756	103.419	No Calc
30	Tsht	14	8.30599	103.419	643
Tsht	40	14	No Calc	103.419	No Calc
40	80	34	No Calc	...	1452
80	90	33	No Calc	...	1838

Minimum

643

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1599.97	8957.51	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	80	7790	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

Step: 12

9:24am

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	2753.88	...	2753.88	...	275.388
20	30	796.904	475860	770.507	476617	79.6904
30	40	918.482	3505262	722.947	3530496	91.8482
40	80	10721.2	8547025	9792.37	8666882	1072.12
80	90	1169.59	721062	1081.87	728459	116.959
Total		16360	13249209.00	15121	13402454.00	1636



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Liqd	...	60	11	...	Liquid: 10
10	Insl	101.009	45	...	...	Insulation
20	Liqd	474.515	104	9.99997	...	Liquid: 20
20	Insl	22.5429	71.5	...	...	Insulation
30	Liqd	3503.12	800	...	...	Liquid: 30
30	Insl	251.966	800	...	...	Insulation
30	Nozl	107.298	700	1139.8	...	N3-Tube Inlet
30	Nozl	153.689	700	-1139.8	...	N4-Tube Outlet
30	Wght	200	600	...	...	pass partition
40	Sadl	499.418	1520	1108.5	...	LH saddle float
40	Sadl	499.418	6080	1108.5	...	RHS saddle fixe
40	Liqd	8541.81	3800	...	...	Liquid: 40
40	Nozl	367.619	3800	1038.2	...	N1-Shell Inlet
40	Nozl	367.619	3800	-1038.2	...	N2-Shell outlet
40	Wght	450	1520	-1420	...	Fire prrof
40	Wght	450	6180	-1420	...	Fire prrof 1
80	Liqd	720.622	189.967	0.72670E-04	...	Liquid: 50
30	FTsh	3063.31	1732	...	...	
30	Tube	26478.2	5100	...	...	Tubes
40	Bafl	185.949	1750	...	...	Baffles
40	Bafl	185.949	3500	...	...	Baffles
40	Bafl	185.949	5250	...	...	Baffles
40	Bafl	186.585	7000	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	998.8
Liquid	13240.1
Insulation	375.5
Nozzles	996.2
Weights	1100.0
Exchanger Components	30285.9
Liquid in Tubes	5113.9

Sum of the Detail Weights 52110.4 kg.

Weight Summation Results: (kg.)



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Element and Detail Weights:

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	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	17996.0	17996.0	17996.0	17996.0	17996.0	17996.0
Saddles	998.8	998.8	998.8	998.8	998.8	998.8
Nozzles	996.2	996.2	996.2	996.2	996.2	996.2
Wld Weights	200.0	200.0	200.0	200.0	200.0	200.0
Exchanger	30285.9	30285.9	30285.9	30285.9	30285.9	30285.9
Insulation	...	...	...	...	375.5	375.5 *
Test Weights	...	900.0	...	...	...	...
Ope Weights	...	...	...	...	...	900.0
Ope. Liquid	...	...	...	...	...	13240.1
Tube Ope Lqd	...	...	...	...	...	5113.9
Test Liquid	...	13241.1	...	...	...	...
Tube Tst Lqd	...	5113.9	...	...	...	...
Totals	50477.0	69732.0	50477.0	50477.0	50852.5	70106.5

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	50477.0 kg.
Shop Test Wt.	- Fabricated weight + water (full)	69732.0 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	50477.0 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	50477.0 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	50852.5 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	70106.5 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	68744.2 kg.
Field Test Wt.	- Empty weight + test medium (full)	70107.5 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	5117031.0 cm ³
Weight of Ope Liq in tubes :	5113.9 kg.
Weight of Water in tubes :	5113.9 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	35319	...
20	30	18359.2	11332.4
30	40	85350.8	83943.4
40	80	414967	398731
80	90	35989.9	32896.5

Total	589985.7	526903.1 cm ²
Total	635.1	567.2 ft ²



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Nozzle Flange MAWP:

Step: 13 9:24am Sep 9,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N1-Shell Inlet	8760.1	10200.0	200	600	GR 1.1	2116.3	8760.1	8760.1	8760.1
N2-Shell outle	8760.1	10200.0	200	600	GR 1.1	2116.3	8760.1	8760.1	8760.1

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 8760.053 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 10200.003 KPa.

Channelside Flange Rating

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N1-Shell Inlet	438.150	0.50	3229	3059	4777.4	8760.1	Pass
N2-Shell outlet	438.150	0.50	3229	3059	4777.4	8760.1	Pass

Evaluating Flange on Nozzle [N1-Shell Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(3229) + 4 \cdot \text{abs}(3059) 438.150 \leq \pi \cdot 438.150^3 [(8760.053 - 4777.354) + .500 \cdot 8760.053]$$

$$= 0.57030E+05 \leq 0.22536E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-Shell Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8760.053, 0.500 \cdot 8760.053 + 8760.053 - 57030.1 / (\pi \cdot 438.150^3))$$

$$= \min(8760.053, 11023.762)$$

$$= 8760.053 \text{ KPa.}$$

Evaluating Flange on Nozzle [N2-Shell outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(3229) + 4 \cdot \text{abs}(3059) 438.150 \leq \pi \cdot 438.150^3 [(8760.053 - 4777.354) + .500 \cdot 8760.053]$$

$$= 0.57030E+05 \leq 0.22536E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-Shell outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8760.053, 0.500 \cdot 8760.053 + 8760.053 - 57030.1 / (\pi \cdot 438.150^3))$$

$$= \min(8760.053, 11023.761)$$

$$= 8760.053 \text{ KPa.}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation:

Step: 14 9:24am Sep 9,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	16000 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.2 \cdot 0.93)/(1 + 1.7 \cdot 3.4 \cdot 0.2)))$$

$$= \min(0.85, 0.89)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.584
Structure Height to Diameter ratio	6.060

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274320 \text{ mm}$

Effective Height [z]:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 1430 + 16000 = 17430 \text{ mm}$$

$$= 57.2 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\text{Because } z (57.2 \text{ ft.}) > 15 \text{ ft.}$$

$$= 2.01 (z / Z_g)^{2/\alpha}$$

$$= 2.01 (57.2/900)^{2/9.5}$$

$$= 1.13$$

Type of Hill: No Hill



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Wind Load Calculation:

Step: 14

9:24am

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Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 1.13 • 1 • 1 • 1 • 60.4²)

= 16 psf [78.1] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.58 • 3.1

= 24.6 lbs. [11.2] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Blind Flg	17430.0	0.000	0.000	0.000	1.125	1.000	78.120
LHS Ch Flg	17430.0	0.000	0.000	0.000	1.125	1.000	78.120
LHS Ch Shell	17430.0	0.000	0.000	0.000	1.125	1.000	78.120
Main Shell	17430.0	0.000	0.000	0.000	1.125	1.000	78.120
Rhs Dish End	17430.0	0.000	0.000	0.000	1.125	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
pass partitio	0.00	17430.00	78.12	0.00
Fire prrof	0.00	16000.00	78.12	0.00
Fire prrof 1	0.00	16000.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	17430	2400	2880	78.12	12.2922
20	30	17430	2136	4442.88	78.12	18.9628
30	40	17430	2229.6	35673.6	78.12	152.259
40	80	17430	2085.6	158506	78.12	676.521
80	90	17430	2083.2	7200.61	78.12	30.7331

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation: Step: 15 9:24am Sep 9,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hnx$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(7.780.75)$
 $= 0.093$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.093, 0.03) \text{ per 12.8.2}$
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.16 \cdot 69108 \cdot 1.25$
 $= 4063.529$ Kgf

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 69108$
 $= 2167.2$ Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	940	9872.52	641.969



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Earthquake Load Calculation:

Step: 15

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20	30	835	9872.52	570.26
30	40	835	9872.52	570.26
40	Sad1	835	9872.52	570.26
Sad1	80	835	9872.52	570.26
40	80	835	9872.52	570.26
80	90	835	9872.52	570.26

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 16 9:24am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5995.175 mm
Center of Gravity of the Liquid	4718.676 mm
Center of Gravity of the Insulation	782.790 mm
Center of Gravity of the Nozzles	4694.726 mm
Center of Gravity of the Added Weights (Operating)	5115.357 mm
Center of Gravity of the Added Weights (Empty)	931.175 mm
Center of Gravity of the Tubesheet(s)	2063.175 mm
Center of Gravity of the Tubes	5431.175 mm
Center of Gravity of the Baffles	6572.421 mm
Center of Gravity of Bare Shell New and Cold	4731.426 mm
Center of Gravity of Bare Shell Corroded	4677.455 mm
Vessel CG in the Operating Condition	4940.928 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4972.933 mm
Vessel CG in the Test Condition	4958.522 mm

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Saddle Calcs: Road Transport:

Step: 17

9:24am

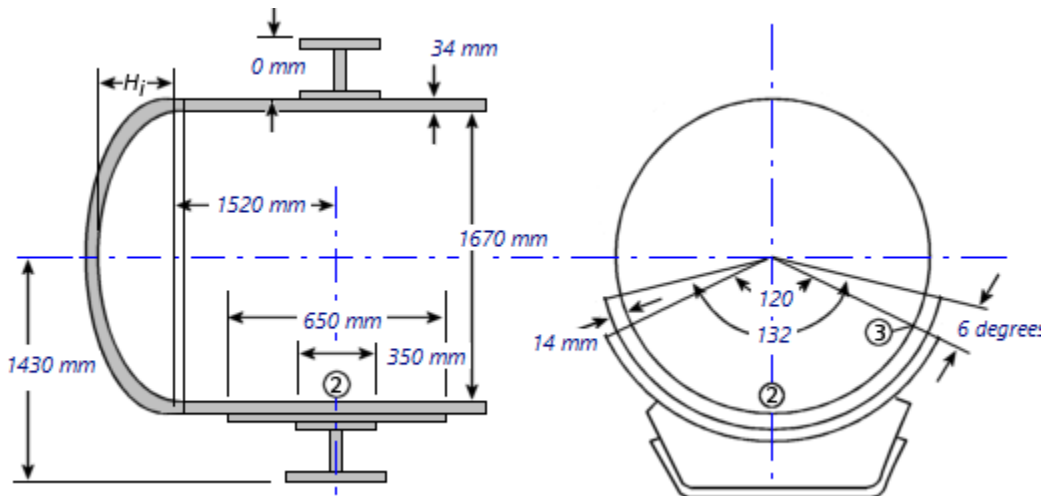
Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	852.00	mm
Shell Thickness used in this Case	t	34.000	mm
Stiffened Vessel Length per 4.15.6	L	7650.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1520.00	mm
Saddle Width	b1 or b	350.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	650.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1430.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		45414.88	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	333.7	Km/hr



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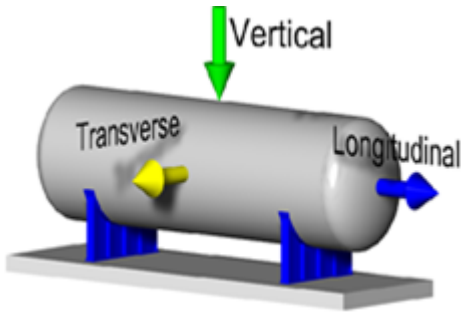
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Saddle Calcs: Road Transport:

Step: 17

9:24am

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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-2.53	-234.00
Long. Stress at Top of Midspan	2.53	138.00
Long. Stress at Bottom of Midspan	2.53	138.00
Long. Stress at Top of Saddles	13.71	138.00
Long. Stress at Bottom of Saddles	-7.60	-234.00
Long. Stress at Bottom of Saddles	7.60	138.00
Tangential Shear in Shell	10.85	110.40
Circ. Stress at Horn of Saddle	19.09	207.00
Circ. Compressive Stress in Shell	1.15	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 225 \cdot 0.00256 \cdot 207^2 \cdot 1 \\
 &= 24730.0 \text{ lbf} \\
 &= 11217.507 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 6.23^2/4 \cdot 0.00256 \cdot 207^2 \cdot 1 \cdot 1.2 \\
 &= 4023.1 \text{ lbf} \\
 &= 1824.9 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$\begin{aligned}
 &= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw} \\
 &= 0.1 \cdot 49478 \cdot 1 + 11218 \\
 &= 16165.3 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$\begin{aligned}
 &= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw} \\
 &= 0.1 \cdot 49478 \cdot 1 + 1825 \\
 &= 6772.7 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$\begin{aligned}
 &= F_t / \text{Num of Saddles} \cdot B / E \\
 &= 16165/2 \cdot 1430/1529 \\
 &= 7557.3 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$\begin{aligned}
 &= F_l \cdot B / L_s \\
 &= 6773 \cdot 1430/4560 \\
 &= 2123.9 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot g_y \\
 &= 36055 \cdot 0.05
 \end{aligned}$$



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= 1802.740 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= 36055 + 1803 + max(2124, 7557)

= 45414.9 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 45914.29 Kgf

Transverse Shear Load 8082.66 Kgf

Longitudinal Shear Load 6772.68 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)

= min(2(1670/2 + 34 + 14)sin(120/2), 2•852)

= 1529.401 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -45415•1520[1-(1-1520/7651+(852²-0²)/

(2•1520•7651))/(1+(4•0)/(3•7651))]

= -11560.2 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 45415•7651/4((1+2(852²-0²)/(7651²))/(1+(4•0)/

(3•7651))-4•1520/7651)

= 19989.3 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 0 • 852/(2•34) - 19989/(π•852²•34)

= -2.53 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 0 • 852/(2 • 34) + 19989/(π • 852² • 34)

= 2.53 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)

= 0•852/(2•34)--11560/(0.11•π•852²•34)

= 13.71 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 0•852/(2•34)+-11560/(0.19•π•852²•34)

= -7.60 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 45415 (7651 - 2 • 1520)/(7651 + (4 • 0/3))

= 27369.5 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 27370 / (852 \cdot 34)$$

$$= 10.85 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(852 \cdot 34)}$$

$$= 132.756 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(350 + 1.56 \cdot \sqrt{(852 \cdot 34)}, 2 \cdot 1520)$$

$$= 615.51 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 45415 \cdot 0.1 / (616(34 + 1 \cdot 14))$$

$$= -1.15 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -45415 / (4(34 + 1 \cdot 14)616) -$$

$$3 \cdot 0.053 \cdot 45415 / (2(34 + 1 \cdot 14)^2)$$

$$= -19.09 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1540.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	375.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	605.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(835 \cdot 34)} + 650)34^3) / 12$$

$$= 298.988 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 299 + 310(154 - 17)^2$$

$$= 58549.211 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	912.8	34.0	17.0	310.4	527627.1	0.585E+05
Wearplate	650.0	14.0	41.0	91.0	373100.0	0.116E+05
Web	10.0	519.0	307.5	51.9	1595925.1	0.239E+05
BasePlate	375.0	28.0	581.0	105.0	6100501.0	0.192E+06
Totals	...	...	...	558.3	8597153.0	0.286E+06

Distance to Centroid [C1]:

$$= AY / A$$



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$$= 8597153/558$$

$$= 153.997 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 45415$$

$$= 9242.9121 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 3.6565 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1430 - 835 \cdot \sin(120/2) / 1.05$$

$$= 739.461 \text{ mm}$$

Bending Moment, M = Fh * d = 6834.9043 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 3.6144 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(45415 + 499)375) / (4 \cdot 1540 \cdot 173)}$$

$$= 21.781 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (835 + 34 + 14)$$

$$= 1529.401 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1529 / (4 - 1)$$

$$= 509.800 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 510 \cdot 375/2$$

$$= 955.876 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 45914 / (1540 \cdot 375)$$

$$= 7.951 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 956 \cdot 7.95$$

$$= 7599.715 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



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$$= 34 + 25.5$$

$$= 59.490 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 7600/59.5$$

$$= 12.528 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	350.0	...	35.0	...	0.357E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 350 - 10$$

$$= 340.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/59.5$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 350/2$$

$$= 175.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3573/59.5)}$$

$$= 77.498 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1430 - \cos(120/2) (835 + 34 + 14) - 28$$

$$= 960.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 960/77.5$$

$$= 12.394$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 6773 / (2 \cdot 1540) \cdot 510 \cdot 960/2$$

$$= 538.376 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($12.4 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.4)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (12.4) / (8 \cdot 123) - (12.4^3) / (8 \cdot 123^3))$$

$$= 151.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 12.5/152 + (538 \cdot 175/35729164) / 173$$



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= 0.232

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	340.0	0.0	34.0	0.0	0.357E+04
Web	509.8	10.0	0.0	51.0	0.0	4.25
Totals	...	...	...	85.0	...	0.358E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/85

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 350/2

= 175.000 mm

Length of Inner Rib [L]:

= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk

= 1430 - $\sqrt{(883 + 14)^2 - (510/2)^2}$ - 28

= 556.592 mm

Radius of Gyration [r]:

= $\sqrt{(Total\ Inertia / Total\ Area)}$

= $\sqrt{(3577/85)}$

= 64.879 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 557/64.9

= 8.579

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 6773/(2 • 1540)

= 2.199 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 2.2 • 510 • 557

= 623.958 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σB,base,c]:

= Bending Moment / Section Modulus

= 624/204405

= 29.935 N./mm²

Compressive Allowable, KL/r < Cc (8.58 < 123) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³)/(8•Cc³)

= (1-(8.58)²/(2 • 123²))260/

(5/3+3•(8.58)/(8• 123)-(8.58³)/(8•123³)

= 153.2 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 17.5/153 + (624 • 175/3577)/173

= 0.287

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



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Vessel Mean Radius	Rm	852.00	mm
Shell Thickness used in this Case	t	34.000	mm
Stiffened Vessel Length per 4.15.6	L	7650.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1520.00	mm
Saddle Width	b1 or b	350.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	650.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	434.00	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1430.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		21651.85	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	333.7	Km/hr

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-0.80	-234.00
Long. Stress at Top of Midspan	0.80	138.00
Long. Stress at Bottom of Midspan	0.80	138.00
Long. Stress at Top of Saddles	9.12	138.00
Long. Stress at Bottom of Saddles	-5.05	-234.00
Long. Stress at Bottom of Saddles	5.05	138.00
Tangential Shear in Shell	4.81	110.40
Circ. Stress at Horn of Saddle	9.10	207.00
Circ. Compressive Stress in Shell	0.55	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 225 \cdot 0.00256 \cdot 207^2 \cdot 1 \\
 &= 24730.0 \text{ lbf} \\
 &= 11217.507 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 6.23^2/4 \cdot 0.00256 \cdot 207^2 \cdot 1 \cdot 1.2 \\
 &= 4023.1 \text{ lbf} \\
 &= 1824.9 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:



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$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + Ftw \\
 &= 0.1 \cdot 49478 \cdot 1 + 11218 \\
 &= 16165.3 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [FI]:

$$\begin{aligned}
 &= gx \cdot \text{Shipping Weight} \cdot gw + Flw \\
 &= 0.1 \cdot 49478 \cdot 1 + 1825 \\
 &= 6772.7 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [Fwt]:

$$\begin{aligned}
 &= Ft/\text{Num of Saddles} \cdot B / E \\
 &= 16165/2 \cdot 1430/1529 \\
 &= 7557.3 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [Fwl]:

$$\begin{aligned}
 &= FI \cdot B / Ls \\
 &= 6773 \cdot 1430/4560 \\
 &= 2123.9 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot gy \\
 &= 13423 \cdot 0.05 \\
 &= 671.167 \text{ Kgf}
 \end{aligned}$$

Load Combination Results due to Transport Loading [Q]:

$$\begin{aligned}
 &= \text{Saddle Load} + Frv + \max(Fwl, Fwt) \\
 &= 13423 + 671 + \max(2124, 7557) \\
 &= 21651.9 \text{ Kgf}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	22151.27	Kgf
Transverse Shear Load	8082.66	Kgf
Longitudinal Shear Load	6772.68	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm) \\
 &= \min(2(1670/2 + 34 + 14) \sin(120/2), 2 \cdot 852) \\
 &= 1529.401 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)] \\
 &= -21652 \cdot 1520 [1 - (1 - 1520/7651 + (852^2 - 434^2)/(2 \cdot 1520 \cdot 7651))/(1 + (4 \cdot 434)/(3 \cdot 7651))] \\
 &= -7685.9 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L) \\
 &= 21652 \cdot 7651/4((1 + 2(852^2 - 434^2)/(7651^2))/(1 + (4 \cdot 434)/(3 \cdot 7651)) - 4 \cdot 1520/7651) \\
 &= 6297.9 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$



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$$= 0 \cdot 852 / (2 \cdot 34) - 6298 / (\pi \cdot 852^2 \cdot 34)$$

$$= -0.80 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 852 / (2 \cdot 34) + 6298 / (\pi \cdot 852^2 \cdot 34)$$

$$= 0.80 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 0 \cdot 852 / (2 \cdot 34) - 7686 / (0.11 \cdot \pi \cdot 852^2 \cdot 34)$$

$$= 9.12 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 852 / (2 \cdot 34) + 7686 / (0.19 \cdot \pi \cdot 852^2 \cdot 34)$$

$$= -5.05 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 21652 (7651 - 2 \cdot 1520) / (7651 + (4 \cdot 434 / 3))$$

$$= 12131.1 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 12131 / (852 \cdot 34)$$

$$= 4.81 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(852 \cdot 34)}$$

$$= 132.756 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(350 + 1.56 \cdot \sqrt{(852 \cdot 34)}, 2 \cdot 1520)$$

$$= 615.51 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 21652 \cdot 0.1 / (616(34 + 1 \cdot 14))$$

$$= -0.55 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -21652 / (4(34 + 1 \cdot 14)616) -$$

$$3 \cdot 0.053 \cdot 21652 / (2(34 + 1 \cdot 14)^2)$$

$$= -9.10 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1540.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	375.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	605.0	mm
Friction Coefficient	mu	0.000	



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Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(835 \cdot 34)} + 650)^3) / 12$$

$$= 298.988 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 299 + 310 (154 - 17)^2$$

$$= 58549.211 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	912.8	34.0	17.0	310.4	527627.1	0.585E+05
Wearplate	650.0	14.0	41.0	91.0	373100.0	0.116E+05
Web	10.0	519.0	307.5	51.9	1595925.1	0.239E+05
BasePlate	375.0	28.0	581.0	105.0	6100501.0	0.192E+06
Totals	...	...	...	558.3	8597153.0	0.286E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 8597153 / 558$$

$$= 153.997 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 21652$$

$$= 4406.6211 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.7432 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1430 - 835 \cdot \sin(120 / 2) / 1.05$$

$$= 739.461 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 3258.5872 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.7232 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(21652 + 499) \cdot 375) / (4 \cdot 1540 \cdot 173)}$$

$$= 15.129 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120 / 2) (835 + 34 + 14)$$

$$= 1529.401 \text{ mm}$$



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Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 1529 / (4 - 1)$$

$$= 509.800 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 510 \cdot 375 / 2$$

$$= 955.876 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 22151 / (1540 \cdot 375)$$

$$= 3.836 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 956 \cdot 3.84$$

$$= 3666.469 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 34 + 25.5$$

$$= 59.490 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 3666 / 59.5$$

$$= 6.044 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	350.0	...	35.0	...	0.357E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 350 - 10$$

$$= 340.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 59.5$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 350 / 2$$

$$= 175.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (3573 / 59.5) }$$

$$= 77.498 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1430 - \cos(120 / 2) (835 + 34 + 14) - 28$$

$$= 960.500 \text{ mm}$$

Intermediate Term [Cc]:



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$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 960 / 77.5$$

$$= 12.394$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 6773 / (2 \cdot 1540) \cdot 510 \cdot 960 / 2$$

$$= 538.376 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($12.4 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.4)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (12.4) / (8 \cdot 123) - (12.4^3) / (8 \cdot 123^3))$$

$$= 151.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 6.04 / 152 + (538 \cdot 175 / 35729164) / 173$$

$$= 0.189$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	340.0	0.0	34.0	0.0	0.357E+04
Web	509.8	10.0	0.0	51.0	0.0	4.25
Totals	...	...	...	85.0	...	0.358E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 85$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 350 / 2$$

$$= 175.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1430 - \sqrt{(883 + 14)^2 - (510/2)^2} - 28$$

$$= 556.592 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3577 / 85)}$$

$$= 64.879 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 557 / 64.9$$

$$= 8.579$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 6773 / (2 \cdot 1540)$$

$$= 2.199 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:



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$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 2.2 \cdot 510 \cdot 557 \\
 &= 623.958 \text{ Kg-m.}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 624 / 204405 \\
 &= 29.935 \text{ N./mm}^2
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($8.58 < 123$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (8.58)^2 / (2 \cdot 123^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (8.58) / (8 \cdot 123) - (8.58^3) / (8 \cdot 123^3)) \\
 &= 153.2 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba} \\
 &= 8.47 / 153 + (624 \cdot 175 / 3577) / 173 \\
 &= 0.228
 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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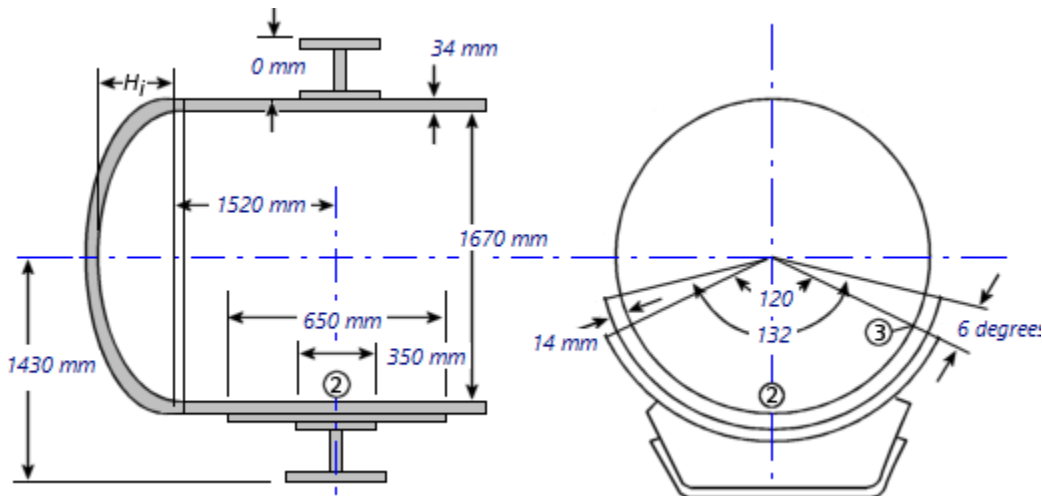
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	853.50	mm
Shell Thickness used in this Case	t	34.000	mm
Stiffened Vessel Length per 4.15.6	L	7650.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1520.00	mm
Saddle Width	b1 or b	350.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	650.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1430.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	57410.51	Kgf
Pressure used in Analysis	P 4558.185	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	59.26	138.00
Long. Stress at Bottom of Midspan	66.25	138.00
Long. Stress at Top of Saddles	81.69	138.00
Long. Stress at Bottom of Saddles	52.26	138.00
Tangential Shear in Shell	15.01	110.40
Circ. Stress at Horn of Saddle	27.22	172.50
Circ. Compressive Stress in Shell	1.58	138.00



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (891/2 + 0) \cdot 1430/1529$$

$$= 1249.3 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(350, 0) \cdot 1430/4560$$

$$= 109.8 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(4064, 0) \cdot 1430/4560$$

$$= 1274.3 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (4064/2 + 0) \cdot 1430/1529$$

$$= 5699.1 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 51711 + \max(110, 1249, 1274, 5699)$$

$$= 57410.5 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (1.9 \cdot 1.2)^2)$$

$$= 3433.324 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	57909.93	Kgf
Transverse Shear Load	2031.76	Kgf
Longitudinal Shear Load	4063.53	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1670/2 + 34 + 14) \sin(120/2), 2 \cdot 854)$$

$$= 1529.401 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -57411 \cdot 1529 [1 - (1 - 1520/7651 + (854^2 - 0^2)/(2 \cdot 1520 \cdot 7651))/(1 + (4 \cdot 0)/(3 \cdot 7651))]$$

$$= -14604.1 \text{ Kg-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 57411 \cdot 7651/4 ((1 + 2(854^2 - 0^2)/(7651^2))/(1 + (4 \cdot 0)/$$



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$$(3 \cdot 7651) - 4 \cdot 1520 / 7651 \\ = 25278.7 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t) \\ = 4558 \cdot 854 / (2 \cdot 31) - 25279 / (\pi \cdot 854^2 \cdot 31) \\ = 59.26 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 4558 \cdot 854 / (2 \cdot 31) + 25279 / (\pi \cdot 854^2 \cdot 31) \\ = 66.25 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t) \\ = 4558 \cdot 854 / (2 \cdot 31) - 14604 / (0.11 \cdot \pi \cdot 854^2 \cdot 31) \\ = 81.69 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 4558 \cdot 854 / (2 \cdot 31) + 14604 / (0.19 \cdot \pi \cdot 854^2 \cdot 31) \\ = 52.26 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 57411 (7651 - 2 \cdot 1520) / (7651 + (4 \cdot 0 / 3)) \\ = 34598.8 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t) \\ = 1.17 \cdot 34599 / (854 \cdot 31) \\ = 15.01 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{854 \cdot 31} \\ = 126.875 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(350 + 1.56 \cdot \sqrt{854 \cdot 31}, 2 \cdot 1520) \\ = 603.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1) \\ = \min(138 / 138, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr)) \\ = -0.76 \cdot 57411 \cdot 0.1 / (604(31 + 1 \cdot 14)) \\ = -1.58 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2) \\ = -57411 / (4(31 + 1 \cdot 14)604) - 3 \cdot 0.053 \cdot 57411 / (2(31 + 1 \cdot 14)^2) \\ = -27.22 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 4560.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.12750 \text{E-04} \cdot 4560 (200 - 21.1)$$



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= 10.400 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1540.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	375.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	225.0	N./mm ²
Height of Web at Center	Hw,c	605.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(838 \cdot 31)} + 650)31^3)/12$$

$$= 223.789 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 224 + 279(160 - 15.5)^2$$

$$= 58431.355 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	901.4	31.0	15.5	279.4	433140.1	0.584E+05
Wearplate	650.0	14.0	38.0	91.0	345800.0	0.135E+05
Web	10.0	519.0	304.5	51.9	1580354.6	0.225E+05
BasePlate	375.0	28.0	578.0	105.0	6068999.5	0.184E+06
Totals	...	...	...	527.3	8428294.0	0.278E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 8428294/527$$

$$= 159.825 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 57411$$

$$= 11684.2852 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh/As) = 4.6223 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 134.9988 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1430 - 838 \cdot \sin(120/2) / 1.05$$

$$= 736.980 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 8611.2500 \text{ Kg-m.}$$



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Bending Stress, $S_b = (M * C_1 / I) = 4.8524 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 * \text{Yield Str} = 149.9987 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{(3(57411 + 499)375 / (4 * 1540 * 150))}$
 $= 26.296 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120/2) (835 + 34 + 14)$
 $= 1529.401 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1529 / (4 - 1)$
 $= 509.800 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 510 * 375 / 2$
 $= 955.876 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 57910 / (1540 * 375)$
 $= 10.028 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p * B_p$
 $= 956 * 10$
 $= 9585.228 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 34 + 25.5$
 $= 59.490 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 9585 / 59.5$
 $= 15.801 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	350.0	...	35.0	...	0.357E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 350 - 10$
 $= 340.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 59.5$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:



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= Saddle Width / 2
= 350/2
= 175.000 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{3573/59.5}$
= 77.498 mm

Length of Outer Rib [L]:
= Saddle Height - $\cos(\theta/2) \cdot (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
= 1430 - $\cos(120/2) \cdot (835 + 34 + 14) - 28$
= 960.500 mm

Intermediate Term [Cc]:
= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199943008/225}$
= 132.447

Slenderness ratio [KL/r]:
= KL/r
= 1 • 960/77.5
= 12.394

Bending Moment [Rm]:
= $Fl / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$
= 4064 / $(2 \cdot 1540) \cdot 510 \cdot 960/2$
= 323.019 Kg-m.

Compressive Allowable, $KL/r < Cc$ (12.4 < 132) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (12.4)^2 / (2 \cdot 132^2)) 225 / (5/3 + 3 \cdot (12.4) / (8 \cdot 132) - (12.4)^3 / (8 \cdot 132^3))$
= 131.6 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
= $S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$
= 15.8/132 + $(323 \cdot 175/35729164) / 150$
= 0.223

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	340.0	0.0	34.0	0.0	0.357E+04
Web	509.8	10.0	0.0	51.0	0.0	4.25
Totals	...	...	...	85.0	...	0.358E+04

Distance to Centroid from Datum [ytot]:
= AY / A
= 0/85
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 350/2
= 175.000 mm

Length of Inner Rib [L]:
= Saddle Height - $\sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$
= 1430 - $\sqrt{(883 + 14)^2 - (510/2)^2} - 28$
= 556.592 mm

Radius of Gyration [r]:



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$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{(3577/85)}$$

$$= 64.879 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 557/64.9$$

$$= 8.579$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 4064 / (2 \cdot 1540)$$

$$= 1.319 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.32 \cdot 510 \cdot 557$$

$$= 374.368 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 374/204405$$

$$= 17.961 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.58 < 132$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.58)^2 / (2 \cdot 132^2)) 225 /$$

$$(5/3 + 3 \cdot (8.58) / (8 \cdot 132) - (8.58)^3 / (8 \cdot 132^3))$$

$$= 132.8 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 22.1/133 + (374 \cdot 175/3577) / 150$$

$$= 0.286$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load Q0 (Weight)	Q0	52210.8 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1274.3 Kg
Maximum Transverse Force	Ft	2031.8 Kg
Maximum Longitudinal Force	F1	4063.5 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 4064 / (4.87 \cdot 6)$$

$$= 13.6 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2032 / (4.87 \cdot 6)$$

$$= 6.8 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$
= $4064 / (82.8 \cdot 6)$
= 0.8021 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1430 \cdot 2032 + 0$
= 2905.48 Kg-m.

Eccentricity (e):
= R_{mom} / QO
= 2905/52211
= 55.65 mm < $B_{\text{plen}}/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0, 0.8, 0]$
= 0.8021 cm²

MDMT Calculations for: LH saddle float, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 14$, $c = 3$ mm, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.96$, Temp. Reduction = 2 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	853.50	mm
Shell Thickness used in this Case	t	34.000	mm
Stiffened Vessel Length per 4.15.6	L	7650.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1520.00	mm
Saddle Width	b1 or b	350.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	650.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	435.50	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1430.00	mm
Coefficient of Friction	mu	0.00	



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Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 25262.61 Kgf
Pressure used in Analysis P 4558.185 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	61.74	138.00
Long. Stress at Bottom of Midspan	63.77	138.00
Long. Stress at Top of Saddles	74.39	138.00
Long. Stress at Bottom of Saddles	56.30	138.00
Tangential Shear in Shell	6.14	110.40
Circ. Stress at Horn of Saddle	11.98	172.50
Circ. Compressive Stress in Shell	0.69	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3 (891/2 + 0) • 1430/1529
= 1249.3 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
= max(Fl,wind; Sum of X Forces) • B / Ls
= max(350, 0) • 1430/4560
= 109.8 Kgf

Saddle Reaction Force due to Earthquake Fl [Fsl]:
= max(Fl,eq; Sum of X Forces) • B / Ls
= max(4064, 0) • 1430/4560
= 1274.3 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3 (4064/2 + 0) • 1430/1529
= 5699.1 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
= 19563 + max(110, 1249, 1274, 5699)
= 25262.6 Kgf

Longitudinal Wind Force [FI]:
= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
= 1.1 • 766 (0 + $\pi/4$ (1.9 • 1.2)²)
= 3433.324 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	25762.03	Kgf
Transverse Shear Load	2031.76	Kgf
Longitudinal Shear Load	4063.53	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)
= min(2(1670/2 + 34 + 14)sin(120/2), 2•854)
= 1529.401 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011



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K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -25263 \cdot 1520 [1 - (1 - 1520/7651 + (854^2 - 436^2)/(2 \cdot 1520 \cdot 7651))/(1 + (4 \cdot 436)/(3 \cdot 7651))]$$

$$= -8972.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L]$$

$$= 25263 \cdot 7651/4 [(1 + 2(854^2 - 436^2)/(7651^2))/(1 + (4 \cdot 436)/(3 \cdot 7651)) - 4 \cdot 1520/7651]$$

$$= 7339.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 4558 \cdot 854/(2 \cdot 31) - 7339/(\pi \cdot 854^2 \cdot 31)$$

$$= 61.74 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 4558 \cdot 854/(2 \cdot 31) + 7339/(\pi \cdot 854^2 \cdot 31)$$

$$= 63.77 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$$

$$= 4558 \cdot 854/(2 \cdot 31) - 8973/(0.11 \cdot \pi \cdot 854^2 \cdot 31)$$

$$= 74.39 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm/(2t) + M1/(K1^* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 4558 \cdot 854/(2 \cdot 31) + 8973/(0.19 \cdot \pi \cdot 854^2 \cdot 31)$$

$$= 56.30 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 25263 (7651 - 2 \cdot 1520)/(7651 + (4 \cdot 436/3))$$

$$= 14150.7 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 14151/(854 \cdot 31)$$

$$= 6.14 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1, x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(854 \cdot 31)}$$

$$= 126.875 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(350 + 1.56 \cdot \sqrt{(854 \cdot 31)}, 2 \cdot 1520)$$

$$= 603.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(138/138, 1)$$

$$= 1.0000$$



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Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma_{6,r}$]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 25263 \cdot 0.1 / (604 (31 + 1 \cdot 14))$$

$$= -0.69 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [$\sigma_{7,r}$]:

$$= -Q / (4(t + \eta \cdot tr) b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -25263 / (4(31 + 1 \cdot 14) 604) -$$

$$3 \cdot 0.053 \cdot 25263 / (2(31 + 1 \cdot 14)^2)$$

$$= -11.98 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1540.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	375.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	225.0	N./mm ²
Height of Web at Center	Hw,c	605.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ (R \cdot t) + \text{WearPlateWidth} }) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ (838 \cdot 31) + 650 }) 31^3) / 12$$

$$= 223.789 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 224 + 279 (160 - 15.5)^2$$

$$= 58431.355 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	901.4	31.0	15.5	279.4	433140.1	0.584E+05
Wearplate	650.0	14.0	38.0	91.0	345800.0	0.135E+05
Web	10.0	519.0	304.5	51.9	1580354.6	0.225E+05
BasePlate	375.0	28.0	578.0	105.0	6068999.5	0.184E+06
Totals	...	...	...	527.3	8428294.0	0.278E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 8428294 / 527$$

$$= 159.825 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 25263$$

$$= 5141.4883 \text{ Kgf}$$



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Tension Stress, $St = (Fh/As) = 2.0340 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 * \text{Yield Str} = 134.9988 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1430 - 838 \cdot \sin(120/2) / 1.05$
 $= 736.980 \text{ mm}$

Bending Moment, $M = Fh * d = 3789.2466 \text{ Kg-m.}$

Bending Stress, $Sb = (M * C1 / I) = 2.1352 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 * \text{Yield Str} = 149.9987 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{(3(25263 + 499)375 / (4 * 1540 * 150))}$
 $= 17.539 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120/2) (835 + 34 + 14)$
 $= 1529.401 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1529 / (4 - 1)$
 $= 509.800 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 510 * 375 / 2$
 $= 955.876 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 25762 / (1540 * 375)$
 $= 4.461 \text{ Kg/cm}^2$

Axial Load [P]:
 $= Ap * Bp$
 $= 956 * 4.46$
 $= 4264.120 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 34 + 25.5$
 $= 59.490 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P/Ar$
 $= 4264/59.5$
 $= 7.029 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	350.0	...	35.0	...	0.357E+04

Rib dimension [D]:



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= Saddle Width - Web Thickness

= 350 - 10

= 340.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/59.5

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 350/2

= 175.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{3573/59.5}$

= 77.498 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 1430 - cos(120/2)(835 + 34 + 14) - 28

= 960.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199943008/225}$

= 132.447

Slenderness ratio [KL/r]:

= KL/r

= 1 • 960/77.5

= 12.394

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 4064 / (2 • 1540) • 510 • 960/2

= 323.019 Kg-m.

Compressive Allowable, KL/r < Cc (12.4 < 132) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (12.4)^2 / (2 \cdot 132^2)) 225 /$

$(5/3 + 3 \cdot (12.4) / (8 \cdot 132) - (12.4^3) / (8 \cdot 132^3))$

= 131.6 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 7.03/132 + (323 • 175/35729164) / 150

= 0.157

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	340.0	0.0	34.0	0.0	0.357E+04
Web	509.8	10.0	0.0	51.0	0.0	4.25
Totals	...	...	...	85.0	...	0.358E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/85

= 0.000 mm

Distance to Centroid [C1]:



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Saddle Calcs: Operating Case: Step: 18 9:24am Sep 9, 2026

= Saddle Width / 2
= 350/2
= 175.000 mm

Length of Inner Rib [L]:
= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk
= 1430 - $\sqrt{(883 + 14)^2 - (510/2)^2}$ - 28
= 556.592 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{3577/85}$
= 64.879 mm

Slenderness ratio [KL/r]:
= KL/r
= 1 * 557/64.9
= 8.579

Unit Force [Force,u]:
= FI / (2 * Baseplate Length)
= 4064 / (2 * 1540)
= 1.319 Kg/mm

Moment at base of inner Rib [Mbase,c]:
= Unit Force * e * L
= 1.32 * 510 * 557
= 374.368 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
= Bending Moment / Section Modulus
= 374/204405
= 17.961 N./mm²

Compressive Allowable, KL/r < Cc (8.58 < 132) per AISC E2-1 [Sca]:
= $(1 - (Klr)^2 / (2 * Cc^2)) Fy / (5/3 + 3 * (Klr) / (8 * Cc) - (Klr^3) / (8 * Cc^3))$
= $(1 - (8.58)^2 / (2 * 132^2)) 225 /$
 $(5/3 + 3 * (8.58) / (8 * 132) - (8.58^3) / (8 * 132^3))$
= 132.8 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
= Sc/Sca + (Mbase,c * C1/I)/Sba
= 9.85/133 + (374 * 175/3577)/150
= 0.194

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load Q0 (Weight)	Q0	20062.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1274.3 Kg
Maximum Transverse Force	Ft	2031.8 Kg
Maximum Longitudinal Force	F1	4063.5 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
= 15663.2 Kg



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Saddle Calcs: Operating Case: Step: 18 9:24am Sep 9, 2026

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:

$$= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 15663 / (4.87 \cdot 6)$$

$$= 52.6 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2032 / (4.87 \cdot 6)$$

$$= 6.8 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 4064 / (82.8 \cdot 6)$$

$$= 0.8021 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1430 \cdot 2032 + 0$$

$$= 2905.48 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 2905 / 20063$$

$$= 144.82 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.8, 0]$$

$$= 0.8021 \text{ cm}^2$$

MDMT Calculations for: RHS saddle fixe, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 14$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.96$, Temp. Reduction = 2 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -35 °C

Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

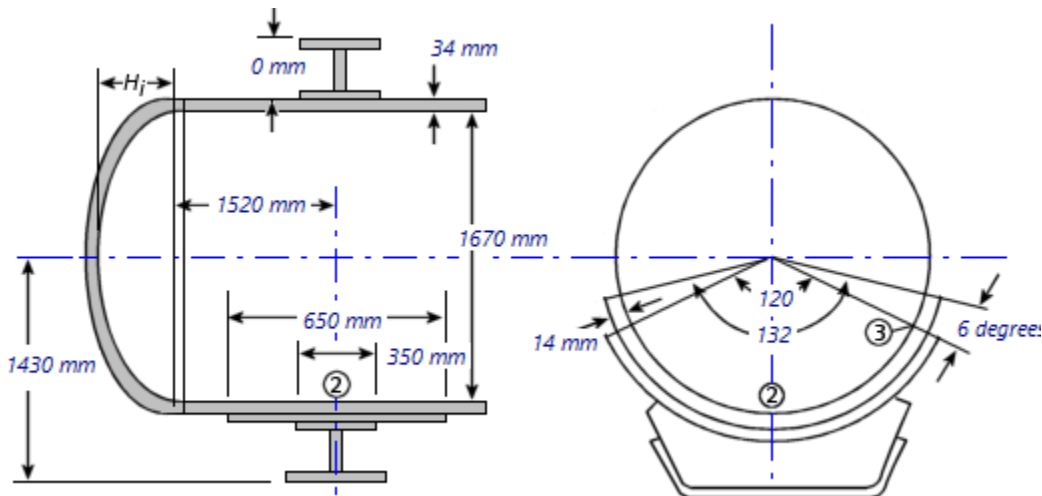
Step: 19 9:24am Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	852.00	mm
Shell Thickness used in this Case	t	34.000	mm
Stiffened Vessel Length per 4.15.6	L	7650.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1520.00	mm
Saddle Width	b1 or b	350.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	650.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1430.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		50629.40	Kgf
Pressure used in Analysis	P	5923.351	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	71.40	234.00
Long. Stress at Bottom of Midspan	77.04	234.00
Long. Stress at Top of Saddles	89.51	234.00
Long. Stress at Bottom of Saddles	65.75	234.00
Tangential Shear in Shell	12.09	138.58
Circ. Stress at Horn of Saddle	21.29	350.99
Circ. Compressive Stress in Shell	1.28	234.00



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (445/2 + 0) \cdot 1430/1529$$

$$= 624.7 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(175, 0) \cdot 1430/4560$$

$$= 54.9 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 50005 + \max(54.9, 625, 0, 0)$$

$$= 50629.4 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(1.9 \cdot 1.2)^2)$$

$$= 3433.324 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	51128.82	Kgf
Transverse Shear Load	222.69	Kgf
Longitudinal Shear Load	175.05	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(1670/2 + 34 + 14)\sin(120/2), 2 \cdot 852)$$

$$= 1529.401 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -50629 \cdot 1520 [1 - (1 - 1520/7651 + (852^2 - 0^2)/(2 \cdot 1520 \cdot 7651))/(1 + (4 \cdot 0)/(3 \cdot 7651))]$$

$$= -12887.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 50629 \cdot 7651/4 ((1 + 2(852^2 - 0^2)/(7651^2))/(1 + (4 \cdot 0)/(3 \cdot 7651)) - 4 \cdot 1520/7651)$$

$$= 22284.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 5923 \cdot 852/(2 \cdot 34) - 22284/(\pi \cdot 852^2 \cdot 34)$$

$$= 71.40 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$



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Saddle Calcs: Test Case: Step: 19 9:24am Sep 9, 2026

$$= 5923 \cdot 852 / (2 \cdot 34) + 22284 / (\pi \cdot 852^2 \cdot 34)$$

$$= 77.04 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 5923 \cdot 852 / (2 \cdot 34) - 12888 / (0.11 \cdot \pi \cdot 852^2 \cdot 34)$$

$$= 89.51 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 5923 \cdot 852 / (2 \cdot 34) + 12888 / (0.19 \cdot \pi \cdot 852^2 \cdot 34)$$

$$= 65.75 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 50629 \cdot (7651 - 2 \cdot 1520) / (7651 + (4 \cdot 0 / 3))$$

$$= 30512.1 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 30512 / (852 \cdot 34)$$

$$= 12.09 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(852 \cdot 34)}$$

$$= 132.756 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(350 + 1.56 \cdot \sqrt{(852 \cdot 34)}, 2 \cdot 1520)$$

$$= 615.51 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 50629 \cdot 0.1 / (616(34 + 1 \cdot 14))$$

$$= -1.28 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -50629 / (4(34 + 1 \cdot 14)616) - 3 \cdot 0.053 \cdot 50629 / (2(34 + 1 \cdot 14)^2)$$

$$= -21.29 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1540.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	375.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	605.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(835 \cdot 34)} + 650)^3) / 12$$

$$= 298.988 \text{ cm}^4$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 299 + 310 (154 - 17)^2$$

$$= 58549.211 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	912.8	34.0	17.0	310.4	527627.1	0.585E+05
Wearplate	650.0	14.0	41.0	91.0	373100.0	0.116E+05
Web	10.0	519.0	307.5	51.9	1595925.1	0.239E+05
BasePlate	375.0	28.0	581.0	105.0	6100501.0	0.192E+06
Totals	...	...	...	558.3	8597153.0	0.286E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 8597153/558$$

$$= 153.997 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 50629$$

$$= 10304.1807 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 4.0763 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1430 - 835 \cdot \sin(120/2) / 1.05$$

$$= 739.461 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 7619.6855 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 4.0294 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(50629 + 499)375) / (4 \cdot 1540 \cdot 173)}$$

$$= 22.985 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (835 + 34 + 14)$$

$$= 1529.401 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 1529 / (4 - 1)$$

$$= 509.800 \text{ mm}$$



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 510 \cdot 375 / 2$$

$$= 955.876 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 51129 / (1540 \cdot 375)$$

$$= 8.853 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 956 \cdot 8.85$$

$$= 8462.820 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 34 + 25.5$$

$$= 59.490 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 8463 / 59.5$$

$$= 13.951 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	350.0	...	35.0	...	0.357E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 350 - 10$$

$$= 340.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 59.5$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 350 / 2$$

$$= 175.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (3573 / 59.5) }$$

$$= 77.498 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1430 - \cos(120 / 2) (835 + 34 + 14) - 28$$

$$= 960.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199943008 / 260) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 960/77.5$$

$$= 12.394$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 175 / (2 \cdot 1540) \cdot 510 \cdot 960/2$$

$$= 13.915 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($12.4 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (12.4)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (12.4) / (8 \cdot 123) - (12.4^3) / (8 \cdot 123^3))$$

$$= 151.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 14/152 + (13.9 \cdot 175/35729164) / 173$$

$$= 0.096$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	340.0	0.0	34.0	0.0	0.357E+04
Web	509.8	10.0	0.0	51.0	0.0	4.25
Totals	...	...	...	85.0	...	0.358E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/85$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 350/2$$

$$= 175.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1430 - \sqrt{ (883 + 14)^2 - (510/2)^2 } - 28$$

$$= 556.592 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{ (3577/85)}$$

$$= 64.879 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 557/64.9$$

$$= 8.579$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 175 / (2 \cdot 1540)$$

$$= 0.057 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.057 \cdot 510 \cdot 557$$

$$= 16.127 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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$$= 16.1/204405$$

$$= 0.774 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.58 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.58)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (8.58) / (8 \cdot 123) - (8.58^3) / (8 \cdot 123^3))$$

$$= 153.2 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 19.5/153 + (16.1 \cdot 175/3577) / 173$$

$$= 0.132$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load QO (Weight)	QO	50504.2 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	54.9 Kg
Maximum Transverse Force	Ft	222.7 Kg
Maximum Longitudinal Force	F1	175.1 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 175 / (4.87 \cdot 6)$$

$$= 0.6 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 223 / (4.87 \cdot 6)$$

$$= 0.7 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 175 / (29 \cdot 6)$$

$$= 0.0987 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1430 \cdot 223 + 0$$

$$= 318.46 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 318/50504$$

$$= 6.31 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$



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Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 0.099, 0]

= 0.0987 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	852.00	mm
Shell Thickness used in this Case	t	34.000	mm
Stiffened Vessel Length per 4.15.6	L	7650.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1520.00	mm
Saddle Width	b1 or b	350.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	650.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	434.00	mm
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1430.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		19353.09	Kgf
Pressure used in Analysis	P	5923.351	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	73.51	234.00
Long. Stress at Bottom of Midspan	74.93	234.00
Long. Stress at Top of Saddles	82.37	234.00
Long. Stress at Bottom of Saddles	69.70	234.00
Tangential Shear in Shell	4.30	138.58
Circ. Stress at Horn of Saddle	8.14	350.99
Circ. Compressive Stress in Shell	0.49	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3 (445/2 + 0) • 1430/1529

= 624.7 Kgf

Longitudinal Saddle Reaction Force [Fwl]:

= max(Fl,wind; Sum of X Forces) • B / Ls

= max(175, 0) • 1430/4560

= 54.9 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:



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$$\begin{aligned}
 &= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst}) \\
 &= 18728 + \max(54.9, 625, 0, 0) \\
 &= 19353.1 \text{ Kg}
 \end{aligned}$$

Longitudinal Wind Force [F_L]:

$$\begin{aligned}
 &= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2)) \\
 &= 1.1 \cdot 766 (0 + \pi/4 (1.9 \cdot 1.2)^2) \\
 &= 3433.324 \text{ N}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	19852.51	Kgf
Transverse Shear Load	222.69	Kgf
Longitudinal Shear Load	175.05	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m) \\
 &= \min(2(1670/2 + 34 + 14)\sin(120/2), 2 \cdot 852) \\
 &= 1529.401 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to R_m/2.

Moment per Equation 4.15.1 [M₁]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)] \\
 &= -19353 \cdot 1520 [1 - (1 - 1520/7651 + (852^2 - 434^2)/(2 \cdot 1520 \cdot 7651))/(1 + (4 \cdot 434)/(3 \cdot 7651))] \\
 &= -6869.9 \text{ Kg-m}
 \end{aligned}$$

Moment per Equation 4.15.2 [M₂]:

$$\begin{aligned}
 &= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L) \\
 &= 19353 \cdot 7651/4 ((1 + 2(852^2 - 434^2)/(7651^2))/(1 + (4 \cdot 434)/(3 \cdot 7651)) - 4 \cdot 1520/7651) \\
 &= 5629.3 \text{ Kg-m}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ₁]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t) \\
 &= 5923 \cdot 852/(2 \cdot 34) - 5629/(\pi \cdot 852^2 \cdot 34) \\
 &= 73.51 \text{ N/mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ₂]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t) \\
 &= 5923 \cdot 852/(2 \cdot 34) + 5629/(\pi \cdot 852^2 \cdot 34) \\
 &= 74.93 \text{ N/mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*₃]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 5923 \cdot 852/(2 \cdot 34) - 6870/(0.11 \cdot \pi \cdot 852^2 \cdot 34) \\
 &= 82.37 \text{ N/mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*₄]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 5923 \cdot 852/(2 \cdot 34) + 6870/(0.19 \cdot \pi \cdot 852^2 \cdot 34) \\
 &= 69.70 \text{ N/mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$



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$$= 19353 (7651 - 2 \cdot 1520) / (7651 + (4 \cdot 434/3))$$

$$= 10843.1 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 10843 / (852 \cdot 34)$$

$$= 4.30 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{852 \cdot 34}$$

$$= 132.756 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(350 + 1.56 \cdot \sqrt{852 \cdot 34}, 2 \cdot 1520)$$

$$= 615.51 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 19353 \cdot 0.1 / (616(34 + 1 \cdot 14))$$

$$= -0.49 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -19353 / (4(34 + 1 \cdot 14)616) - 3 \cdot 0.053 \cdot 19353 / (2(34 + 1 \cdot 14)^2)$$

$$= -8.14 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1540.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	375.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	605.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{835 \cdot 34} + 650)34^3) / 12$$

$$= 298.988 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 299 + 310(154 - 17)^2$$

$$= 58549.211 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	912.8	34.0	17.0	310.4	527627.1	0.585E+05
Wearplate	650.0	14.0	41.0	91.0	373100.0	0.116E+05
Web	10.0	519.0	307.5	51.9	1595925.1	0.239E+05
BasePlate	375.0	28.0	581.0	105.0	6100501.0	0.192E+06
Totals	...	...	...	558.3	8597153.0	0.286E+06



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Distance to Centroid [C1]:

$$= AY / A$$

$$= 8597153/558$$

$$= 153.997 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 19353$$

$$= 3938.7734 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.5582 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1430 - 835 \cdot \sin(120/2) / 1.05$$

$$= 739.461 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2912.6252 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.5402 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(19353 + 499) \cdot 375) / (4 \cdot 1540 \cdot 173)}$$

$$= 14.323 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (835 + 34 + 14)$$

$$= 1529.401 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 1529 / (4 - 1)$$

$$= 509.800 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 510 \cdot 375 / 2$$

$$= 955.876 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 19853 / (1540 \cdot 375)$$

$$= 3.438 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 956 \cdot 3.44$$

$$= 3285.979 \text{ Kgf}$$



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Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= 34 + 25.5

= 59.490 cm²

Compressive Stress [Sc]:

= P/Ar

= 3286/59.5

= 5.417 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	350.0	...	35.0	...	0.357E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 350 - 10

= 340.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/59.5

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 350/2

= 175.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{3573/59.5}$

= 77.498 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 1430 - cos(120/2)(835 + 34 + 14) - 28

= 960.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199943008/260}$

= 123.211

Slenderness ratio [KL/r]:

= KL/r

= 1 • 960/77.5

= 12.394

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 175 / (2 • 1540) • 510 • 960/2

= 13.915 Kg-m.

Compressive Allowable, KL/r < Cc (12.4 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (12.4)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (12.4) / (8 \cdot 123) - (12.4^3) / (8 \cdot 123^3))$

= 151.8 N./mm²



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AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 5.42/152 + (13.9 \cdot 175/35729164) / 173$$

$$= 0.040$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	340.0	0.0	34.0	0.0	0.357E+04
Web	509.8	10.0	0.0	51.0	0.0	4.25
Totals	...	...	...	85.0	...	0.358E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/85$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 350/2$$

$$= 175.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2} - Bpdthk$$

$$= 1430 - \sqrt{(883 + 14)^2 - (510/2)^2} - 28$$

$$= 556.592 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3577/85)}$$

$$= 64.879 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 557/64.9$$

$$= 8.579$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 175 / (2 \cdot 1540)$$

$$= 0.057 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.057 \cdot 510 \cdot 557$$

$$= 16.127 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 16.1/204405$$

$$= 0.774 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ ($8.58 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (8.58)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (8.58) / (8 \cdot 123) - (8.58^3) / (8 \cdot 123^3))$$

$$= 153.2 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1 / I) / Sba$$

$$= 7.59/153 + (16.1 \cdot 175/3577) / 173$$

$$= 0.054$$



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Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load Q0 (Weight)	Q0	19227.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	54.9 Kg
Maximum Transverse Force	Ft	222.7 Kg
Maximum Longitudinal Force	Fl	175.1 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 175 / (4.87 \cdot 6)$$

$$= 0.6 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 223 / (4.87 \cdot 6)$$

$$= 0.7 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 175 / (29 \cdot 6)$$

$$= 0.0987 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1430 \cdot 223 + 0$$

$$= 318.46 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 318 / 19228$$

$$= 16.56 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.099, 0]$$

$$= 0.0987 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N3-Tube Inlet

Nozl: 9 9:24am

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Input, Nozzle Desc: N3-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	1095.335	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1670.00	mm
Design Length of Section	L	1599.9683	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1031.17	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Hub Height of Integral Nozzle	h	17.0000	mm
Height of Beveled Transition	L'	28.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



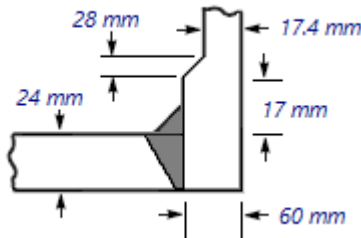
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Nozzle Calcs.: N3-Tube Inlet

Nozl: 9 9:24am Sep 9, 2026



Reinforcement CALCULATION, Description: N3-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.687 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1095 \cdot 838 / (138 \cdot 1 - 0.6 \cdot 1095)$
 $= 6.6837 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1095 \cdot 305 / (138 \cdot 1 + 0.4 \cdot 1095)$
 $= 2.4118 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.6373 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 42.6 / 0.5773$
 $= 73.6991 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1161.4008 mm
Parallel to Vessel Wall, opening length	d	580.7004 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	52.5000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



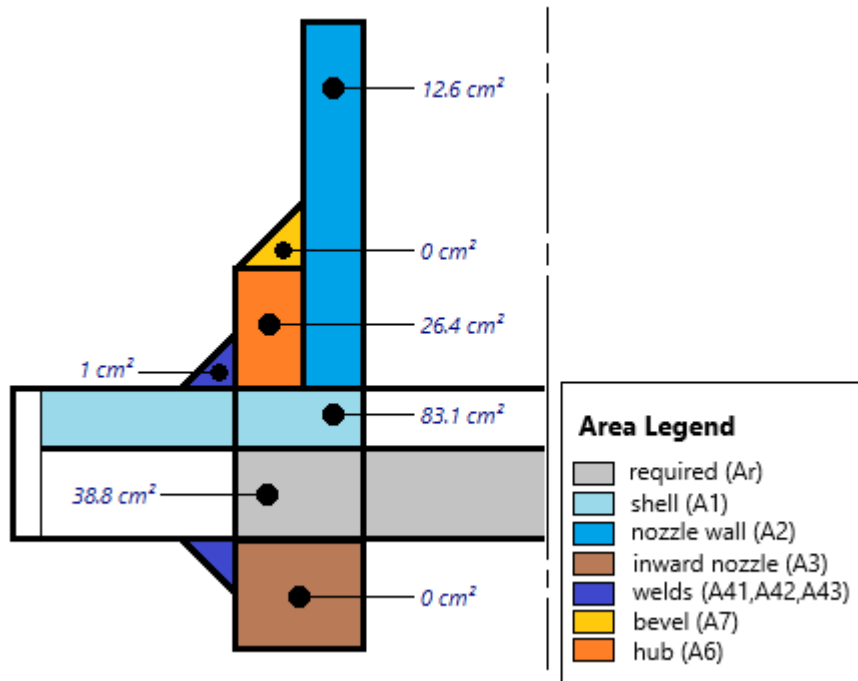
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Nozzle Calcs.: N3-Tube Inlet

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	38.812	15.406	NA
Area in Shell	A1	83.135	91.135	NA
Area in Nozzle Wall	A2	12.640	13.453	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	26.381	26.381	NA
TOTAL AREA AVAILABLE	Atot	123.156	131.969	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (581 \cdot 6.68 \cdot 1 + 2 \cdot 57 \cdot 6.68 \cdot 1 \cdot (1 - 1))$$

$$= 38.812 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 581(1 \cdot 21 - 1 \cdot 6.68) - 2 \cdot 57$$

$$(1 \cdot 21 - 1 \cdot 6.68) \cdot (1 - 1)$$

$$= 83.135 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 52.5)(14.4 - 2.41)1$$

$$= 12.640 \text{ cm}^2$$



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Nozzle Calcs.: N3-Tube Inlet

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(52.5, 400, 17)) \cdot (60 - 17.4) \cdot 1$$

$$= 26.381 \text{ cm}^2 \text{ Includes Bevel Area of } 11.914 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.4118 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 9.6837 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 9.6837 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.6283 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(9.68, 4.5)]$$

$$= 9.6837 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.41, 9.68)$$

$$= 9.6837 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 17.450 = 15.269 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 27.2,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 317.8 N./mm ²	Passed
Occasional	: 10.7,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 12.6,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 24, tr = 6.68, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.32$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 42.6 / 0.5773$$

$$= 73.6991 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).



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Nozzle Calcs.: N3-Tube Inlet

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UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 871.0506 mm

Parallel to Vessel Wall, opening length d 435.5253 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 52.5000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	25.876	25.876
Area Available in Shell	A1	41.567	41.567
Area Available in Nozzle Wall	A2	12.640	12.640
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	1.000	1.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	26.381	26.381
TOTAL AREA AVAILABLE	Atot	81.588	81.588

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N3-Tube Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (38.8 - 83.1 + 2 \cdot 57 \cdot 1 \cdot (1 \cdot 21 - 6.68)) \cdot 138)$$

$$= \max(0, -39402.88) \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (39 + 1 - 0 \cdot 1) \cdot 138$$

$$= 56315.88 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (39 + 0 + 1 + (23.9)) \cdot 138$$

$$= 90003.22 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (39 + 0 + 1 + 0 + (23.9)) \cdot 138$$

$$= 90003.22 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 138$$



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Nozzle Calcs.: N3-Tube Inlet

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= 66024. Kgf

Shear, Nozzle Wall [Snw]:

= $(\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$

= $(3.14 \cdot 298) \cdot (60 - 3) \cdot 0.7 \cdot 138$

= 524882. Kgf

Tension, Shell Groove Weld [Tngw]:

= $(\pi / 2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$

= $(3.14 / 2.0) \cdot 610 \cdot (14 - 0) \cdot 0.74 \cdot 138$

= 139594. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= $(SONW + SNW) = (66024 + 524882) = 590907 \text{ Kgf}$

Failure Path [Path 2-2]:

= $(Sonw + Tpgw + Tngw + Sinw)$

= $(66024 + 0 + 139594 + 0) = 205618 \text{ Kgf}$

Failure Path [Path 3-3]:

= $(Sonw + Tngw + Sinw)$

= $(66024 + 139594 + 0) = 205618 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 590906 Kgf, must exceed W = 0 Kgf or W1 = 56315 Kgf

Path 2-2 = 205618 Kgf, must exceed W = 0 Kgf or W2 = 90003 Kgf

Path 3-3 = 205618 Kgf, must exceed W = 0 Kgf or W3 = 90003 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1781 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 57.6185 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 481.6185 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N3-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1670.000	mm
Shell Thickness	24.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	694.700	mm
Nozzle Thickness	60.0000	mm



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Nozzle Calcs.: N3-Tube Inlet Nozl: 9 9:24am Sep 9, 2026

Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1095.33	KPa.
Radial Load	P	-4589.03	Kgf
Circumferential Shear	Vc	3059.35	Kgf
Longitudinal Shear	Vl	3059.35	Kgf
Circumferential Moment	Mc	5137.73	Kg-m.
Longitudinal Moment	Ml	5137.73	Kg-m.
Torsional Moment	Mt	7339.62	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 694.7 + 2 \cdot 1.65 \cdot \sqrt{848.5(24.0 - 3.0)}$$

$$= 1135.204 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1697.000 mm
Nozzle Outside Diameter	(d) =	694.700 mm
Vessel Thickness used	(T) =	21.000 mm
Nozzle Thickness used	(t) =	57.000 mm
T / t	=	0.368
d / t	=	12.188

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.680$$

Nr/P	=	0.054
Mr/P	=	0.088
M0/P	=	0.029
N0/P	=	0.074
MrD/Mc	=	0.258
NrDL/Mc	=	0.090
M0d/Mc	=	0.078
N0DL/Mc	=	0.088
MrD/Ml	=	0.079
NrDL/Ml	=	0.047
M0D/Ml	=	0.023
N0DL/Ml	=	0.083

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0



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Nozzle Calcs.: N3-Tube Inlet Nozl: 9 9:24am Sep 9,2026

Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-23	23	23	-23
Normal Pressure Stress	43	44	43	44

Outplane Stress Summary	31	43	105	24
-------------------------	----	----	-----	----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	53	-53	53	-53
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-78	78	78	-78
Inplane Pressure Stress	21	21	21	21

Inplane Stress Summary	-5	43	166	-97
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	5	5	3	3
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	38	49	166	122
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-14	-14	14	14
Outplane Bending (Mc)	-76	76	76	-76
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	21	21	21	21

Outplane Stress Summary	-44	73	138	-50
-------------------------	-----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	53	-53	53	-53
Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-254	254	254	-254
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0



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Nozzle Calcs.: N3-Tube Inlet Nozl: 9 9:24am Sep 9,2026

Inplane Pressure Stress	43	44	43	44
-------------------------	----	----	----	----

Inplane Stress Summary	-166	235	371	-243
------------------------	------	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	5	5
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	166	235	371	243
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7

Hoop Stress Summary	0	1	27	28
---------------------	---	---	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-7	7	7	-7
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-1	2	18	-6
----------------------	----	---	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	2	2	1	1
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------



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Nozzle Calcs.: N3-Tube Inlet

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Two * Max Shear Stress	4	4	27	34
------------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-14	-14	14	14
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7

Hoop Stress Summary	0	0	28	29
---------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-28	28	28	-28
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-22	23	39	-27
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	2	2
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	23	23	40	56
------------------------	----	----	----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	43	44	43	44	43	44	43	44
Circ. Pl (SUS)	-6	-6	20	20	-9	-9	19	19
Circ. Q (SUS)	-6	6	40	-40	-201	201	307	-307
Long. Pm (SUS)	21	21	21	21	21	21	21	21
Long. Pl (SUS)	-2	-2	12	12	-7	-7	21	21
Long. Q (SUS)	-25	25	131	-131	-59	59	93	-93



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Nozzle Calcs.: N3-Tube Inlet

Nozl: 9 9:24am

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Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	43.0	44.0	43.0	44.0	43.0	44.0	43.0	44.0
Pm+P1 (SUS)	37.1	38.1	63.0	64.0	34.0	35.0	62.0	63.0
Pm+P1+Q (Total)	38.3	49.0	164.1	122.1	167.1	236.1	369.1	244.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	44.00	138.00	Passed
Pm+P1 (SUS)	64.03	207.00	Passed
Pm+P1+Q (TOTAL)	369.11	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	7	6	7	6	7	6	7
Circ. P1 (SUS)	-6	-6	20	20	-7	-7	21	21
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. P1 (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-2	2	12	-12	-23	23	33	-33
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	6.0	7.0	6.0	7.0	6.0	7.0	6.0	7.0
Pm+P1 (SUS)	1.0	1.0	26.0	27.0	2.0	1.0	27.0	28.0
Pm+P1+Q (Total)	2.2	3.4	26.1	34.1	22.0	24.1	38.1	56.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	138.00	Passed
Pm+P1 (SUS)	28.00	207.00	Passed
Pm+P1+Q (TOTAL)	56.04	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: N4-Tube Outlet

Nozl: 10

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Input, Nozzle Desc: N4-Tube Outlet

From: 30

Pressure for Reinforcement Calculations	P	1111.704	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1670.00	mm
Design Length of Section	L	1599.9683	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1031.17	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Hub Height of Integral Nozzle	h	17.0000	mm
Height of Beveled Transition	L'	28.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



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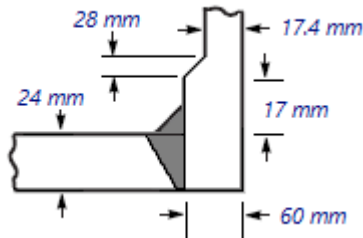
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Nozzle Calcs.: N4-Tube Outlet

Noz1: 10

9:24am

Sep 9, 2026



Reinforcement CALCULATION, Description: N4-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.687 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1112 \cdot 838 / (138 \cdot 1 - 0.6 \cdot 1112)$
 $= 6.7841 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1112 \cdot 305 / (138 \cdot 1 + 0.4 \cdot 1112)$
 $= 2.4477 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.6373 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 42.6 / 0.5773$
 $= 73.6991 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1161.4008 mm
Parallel to Vessel Wall, opening length	d	580.7004 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	52.5000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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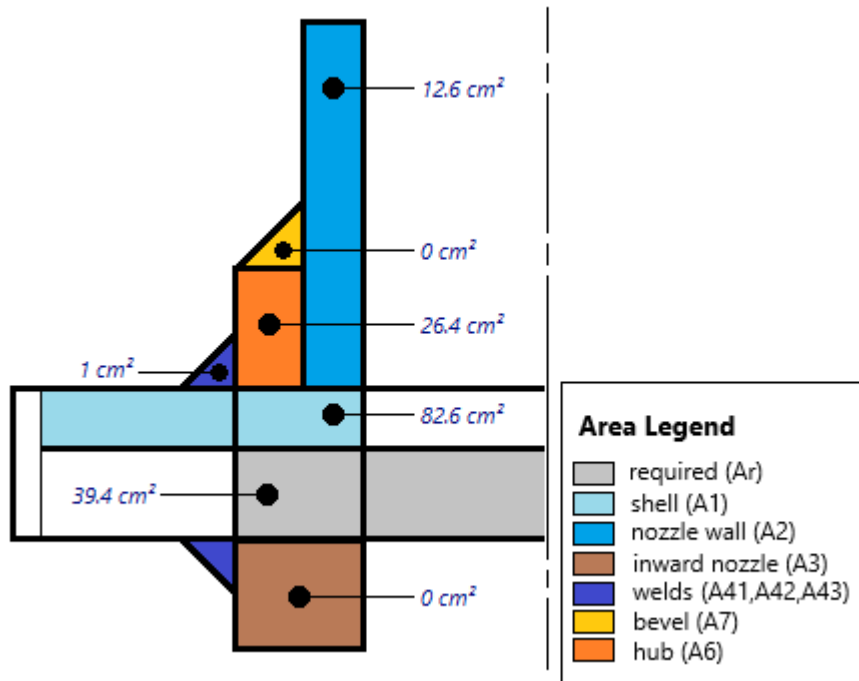
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Nozzle Calcs.: N4-Tube Outlet

Nozl: 10

9:24am

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	39.395	15.406	NA
Area in Shell	A1	82.552	91.135	NA
Area in Nozzle Wall	A2	12.602	13.453	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	26.381	26.381	NA
TOTAL AREA AVAILABLE	Atot	122.535	131.969	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (581 \cdot 6.78 \cdot 1 + 2 \cdot 57 \cdot 6.78 \cdot 1 \cdot (1 - 1))$$

$$= 39.395 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 581(1 \cdot 21 - 1 \cdot 6.78) - 2 \cdot 57$$

$$(1 \cdot 21 - 1 \cdot 6.78) \cdot (1 - 1)$$

$$= 82.552 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 52.5)(14.4 - 2.45)1$$

$$= 12.602 \text{ cm}^2$$



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Nozzle Calcs.: N4-Tube Outlet

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - c_n / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(52.5, 400, 17)) \cdot (60 - 17.4) \cdot 1$$

$$= 26.381 \text{ cm}^2 \text{ Includes Bevel Area of } 11.914 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.4477 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 9.7841 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 9.7841 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.6283 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(9.78, 4.5)]$$

$$= 9.7841 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.45, 9.78)$$

$$= 9.7841 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 17.450 = 15.269 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 27.4,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 317.6 N./mm ²	Passed
Occasional	: 10.9,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 12.6,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 24, tr = 6.78, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.32$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 42.6 / 0.5773$$

$$= 73.6991 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).



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Nozzle Calcs.: N4-Tube Outlet

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UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 871.0506 mm

Parallel to Vessel Wall, opening length d 435.5253 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 52.5000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	26.265	26.265
Area Available in Shell	A1	41.276	41.276
Area Available in Nozzle Wall	A2	12.602	12.602
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	1.000	1.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	26.381	26.381
TOTAL AREA AVAILABLE	Atot	81.259	81.259

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N4-Tube Outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (39.4 - 82.6 + 2 \cdot 57 \cdot 1 \cdot (1 \cdot 21 - 6.78)) \cdot 138)$$

$$= \max(0, -37923.55) \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_r2) \cdot S_v$$

$$= (39 + 1 - 0 \cdot 1) \cdot 138$$

$$= 56262.80 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$$

$$= (39 + 0 + 1 + (23.9)) \cdot 138$$

$$= 89950.14 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (39 + 0 + 1 + 0 + (23.9)) \cdot 138$$

$$= 89950.14 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 138$$



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Nozzle Calcs.: N4-Tube Outlet

Nozl: 10

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= 66024. Kgf

Shear, Nozzle Wall [Snw]:

= $(\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$

= $(3.14 \cdot 298) \cdot (60 - 3) \cdot 0.7 \cdot 138$

= 524882. Kgf

Tension, Shell Groove Weld [Tngw]:

= $(\pi / 2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$

= $(3.14 / 2.0) \cdot 610 \cdot (14 - 0) \cdot 0.74 \cdot 138$

= 139594. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= $(SONW + SNW) = (66024 + 524882) = 590907 \text{ Kgf}$

Failure Path [Path 2-2]:

= $(Sonw + Tpgw + Tngw + Sinw)$

= $(66024 + 0 + 139594 + 0) = 205618 \text{ Kgf}$

Failure Path [Path 3-3]:

= $(Sonw + Tngw + Sinw)$

= $(66024 + 139594 + 0) = 205618 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 590906 Kgf, must exceed W = 0 Kgf

or W1 = 56262 Kgf

Path 2-2 = 205618 Kgf, must exceed W = 0 Kgf

or W2 = 89950 Kgf

Path 3-3 = 205618 Kgf, must exceed W = 0 Kgf

or W3 = 89950 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1781 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 57.6185 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 481.6185 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N4-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1670.000	mm
Shell Thickness	24.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	694.700	mm
Nozzle Thickness	60.0000	mm



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Nozzle Calcs.: N4-Tube Outlet Nozl: 10 9:24am Sep 9, 2026

Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1111.70	KPa.
Radial Load	P	-4589.03	Kgf
Circumferential Shear	Vc	3059.35	Kgf
Longitudinal Shear	Vl	3059.35	Kgf
Circumferential Moment	Mc	5137.73	Kg-m.
Longitudinal Moment	Ml	5137.73	Kg-m.
Torsional Moment	Mt	7339.62	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 694.7 + 2 \cdot 1.65 \cdot \sqrt{848.5(24.0 - 3.0)}$$

$$= 1135.204 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1697.000 mm
Nozzle Outside Diameter	(d) =	694.700 mm
Vessel Thickness used	(T) =	21.000 mm
Nozzle Thickness used	(t) =	57.000 mm
T / t	=	0.368
d / t	=	12.188

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.680$$

Nr/P	=	0.054
Mr/P	=	0.088
M0/P	=	0.029
N0/P	=	0.074
MrD/Mc	=	0.258
NrDL/Mc	=	0.090
M0d/Mc	=	0.078
N0DL/Mc	=	0.088
MrD/Ml	=	0.079
NrDL/Ml	=	0.047
M0D/Ml	=	0.023
N0DL/Ml	=	0.083

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0



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Nozzle Calcs.: N4-Tube Outlet Nozl: 10 9:24am Sep 9,2026

Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-23	23	23	-23
Normal Pressure Stress	43	44	43	44

Outplane Stress Summary	32	44	105	25
-------------------------	----	----	-----	----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	53	-53	53	-53
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-78	78	78	-78
Inplane Pressure Stress	21	21	21	21

Inplane Stress Summary	-4	44	167	-96
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	5	5	3	3
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	39	50	167	122
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-14	-14	14	14
Outplane Bending (Mc)	-76	76	76	-76
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	21	21	21	21

Outplane Stress Summary	-44	74	138	-50
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	53	-53	53	-53
Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-254	254	254	-254
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0



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Nozzle Calcs.: N4-Tube Outlet Nozl: 10 9:24am Sep 9,2026

Inplane Pressure Stress	43	44	43	44
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Inplane Stress Summary	-165	236	371	-242
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
---------------------	---	---	---	---

Outplane Shear (Vl)	-1	-1	1	1
---------------------	----	----	---	---

Torsional Shear (Mt)	4	4	4	4
----------------------	---	---	---	---

Shear Stress Summary	3	3	5	5
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	165	236	371	242
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	7	7	7	7
--------------------	---	---	---	---

Hoop Bending (P)	0	0	0	0
-------------------	---	---	---	---

Hoop Membrane (Mc)	0	0	0	0
--------------------	---	---	---	---

Hoop Bending (Mc)	0	0	0	0
-------------------	---	---	---	---

Hoop Membrane (ML)	-13	-13	13	13
--------------------	-----	-----	----	----

Hoop Bending (ML)	0	0	0	0
-------------------	---	---	---	---

Hoop Pressure Stress	6	7	6	7
----------------------	---	---	---	---

Hoop Stress Summary	0	1	27	28
---------------------	---	---	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
---------------------	---	---	---	---

Axial Bending (P)	5	-5	5	-5
--------------------	---	----	---	----

Axial Membrane (Mc)	0	0	0	0
---------------------	---	---	---	---

Axial Bending (Mc)	0	0	0	0
--------------------	---	---	---	---

Axial Membrane (ML)	-2	-2	2	2
---------------------	----	----	---	---

Axial Bending (ML)	-7	7	7	-7
--------------------	----	---	---	----

Axial Pressure Stress	3	3	3	3
-----------------------	---	---	---	---

Axial Stress Summary	-1	2	18	-6
----------------------	----	---	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
-------------------	---	---	---	---

Shear due to (Vl)	0	0	0	0
-------------------	---	---	---	---

Shear due to Torsion	1	1	1	1
----------------------	---	---	---	---

Shear Stress Summary	2	2	1	1
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------



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Nozzle Calcs.: N4-Tube Outlet

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Two * Max Shear Stress	4	4	27	34
------------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-14	-14	14	14
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7
Hoop Stress Summary	0	0	28	29

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-28	28	28	-28
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-22	23	39	-27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	2	2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	23	24	40	56

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	43	44	43	44	43	44	43	44
Circ. Pl (SUS)	-6	-6	20	20	-9	-9	19	19
Circ. Q (SUS)	-6	6	40	-40	-201	201	307	-307
Long. Pm (SUS)	21	21	21	21	21	21	21	21
Long. Pl (SUS)	-2	-2	12	12	-7	-7	21	21
Long. Q (SUS)	-25	25	131	-131	-59	59	93	-93



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Nozzle Calcs.: N4-Tube Outlet

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Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	43.0	44.0	43.0	44.0	43.0	44.0	43.0	44.0
Pm+P1 (SUS)	37.1	38.1	63.0	64.0	34.0	35.0	62.0	63.0
Pm+P1+Q (Total)	38.3	49.0	164.1	122.1	167.1	236.1	369.1	244.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	44.00	138.00	Passed
Pm+P1 (SUS)	64.03	207.00	Passed
Pm+P1+Q (TOTAL)	369.11	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	7	6	7	6	7	6	7
Circ. P1 (SUS)	-6	-6	20	20	-7	-7	21	21
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. P1 (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-2	2	12	-12	-23	23	33	-33
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	6.0	7.0	6.0	7.0	6.0	7.0	6.0	7.0
Pm+P1 (SUS)	1.0	1.0	26.0	27.0	2.0	1.0	27.0	28.0
Pm+P1+Q (Total)	2.2	3.4	26.1	34.1	22.0	24.1	38.1	56.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	138.00	Passed
Pm+P1 (SUS)	28.00	207.00	Passed
Pm+P1+Q (TOTAL)	56.04	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: N1-Shell Inlet

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Input, Nozzle Desc: N1-Shell Inlet

From: 40

Pressure for Reinforcement Calculations	P	4777.354	KPa.
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516	70
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1670.00	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5995.18	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266	2
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	17.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	31.3064	mm
Height of Beveled Transition	L'	53.6936	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



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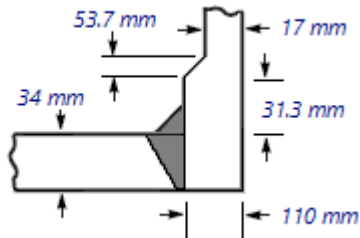
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Nozzle Calcs.: N1-Shell Inlet

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Reinforcement CALCULATION, Description: N1-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	16.000 in.
Actual Thickness Used in Calculation	0.669 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4777 \cdot 838 / (138 \cdot 1 - 0.6 \cdot 4777)$
 $= 29.6280 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 4777 \cdot 203 / (138 \cdot 1 + 0.4 \cdot 4777)$
 $= 6.9389 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 93 / 0.5773$
 $= 161.0808 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	756.8000 mm
Parallel to Vessel Wall, opening length	d	378.4000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	77.5000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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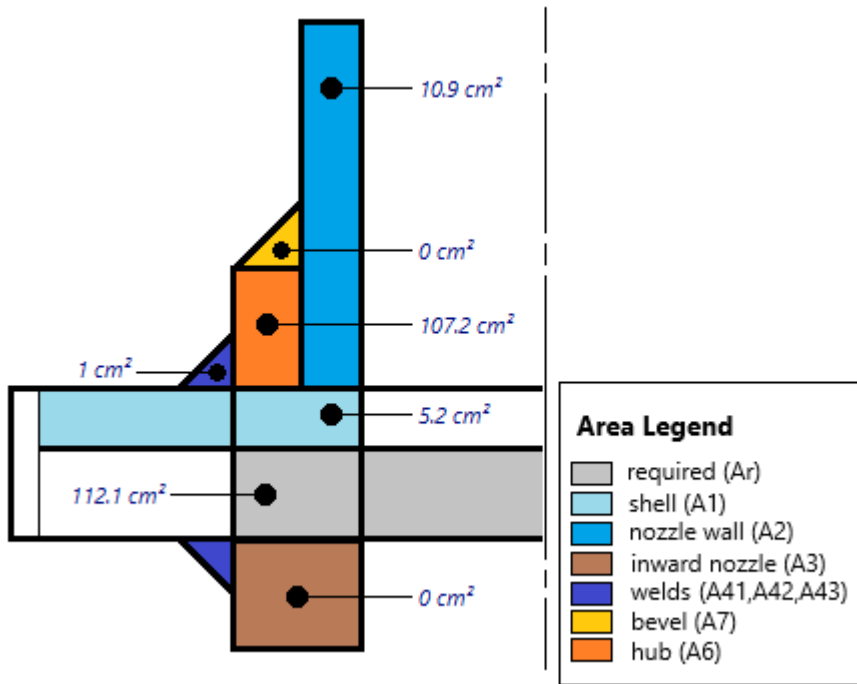
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Nozzle Calcs.: N1-Shell Inlet

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	112.112	NA	NA
Area in Shell	A1	5.192	NA	NA
Area in Nozzle Wall	A2	10.945	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	107.191	NA	NA
TOTAL AREA AVAILABLE	Atot	124.327	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (378 \cdot 29.6 \cdot 1 + 2 \cdot 107 \cdot 29.6 \cdot 1 \cdot (1 - 1))$$

$$= 112.112 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 378(1 \cdot 31 - 1 \cdot 29.6) - 2 \cdot 107$$

$$(1 \cdot 31 - 1 \cdot 29.6) \cdot (1 - 1)$$

$$= 5.192 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 77.5)(14 - 6.94)1$$

$$= 10.945 \text{ cm}^2$$



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Nozzle Calcs.: N1-Shell Inlet

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - c_{an}/0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(77.5, 300, 31.3)) \cdot (110 - 17) \cdot 1$$

$$= 107.191 \text{ cm}^2 \text{ Includes Bevel Area of } 48.961 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 9.9389 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 32.6280 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 32.6280 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(32.6, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(9.94, 11.3)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 17.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 48.7,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 296.3 N./mm ²	Passed
Occasional	: 31.1,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 13.6,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 17, tr = 6.94, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.5, Temp. Reduction = 33 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	8 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHt credit	-9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 34, tr = 29.6, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.96, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	5 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 5 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C



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Nozzle Calcs.: N1-Shell Inlet Nozl: 11 9:24am Sep 9, 2026

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4777.35/10200.00 = 0.468$

Weld Size Calculations, Description: N1-Shell Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (112 - 5.19 + 2 \cdot 107 \cdot 1 \cdot (1 \cdot 31 - 29.6)) \cdot 138)$
 $= 154585.52 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A4 - (W_i - \text{Can}) / .707)^2 \cdot f_r2 \cdot S_v$
 $= (118 + 1 - 0 \cdot 1) \cdot 138$
 $= 167642.11 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (118 + 0 + 1 + (66.3)) \cdot 138$
 $= 260992.92 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (118 + 0 + 1 + 0 + (66.3)) \cdot 138$
 $= 260992.92 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 406 \cdot 10 \cdot 0.49 \cdot 138$
 $= 44016. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 196) \cdot (110 - 3) \cdot 0.7 \cdot 138$
 $= 649641. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 406 \cdot (34 - 3) \cdot 0.74 \cdot 138$
 $= 206068. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (44016 + 649641) = 693657 \text{ Kgf}$

Failure Path [Path 2-2]:



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Nozzle Calcs.: N1-Shell Inlet Nozl: 11 9:24am Sep 9, 2026

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (44016 + 0 + 206068 + 0) = 250084 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (44016 + 206068 + 0) = 250084 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 693656 Kg, must exceed W = 154585 Kg or W1 = 167642 Kg

Path 2-2 = 250083 Kg, must exceed W = 154585 Kg or W2 = 260992 Kg

Path 3-3 = 250083 Kg, must exceed W = 154585 Kg or W3 = 260992 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4977 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 25.1020 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 359.1020 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm, while the distance from the center of the nozzle to the edge of the diameter limit is: 378.40 mm

Input Echo, WRC297 Item 1, Description: N1-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1670.000	mm
Shell Thickness	34.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	592.400	mm
Nozzle Thickness	110.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	4777.35	KPa.
Radial Load	P	-3059.35	Kgf
Circumferential Shear	Vc	2039.57	Kgf
Longitudinal Shear	Vl	2039.57	Kgf
Circumferential Moment	Mc	2283.43	Kg-m.
Longitudinal Moment	Ml	2283.43	Kg-m.
Torsional Moment	Mt	3262.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No



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Nozzle Calcs.: N1-Shell Inlet

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Local Loads applied at end of Nozzle/Attachment

No

*Warning - The ratio, Dn/Tn (5.385) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 592.4 + 2 \cdot 1.65 \cdot \sqrt{853.5(34.0 - 3.0)}$$

$$= 1129.180 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1707.000 mm
Nozzle Outside Diameter	(d) =	592.400 mm
Vessel Thickness used	(T) =	31.000 mm
Nozzle Thickness used	(t) =	107.000 mm
T / t	=	0.290
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.575$$

Nr/P	=	0.062
Mr/P	=	0.119
M0/P	=	0.036
N0/P	=	0.074
MrD/Mc	=	0.276
NrDL/Mc	=	0.094
M0d/Mc	=	0.083
N0DL/Mc	=	0.083
MrD/Ml	=	0.095
NrDL/Ml	=	0.068
M0D/Ml	=	0.028
N0DL/Ml	=	0.093

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	126	131	126	131

Outplane Stress Summary	125	129	146	124
-------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	22	-22	22	-22
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-22	22	22	-22
Inplane Pressure Stress	63	63	63	63

Inplane Stress Summary	62	62	112	23
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			1	1	1	1
Shear Stress Summary			2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			125	130	146	124

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			2	2	2	2
Outplane Bending (P)			6	-6	6	-6
Outplane Membrane (Mc)			-3	-3	3	3
Outplane Bending (Mc)			-19	19	19	-19
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			63	63	63	63
Outplane Stress Summary			49	75	95	42

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			1	1	1	1
Inplane Bending (P)			22	-22	22	-22
Inplane Membrane (Mc)			-3	-3	3	3
Inplane Bending (Mc)			-65	65	65	-65
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			126	131	126	131
Inplane Stress Summary			82	172	219	49

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Torsional Shear (Mt)			1	1	1	1
Shear Stress Summary			1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside



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Two * Max Shear Stress 82 172 219 50

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses in the	Au	Al	Bu	Bl
hoop direction)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	13	7	13
Hoop Stress Summary	6	12	13	19

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to	Au	Al	Bu	Bl
pipe cross-section)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	2	2	8	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	6	12	13	19

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu	Cl	Du	Dl
hoop direction)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	13	7	13
Hoop Stress Summary	6	12	13	19

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
	Cu	Cl	Du	Dl



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(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-4	4	4	-4
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	0	5	11	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	7	12	13	19

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	126	131	126	131	126	131	126	131
Circ. Pl (SUS)	-1	-1	5	5	-2	-2	4	4
Circ. Q (SUS)	0	0	12	-12	-43	43	87	-87
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	-1	-1	3	3	-1	-1	5	5
Long. Q (SUS)	0	0	44	-44	-13	13	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	126.0	131.0	126.0	131.0	126.0	131.0	126.0	131.0
Pm+Pl (SUS)	125.0	130.0	131.0	136.0	124.0	129.0	130.0	135.0
Pm+Pl+Q (Total)	125.0	130.0	143.0	124.0	81.0	172.0	217.0	48.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	131.00	138.00	Passed
Pm+Pl (SUS)	136.00	207.00	Passed
Pm+Pl+Q (TOTAL)	217.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	13	7	13	7	13	7	13
Circ. Pl (SUS)	-1	-1	5	5	-1	-1	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	2	-2	-3	3	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	7.0	13.0	7.0	13.0	7.0	13.0	7.0	13.0
Pm+Pl (SUS)	6.0	12.0	12.0	18.0	6.0	12.0	12.0	18.0
Pm+Pl+Q (Total)	6.0	12.0	12.0	18.0	7.0	12.0	12.0	19.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	13.00	138.00	Passed
Pm+Pl (SUS)	18.00	207.00	Passed
Pm+Pl+Q (TOTAL)	19.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N2-Shell outlet

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Input, Nozzle Desc: N2-Shell outlet

From: 40

Pressure for Reinforcement Calculations	P	4793.724	KPa.
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1670.00	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5995.18	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	17.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	26.3065	mm
Height of Beveled Transition	L'	53.6936	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



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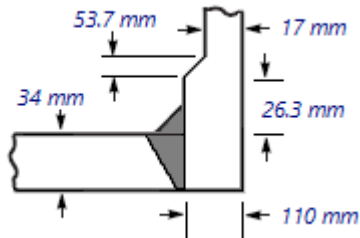
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Nozzle Calcs.: N2-Shell outlet

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Reinforcement CALCULATION, Description: N2-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	16.000 in.
Actual Thickness Used in Calculation	0.669 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4794 \cdot 838 / (138 \cdot 1 - 0.6 \cdot 4794)$
 $= 29.7317 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 4794 \cdot 203 / (138 \cdot 1 + 0.4 \cdot 4794)$
 $= 6.9624 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 93 / 0.5773$
 $= 161.0808 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	756.8000 mm
Parallel to Vessel Wall, opening length	d	378.4000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	77.5000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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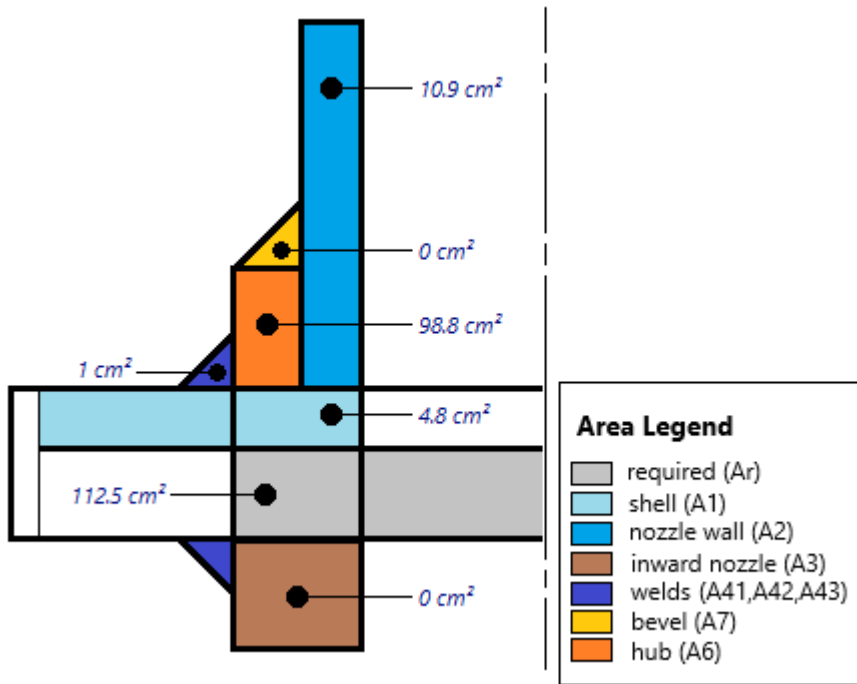
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Nozzle Calcs.: N2-Shell outlet

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	112.505	NA	NA
Area in Shell	A1	4.799	NA	NA
Area in Nozzle Wall	A2	10.908	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	98.757	NA	NA
TOTAL AREA AVAILABLE	Atot	115.465	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (378 \cdot 29.7 \cdot 1 + 2 \cdot 107 \cdot 29.7 \cdot 1 \cdot (1 - 1))$$

$$= 112.505 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 378(1 \cdot 31 - 1 \cdot 29.7) - 2 \cdot 107$$

$$(1 \cdot 31 - 1 \cdot 29.7) \cdot (1 - 1)$$

$$= 4.799 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 77.5)(14 - 6.96)1$$

$$= 10.908 \text{ cm}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - c_{an}/0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(77.5, 300, 26.3)) \cdot (110 - 17) \cdot 1$$

$$= 98.757 \text{ cm}^2 \text{ Includes Bevel Area of } 49.827 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 9.9624 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 32.7317 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 32.7317 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(32.7, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(9.96, 11.3)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 17.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 48.8,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 296.1 N./mm ²	Passed
Occasional	: 31.2,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 13.6,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 17, tr = 6.96, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.5, Temp. Reduction = 33 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	8 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHt credit	-9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 34, tr = 29.7, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.96, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	5 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 5 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C



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Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4793.72/10200.00 = 0.470$

Weld Size Calculations, Description: N2-Shell outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (113 - 4.8 + 2 \cdot 107 \cdot 1 \cdot (1 \cdot 31 - 29.7)) \cdot 138)$
 $= 155377.44 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A4 - (W_i - \text{Can}) / .707)^2 \cdot f_r2 \cdot S_v$
 $= (110 + 1 - 0 \cdot 1) \cdot 138$
 $= 155723.30 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (110 + 0 + 1 + (66.3)) \cdot 138$
 $= 249074.08 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (110 + 0 + 1 + 0 + (66.3)) \cdot 138$
 $= 249074.08 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 406 \cdot 10 \cdot 0.49 \cdot 138$
 $= 44016. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 196) \cdot (110 - 3) \cdot 0.7 \cdot 138$
 $= 649641. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 406 \cdot (34 - 3) \cdot 0.74 \cdot 138$
 $= 206068. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (44016 + 649641) = 693657 \text{ Kgf}$

Failure Path [Path 2-2]:



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$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (44016 + 0 + 206068 + 0) = 250084 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (44016 + 206068 + 0) = 250084 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 693656 Kg, must exceed W = 155377 Kg or W1 = 155723 Kg

Path 2-2 = 250083 Kg, must exceed W = 155377 Kg or W2 = 249074 Kg

Path 3-3 = 250083 Kg, must exceed W = 155377 Kg or W3 = 249074 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4853 KPa.

The Drop for this Nozzle is : 25.1020 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 359.1020 mm

Input Echo, WRC297 Item 1, Description: N2-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1670.000	mm
Shell Thickness	34.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	592.400	mm
Nozzle Thickness	110.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTained loads.

Design Internal Pressure	Dp	4793.72	KPa.
Radial Load	P	-3059.36	Kgf
Circumferential Shear	Vc	2039.57	Kgf
Longitudinal Shear	Vl	2039.57	Kgf
Circumferential Moment	Mc	2283.43	Kg-m.
Longitudinal Moment	Ml	2283.43	Kg-m.
Torsional Moment	Mt	3262.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.385) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:



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Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 592.4 + 2 \cdot 1.65 \cdot \sqrt{853.5(34.0 - 3.0)}$$

$$= 1129.180 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1707.000 mm

Nozzle Outside Diameter (d) = 592.400 mm

Vessel Thickness used (T) = 31.000 mm

Nozzle Thickness used (t) = 107.000 mm

T / t = 0.290

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.575$$

Nr/P = 0.062

Mr/P = 0.119

M0/P = 0.036

N0/P = 0.074

MrD/Mc = 0.276

NrDL/Mc = 0.094

M0d/Mc = 0.083

N0DL/Mc = 0.083

MrD/Ml = 0.095

NrDL/Ml = 0.068

M0D/Ml = 0.028

N0DL/Ml = 0.093

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	127	132	127	132
Outplane Stress Summary	126	130	146	124

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	22	-22	22	-22
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-22	22	22	-22
Inplane Pressure Stress	63	63	63	63
Inplane Stress Summary	62	63	113	23

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------



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Nozzle Calcs.: N2-Shell outlet Nozl: 12 9:24am Sep 9,2026

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	1	1
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	126	130	146	124
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-19	19	19	-19
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	63	63	63

Outplane Stress Summary	49	75	95	42
-------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	22	-22	22	-22
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-65	65	65	-65
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	127	132	127	132

Inplane Stress Summary	82	172	220	50
------------------------	----	-----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	2	2
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	82	172	220	51
------------------------	----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside



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Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	13	7	13

Hoop Stress Summary	6	12	13	19
---------------------	---	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	2	2	8	2
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	6	12	13	19
------------------------	---	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	13	7	13

Hoop Stress Summary	6	12	13	19
---------------------	---	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1



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Nozzle Calcs.: N2-Shell outlet Nozl: 12 9:24am Sep 9, 2026

Axial Bending (Mc)	-4	4	4	-4
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	0	5	11	0
----------------------	---	---	----	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	7	12	13	19

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	127	132	127	132	127	132	127	132
Circ. Pl (SUS)	-1	-1	5	5	-2	-2	4	4
Circ. Q (SUS)	0	0	12	-12	-43	43	87	-87
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	-1	-1	3	3	-1	-1	5	5
Long. Q (SUS)	0	0	44	-44	-13	13	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	127.0	132.0	127.0	132.0	127.0	132.0	127.0	132.0
Pm+Pl (SUS)	126.0	131.0	132.0	137.0	125.0	130.0	131.0	136.0
Pm+Pl+Q (Total)	126.0	131.0	144.0	125.0	82.0	173.0	218.0	49.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	132.00	138.00	Passed
Pm+Pl (SUS)	137.00	207.00	Passed
Pm+Pl+Q (TOTAL)	218.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):



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Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	13	7	13	7	13	7	13
Circ. Pl (SUS)	-1	-1	5	5	-1	-1	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	2	-2	-3	3	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	7.0	13.0	7.0	13.0	7.0	13.0	7.0	13.0
Pm+Pl (SUS)	6.0	12.0	12.0	18.0	6.0	12.0	12.0	18.0
Pm+Pl+Q (Total)	6.0	12.0	12.0	18.0	7.0	12.0	12.0	19.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	13.00	138.00	Passed
Pm+Pl (SUS)	18.00	207.00	Passed
Pm+Pl+Q (TOTAL)	19.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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ASME TS Calc:

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	4550.00 KPa.
Shell Maximum Operating Pressure	Psox,max	4550.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	34.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1670.000 mm
Shell Temperature for Internal Pressure	Ts	200.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	1000.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1670.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1996
Tube Wall Thickness	et	2.7700 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	7000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6736.00 mm
Design Temperature of the Tubes		200.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	154.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	33.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.7700 mm
Groove Weld Leg	ag	2.7700 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	811.245 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm



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ASME TS Calc:

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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	18959.3 cm ²
Perimeter of Tube Layout	Cp	5147.1 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	200.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

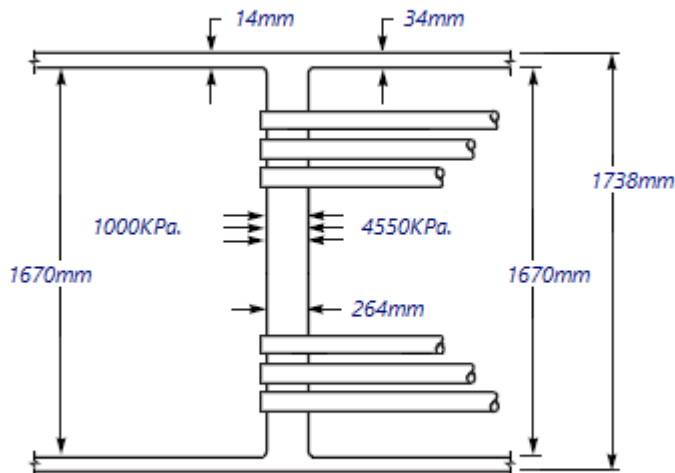
Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	264.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1738.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Junction Stress Reduction option	Analyze as Simply Supported
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes
Channel Side Min. Design Pressure	Ptd,min 103.4190 KPa.

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%



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ASME TS Calc:

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Elastic Mod. at Design Temperature	200.0 °C	0.19258E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	200.0 °C	0.19258E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	200.0 °C	0.19120E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (1000 \cdot (25.4/2 - 0)) / (92.4 \cdot 1 + 0.4 \cdot 1000)$
 $= 0.1369 + 0.0000 = 0.1369 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 200.00 °C
 Elastic Modulus for Material 193602528.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.7700	25.40	7547.24	9.17	50.0000	0.0130823	90.70

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 13879 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.1184	25.40	7547.24	22.71	50.0000	0.0021328	79.28

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 79.3) / (3 \cdot 22.7) = 4654 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.1184 mm
Allowable Internal Pressure at Corroded thickness	22078.03 KPa.
Required Internal Design Pressure	1000.00 KPa.
Allowable External Pressure at Corroded thickness	13878.92 KPa.
Required External Design Pressure	4653.42 KPa.
Required Thickness due to Shell Side pressure	1.1184 mm

Detailed Results for load Case D1 un-corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

Note:

The Tubesheet stress reduction option was Simply Supported. Performing the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:



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ASME TS Calc:

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Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned}
 &= 2 \cdot r_o + dt \\
 &= 2 \cdot 811 + 25.4 \\
 &= 1647.890 \text{ mm}
 \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned}
 &= (p - dt) / p \\
 &= (33 - 25.4) / 33 \\
 &= 0.230
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned}
 &= D_s / D_o \text{ (Configurations a, b, c)} \\
 &= 1670 / 1648 \\
 &= 1.0134
 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned}
 &= D_c / D_o \text{ (Configurations a, e, f)} \\
 &= 1670 / 1648 \\
 &= 1.0134
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned}
 &= l_{tx} / h \\
 &= 0 / 264 \\
 &= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}
 \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned}
 &= \max(dt - 2t(E_t/E) \cdot (S_t/S) \cdot (\rho); dt - 2t) \\
 &= \max(25.4 - 2 \cdot 2.77 (192580832 / 191201920) \cdot \\
 &\quad (92.4 / 138) \cdot (0); 25.4 - 2 \cdot 2.77) \\
 &= 25.4000 \text{ mm}
 \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned}
 &= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot D_o \cdot \rho) / (\pi \cdot D_o^2)) } \\
 &= 33 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1648 \cdot 33) / (\pi \cdot 1648^2)) } \\
 &= 33.0000 \text{ mm}
 \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned}
 &= (p^* - d^*) / p^* \\
 &= (33 - 25.4) / 33 \\
 &= 0.2303
 \end{aligned}$$

$h/p = 8.000000$; $\mu^* = 0.230303$; $E^* = 36704808 \text{ KPa}$.

$E^*/E = 0.191969$; $v^* = 0.418359$

Note: As h/p (8.000) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned}
 &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s } \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (1670 + 34) 34 } \\
 &= 0.0076 \text{ 1/mm}
 \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned}
 &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2)) \\
 &= 0.0076 \cdot 192580832 \cdot 34^3 / (6(1 - 0.3^2)) \\
 &= 10469806080.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$



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ASME TS Calc:

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$$= 6(1670 + 34) \cdot 10.47E+09/264^3 \cdot (1 + 264 \cdot 0.0076 + (264 \cdot 0.0076)^2/2)$$

$$= 28980458.0 \text{ KPa.}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 1670^2 / (4 \cdot 192580832 \cdot 34) \cdot (1 - 1670 \cdot 0.3 / (2(1670 + 34)))$$

$$= 0.90824E-01 \text{ mm/N./mm}^2$$

Parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.01 \cdot 10.47E+09 \cdot 0.0076 \cdot 0.090824112 (1 + 264 \cdot 0.0076)$$

$$= 21790.1309 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1670 + 14)^{1/4})$$

$$= 0.0118 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.012 \cdot 199214080 \cdot 14^3 / (6(1 - 0.3^2))$$

$$= 1185310720.0000 \text{ KPa.} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot Cq\Delta)$$

$$= 6(1670 + 14 / \cos(0)) \cdot 1185310720/264^3 \cdot (1 + 264 \cdot 0.012 \cdot (1 + 1)/2 + (264 \cdot 0.012)^2/2 \cdot 1)$$

$$= 5864617.5 \text{ KPa.}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1670^2 / (4 \cdot 199214080 \cdot 14) \cdot (1 - 1670 \cdot 0.3 / (2(1670 + 14)))$$

$$= 0.21279E+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.01 \cdot 1185310720 [0.21279198 (1 + 0.012 \cdot 1 \cdot 264) - 0.000 (1/264 + 1 \cdot 0.012/2)]$$

$$= 12485.3818 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1738/1648$$

$$= 1.0547$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.42) / 36704808 \cdot (28980458 + 5864617 + 191201920 \cdot \ln(1.05))$$

$$= 0.7135$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2/16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t]$$



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ASME TS Calc:

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$$= 1648^2/16 \cdot [(1.01 - 1) \cdot (1.01^2 + 1) \cdot 0 - (1.01 - 1) \cdot (1.01^2 + 1) \cdot 1000]$$

$$= -4615861.5000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega c \cdot Pt - \omega s \cdot Ps$$

$$= -4615861 + 12485 \cdot 1000 - 21790 \cdot 0$$

$$= 7869521.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M*) - Do^2/32 \cdot F (Ps - Pt)) / (1 + F)$$

$$= ((7869521) - 1648^2/32 \cdot 0.71 (0 - 1000)) / (1 + 0.71)$$

$$= 39928020.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu) \cdot (Ps - Pt)$$

$$= 39928020 + 1648^2/64 \cdot (3 + 0.42) \cdot (0 - 1000)$$

$$= -105114128.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp | ; | Mo |)$$

$$= \max(| 39928020 | ; | -105114128 |)$$

$$= 105114128.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 105114128 / (0.23 (264 - 8)^2)$$

$$= 41.7886 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 73840384 / (0.23 (91.6 - 8)^2)$$

$$= 275.5757 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 91.6 + 0 + 0$$

$$= 91.5536 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB ; HreqS)$$

$$= \max(91.6 ; 14.5)$$

$$= 91.5536 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | Ps - Pt |$$

$$= | 0 - 1000 |$$

$$= 1000.000 \text{ KPa}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S ; 0.533 \cdot Sy)$$

$$= 4 \cdot 0.23 \cdot 264 / 1648 \cdot \min(0.8 \cdot 138 ; 0.533 \cdot 213)$$

$$= 16.293 \text{ N./mm}^2$$



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| Ps - Pt | is ≤ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.23 \cdot 110) \cdot (4 \cdot 18959 / 5147) \cdot | 0 - 1000 | + 0$$

$$= 14.4884 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 1670^2 / (4 \cdot 34 (1670 + 34)) \cdot 0$$

$$= 0.0000 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P'_s + 6(1 - \nu^*) / E^* \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 2353759 / 34^2 \cdot [0.0076 \cdot 0 + 6(1 - 0.42) / 36704808 \cdot 1648 / 264^3 \cdot (1 + 264 \cdot 0.0076 / 2) \cdot (39928020 + 1648^2 / 32 \cdot (0 - 1000))]$$

$$= -41.522 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq SPS_s \text{)}$$

$$= 41.522 \text{ N./mm}^2 \text{ (Should be } \leq 450.02 \text{)}$$

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 1670^2 / (4 \cdot 14 (1670 + 14) \cdot \cos(0)) \cdot 1000$$

$$= 29.575 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 266475 / 14^2 \cdot [(0.21279198 \cdot 1 \cdot 0.012 - 0.000 \cdot 1) \cdot 1000 - 6((1 - 0.42) / 36704808) \cdot 1648 / 264^3 \cdot (1 + 0.012 \cdot 264 / 2 \cdot 1) \cdot (39928020 + 1648^2 / 32 \cdot (0 - 1000))]$$

$$= 127.004 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq SPS_c \text{)}$$

$$= 156.579 \text{ N./mm}^2 \text{ (Should be } \leq 485.22 \text{)}$$

Note: The Cylinder-to-Tubesheet Junction stress is < SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate Shell Side parameters for the Simply Supported case:

$$\beta_s, k_s, \lambda_s, \delta_s \text{ are all } = 0$$

Intermediate Channel side parameters for the Simply Supported condition:

$$\beta_c, k_c, \lambda_c, \delta_c \text{ are all } = 0$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:



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ASME TS Calc:

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Coefficient [F]:

$$\begin{aligned}
 &= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\
 &= (1 - 0.42) / 36704808 \cdot (0 + 0 + 191201920 \cdot \ln(1.05)) \\
 &= 0.1613
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 &= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t] \\
 &= 1648^2 / 16 \cdot [(1.01 - 1) \cdot (1.01^2 + 1) \cdot 0 - (1.01 - 1) \cdot (1.01^2 + 1) \cdot 1000] \\
 &= -4615861.5000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned}
 &= M_{ts} + w_c \cdot P_t - w_s \cdot P_s \\
 &= -4615861 + 0 \cdot 1000 - 0 \cdot 0 \\
 &= -4615861.5000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= M_p + D_o^2 / 64 \cdot (3 + v^*) \cdot (P_s - P_t) \\
 &= 7812624 + 1648^2 / 64 \cdot (3 + 0.42) \cdot (0 - 1000) \\
 &= -137229520.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= |M_o| \\
 &= | -137229520 | \\
 &= 137229520.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu^* (h - h_g')^2) \\
 &= 6 \cdot 137229520 / (0.23 (264 - 8)^2) \\
 &= 54.5562 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 \cdot S \\
 &= 2 \cdot 138 \\
 &= 275.99 \text{ N./mm}^2
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot M / (\mu^* (h - h_g')^2) \\
 &= 6 \cdot 137229520 / (0.23 (122 - 8)^2) \\
 &= 275.9272 \text{ N./mm}^2
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + C_{ats} + C_{atc} \\
 &= 122 + 0 + 0 \\
 &= 121.8321 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(H_{reqB}; H_{reqS}) \\
 &= \max(122; 16.2) \\
 &= 121.8321 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$\begin{aligned}
 &= |P_s - P_t| \\
 &= |0 - 1000| \\
 &= 1000.000 \text{ KPa}
 \end{aligned}$$



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Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.23 \cdot 264 / 1648 \cdot \min(0.8 \cdot 138; 0.533 \cdot 213)$$

$$= 16.293 \text{ N./mm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.23 \cdot 110) \cdot (4 \cdot 18959 / 5147) \cdot |0 - 1000| + 0$$

$$= 14.4884 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 1670^2 / (4 \cdot 34 (1670 + 34)) \cdot 0$$

$$= 0.0000 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 34^2 \cdot [0 \cdot 0 + 6(1 - 0.42) / 36704808 \cdot 1648 / 264^3 \cdot (1 + 264 \cdot 0 / 2) \cdot (7812624 + 1648^2 / 32 \cdot (0 - 1000))]$$

$$= -0.000 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= |\sigma_s, m| + |\sigma_s, b| \text{ (Should be } \leq \text{SPS}, s)$$

$$= 0.000 \text{ N./mm}^2 \text{ (Should be } \leq 450.02)$$

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c)) \cdot \cos(\theta) \cdot P_t$$

$$= 1670^2 / (4 \cdot 14 (1670 + 14)) \cdot \cos(0) \cdot 1000$$

$$= 29.575 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 14^2 \cdot [(0.000 \cdot 1 \cdot 0 - 0.000 \cdot 1) \cdot 1000 - 6((1 - 0.42) / 36704808) \cdot 1648 / 264^3 \cdot (1 + 0 \cdot 264 / 2 \cdot 1) \cdot (7812624 + 1648^2 / 32 \cdot (0 - 1000))]$$

$$= 0.000 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_c, m| + |\sigma_c, b| \text{ (Should be } \leq \text{SPS}, c)$$

$$= 29.575 \text{ N./mm}^2 \text{ (Should be } \leq 485.22)$$

Analysis Finished.

Tube Metal Area [A_t]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.77) 2.77$$

$$= 1.969 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:



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ASME TS Calc:

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$$\begin{aligned}
 &= \pi \cdot t (d_o - t) S \\
 &= \pi \cdot 2.77 (25.4 - 2.77) 92.4 \\
 &= 1855.452 \text{ Kgf}
 \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned}
 &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\
 &= 0.85 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 92.4 \\
 &= 1855.4518 \text{ Kgf}
 \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned}
 &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\
 &= 92.4/92.4 \\
 &= 1.000
 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned}
 &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t) \\
 &= 0.55 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 92.4 \\
 &= 1229.1031 \text{ Kgf}
 \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned}
 &= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right) \\
 &= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1 \cdot 1} - 0.75 \cdot 25.4 \right) \\
 &= 3.372 \text{ mm}
 \end{aligned}$$

Minimum Required Fillet Weld Leg [afr]: 1.6859 mm
Minimum Required Groove Weld Leg [agr]: 1.6859 mm

Tube-Tubesheet Jt allowable, 1855 is $\geq$ tube strength 1855 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 un-corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	54.6 $\leq$	276.0 N./mm ²	Ok
Tubesheet shear stress	6.1 $\leq$	110.4 N./mm ²	Ok
Stress in Shell at Tubesheet	0.0 $\leq$	450.0 N./mm ²	Ok
Stress in Channel at Tubesheet	29.6 $\leq$	485.2 N./mm ²	Ok

Thickness results for loadcase D1 un-corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	121.832	264.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.686	2.770	Ok
Tube-Tubesheet Groove Weld Leg :	1.686	2.770	Ok

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

Note:
The Tubesheet stress reduction option was Simply Supported. Performing



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ASME TS Calc:

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the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 811 + 25.4 \\ &= 1647.890 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (33 - 25.4) / 33 \\ &= 0.230 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 1670 / 1648 \\ &= 1.0134 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 1670 / 1648 \\ &= 1.0134 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned} &= l_{tx} / h \\ &= 0 / 264 \\ &= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2tt(E_tT/E) \cdot (StT/S) \cdot (p); dt - 2tt) \\ &= \max(25.4 - 2 \cdot 2.77 (192580832 / 191201920) \cdot \\ &\quad (92.4 / 138) \cdot (0); 25.4 - 2 \cdot 2.77) \\ &= 25.4000 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\ &= 33 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1648 \cdot 33) / (\pi \cdot 1648^2)) } \\ &= 33.0000 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (33 - 25.4) / 33 \\ &= 0.2303 \end{aligned}$$

$h/p = 8.000000$; $\mu^* = 0.230303$; $E^* = 36704808 \text{ KPa}$.

$E^*/E = 0.191969$; $v^* = 0.418359$

Note: As h/p (8.000) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (1670 + 34) 34 } \\ &= 0.0076 \text{ 1/mm} \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned} &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2)) \\ &= 0.0076 \cdot 192580832 \cdot 34^3 / (6(1 - 0.3^2)) \end{aligned}$$



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ASME TS Calc:

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$$= 10469806080.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1670 + 34) \cdot 10.47E+09 / 264^3 \cdot (1 + 264 \cdot 0.0076 + (264 \cdot 0.0076)^2 / 2)$$

$$= 28980458.0 \text{ KPa}.$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 1670^2 / (4 \cdot 192580832 \cdot 34) \cdot (1 - 1670 \cdot 0.3 / (2(1670 + 34)))$$

$$= 0.90824E-01 \text{ mm/N./mm}^2$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.01 \cdot 10.47E+09 \cdot 0.0076 \cdot 0.090824112 (1 + 264 \cdot 0.0076)$$

$$= 21790.1309 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (1670 + 14) 14 }$$

$$= 0.0118 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.012 \cdot 199214080 \cdot 14^3 / (6(1 - 0.3^2))$$

$$= 1185310720.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1670 + 14 / \cos(0)) \cdot 1185310720 / 264^3 \cdot (1 + 264 \cdot 0.012 \cdot (1 + 1) / 2 + (264 \cdot 0.012)^2 / 2 \cdot 1)$$

$$= 5864617.5 \text{ KPa}.$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1670^2 / (4 \cdot 199214080 \cdot 14) \cdot (1 - 1670 \cdot 0.3 / (2(1670 + 14)))$$

$$= 0.21279E+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [w_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.01 \cdot 1185310720 [0.21279198 (1 + 0.012 \cdot 1 \cdot 264) - 0.000 (1 / 264 + 1 \cdot 0.012 / 2)]$$

$$= 12485.3818 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1738 / 1648$$

$$= 1.0547$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.42) / 36704808 \cdot (28980458 + 5864617 + 191201920 \cdot \ln(1.05))$$

$$= 0.7135$$



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ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= \frac{D_o^2}{16} \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= \frac{1648^2}{16} \cdot [(1.01 - 1) \cdot (1.01^2 + 1) \cdot 4550 - (1.01 - 1) \cdot (1.01^2 + 1) \cdot 1000]$$

$$= 16386309.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + w_c \cdot P_t - w_s \cdot P_s$$

$$= 16386309 + 12485 \cdot 1000 - 21790 \cdot 4550$$

$$= -70273416.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M^*) - \frac{D_o^2}{32} \cdot F (P_s - P_t)) / (1 + F)$$

$$= ((-70273416) - \frac{1648^2}{32} \cdot 0.71 (4550 - 1000)) / (1 + 0.71)$$

$$= -166452480.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + \frac{D_o^2}{64} \cdot (3 + v^*) \cdot (P_s - P_t)$$

$$= -166452480 + \frac{1648^2}{64} \cdot (3 + 0.42) \cdot (4550 - 1000)$$

$$= 348447168.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| M_p | ; | M_o |)$$

$$= \max(| -166452480 | ; | 348447168 |)$$

$$= 348447168.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 348447168 / (0.23 (264 - 8)^2)$$

$$= 138.5268 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 323815872 / (0.23 (183 - 8)^2)$$

$$= 274.9722 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 183 + 0 + 0$$

$$= 183.1633 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(183 ; 51.4)$$

$$= 183.1633 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | P_s - P_t |$$

$$= | 4550 - 1000 |$$

$$= 3550.000 \text{ KPa.}$$

Shear Stress check [τ_{limit}]:



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$$\begin{aligned}
 &= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy) \\
 &= 4 \cdot 0.23 \cdot 264 / 1648 \cdot \min(0.8 \cdot 138; 0.533 \cdot 213) \\
 &= 16.293 \text{ N./mm}^2
 \end{aligned}$$

| Ps - Pt | is ≤ the Shear Stress limit (τlimit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned}
 &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot |Ps - Pt| + Cats + Catc \\
 &= 1 / (4 \cdot 0.23 \cdot 110) \cdot (4 \cdot 18959 / 5147) \cdot |4550 - 1000| + 0 \\
 &= 51.4338 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σs,m]:

$$\begin{aligned}
 &= Ds^2 / (4 \cdot ts (Ds + ts)) \cdot Ps \\
 &= 1670^2 / (4 \cdot 34 (1670 + 34)) \cdot 4550 \\
 &= 54.7597 \text{ N./mm}^2
 \end{aligned}$$

Axial Shell Bending Stress [σs,b]:

$$\begin{aligned}
 &= 6 \cdot ks / ts^2 \cdot [\beta S \cdot Ps + 6(1 - \nu^*) / E^* \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))] \\
 &= 6 \cdot 2353759 / 34^2 \cdot [0.0076 \cdot 0.41 + 6(1 - 0.42) / 36704808 \cdot 1648 / 264^3 \cdot (1 + 264 \cdot 0.0076 / 2) \cdot (-166452480 + 1648^2 / 32 \cdot (4550 - 1000))] \\
 &= 294.190 \text{ N./mm}^2
 \end{aligned}$$

Shell Membrane + Bending Stress [σs]:

$$\begin{aligned}
 &= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq \text{SPS,s)} \\
 &= 348.950 \text{ N./mm}^2 \text{ (Should be } \leq 450.02)
 \end{aligned}$$

Axial Channel Membrane Stress [σc,m]:

$$\begin{aligned}
 &= Dc^2 / (4 \cdot tc (Dc + tc)) \cdot \cos(\theta) \cdot Pt \\
 &= 1670^2 / (4 \cdot 14 (1670 + 14)) \cdot \cos(0) \cdot 1000 \\
 &= 29.575 \text{ N./mm}^2
 \end{aligned}$$

Axial Channel Bending Stress [σc,b]:

$$\begin{aligned}
 &= 6 \cdot kc / tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\theta) \cdot Pt - 6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (Cm\theta + h \cdot \beta_c / 2 \cdot Cm\Delta) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))] \\
 &= 6 \cdot 266475 / 14^2 \cdot [(0.21279198 \cdot 1 \cdot 0.012 - 0.000 \cdot 1) \cdot 1000 - 6((1 - 0.42) / 36704808) \cdot 1648 / 264^3 \cdot (1 + 0.012 \cdot 264 / 2 \cdot 1) \cdot (-166452480 + 1648^2 / 32 \cdot (4550 - 1000))] \\
 &= -15.326 \text{ N./mm}^2
 \end{aligned}$$

Channel Membrane + Bending Stress [σc]:

$$\begin{aligned}
 &= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq \text{SPS,c)} \\
 &= 44.901 \text{ N./mm}^2 \text{ (Should be } \leq 485.22)
 \end{aligned}$$

Note: The Cylinder-to-Tubesheet Junction stress is < SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate Shell Side parameters for the Simply Supported case:

βs, ks, λs, δs are all = 0



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ASME TS Calc:

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Intermediate Channel side parameters for the Simply Supported condition:

$\beta_c, k_c, \lambda_c, \delta_c$ are all = 0

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.42) / 36704808 \cdot (0 + 0 + 191201920 \cdot \ln(1.05))$$

$$= 0.1613$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 1648^2 / 16 \cdot [(1.01 - 1) \cdot (1.01^2 + 1) \cdot 4550 - (1.01 - 1) \cdot (1.01^2 + 1) \cdot 1000]$$

$$= 16386309.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= 16386309 + 0 \cdot 1000 - 0 \cdot 4550$$

$$= 16386309.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= -27734818 + 1648^2 / 64 \cdot (3 + 0.42) \cdot (4550 - 1000)$$

$$= 487164832.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= |M_o|$$

$$= |487164832|$$

$$= 487164832.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 487164832 / (0.23 (264 - 8)^2)$$

$$= 193.6747 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 487164832 / (0.23 (222 - 8)^2)$$

$$= 275.9886 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 222 + 0 + 0$$

$$= 222.4523 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(222; 57.5)$$

$$= 222.4523 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:



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ASME TS Calc:

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Absolute pressure differential [| Ps - Pt |]:

$$= | P_s - P_t |$$

$$= | 4550 - 1000 |$$

$$= 3550.000 \text{ KPa.}$$

Shear Stress check [rlimit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.23 \cdot 264 / 1648 \cdot \min(0.8 \cdot 138; 0.533 \cdot 213)$$

$$= 16.293 \text{ N./mm}^2$$

| Ps - Pt | is ≤ the Shear Stress limit (rlimit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.23 \cdot 110) \cdot (4 \cdot 18959 / 5147) \cdot | 4550 - 1000 | + 0$$

$$= 51.4338 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σs,m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 1670^2 / (4 \cdot 34 (1670 + 34)) \cdot 4550$$

$$= 54.7597 \text{ N./mm}^2$$

Axial Shell Bending Stress [σs,b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P's + 6(1 - \nu^*) / E^* \cdot$$

$$Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 34^2 \cdot [0 \cdot 0.41 + 6(1 - 0.42) / 36704808 \cdot$$

$$1648 / 264^3 \cdot (1 + 264 \cdot 0 / 2) \cdot (-27734818 + 1648^2 / 32 \cdot$$

$$(4550 - 1000))]$$

$$= 0.000 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σs]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq \text{SPS,s)}$$

$$= 54.760 \text{ N./mm}^2 \text{ (Should be } \leq 450.02)$$

Axial Channel Membrane Stress [σc,m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 1670^2 / (4 \cdot 14 (1670 + 14) \cdot \cos(0)) \cdot 1000$$

$$= 29.575 \text{ N./mm}^2$$

Axial Channel Bending Stress [σc,b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta C / 2 \cdot C_m \Delta) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 14^2 \cdot [(0.000 \cdot 1 \cdot 0 - 0.000 \cdot 1) \cdot 1000 -$$

$$6((1 - 0.42) / 36704808) \cdot 1648 / 264^3 \cdot (1 +$$

$$0 \cdot 264 / 2 \cdot 1) \cdot (-27734818 + 1648^2 / 32 \cdot (4550 - 1000))]$$

$$= -0.000 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σc]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq \text{SPS,c)}$$

$$= 29.575 \text{ N./mm}^2 \text{ (Should be } \leq 485.22)$$

Analysis Finished.

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.77) 2.77$$

$$= 1.969 \text{ cm}^2$$



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ASME TS Calc:

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Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.77 (25.4 - 2.77) 92.4$$

$$= 1855.452 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 92.4$$

$$= 1855.4518 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 92.4/92.4$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 92.4$$

$$= 1229.1031 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) \cdot 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$$

$$= 3.372 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.6859 mm

Minimum Required Groove Weld Leg [agr]: 1.6859 mm

Tube-Tubesheet Jt allowable, 1855 is $\geq$ tube strength 1855 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	193.7 $\leq$	276.0 N./mm ²	Ok
Tubesheet shear stress	21.5 $\leq$	110.4 N./mm ²	Ok
Stress in Shell at Tubesheet	54.8 $\leq$	450.0 N./mm ²	Ok
Stress in Channel at Tubesheet	29.6 $\leq$	485.2 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	222.452	264.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.686	2.770	Ok
Tube-Tubesheet Groove Weld Leg :	1.686	2.770	Ok

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :



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ASME TS Calc:

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Note:

The Tubesheet stress reduction option was Simply Supported. Performing the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 811 + 25.4 \\ &= 1647.890 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (33 - 25.4) / 33 \\ &= 0.230 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 1676 / 1648 \\ &= 1.0171 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 1676 / 1648 \\ &= 1.0171 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned} &= l_{tx} / h \\ &= 0 / 258 \\ &= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2t_t (E_t T / E) \cdot (S_t T / S) \cdot (\rho); dt - 2t_t) \\ &= \max(25.4 - 2 \cdot 2.77 (192580832 / 191201920) \cdot \\ &\quad (92.4 / 138) \cdot (0); 25.4 - 2 \cdot 2.77) \\ &= 25.4000 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{(1 - 4 \cdot \min(AI; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2))} \\ &= 33 / \sqrt{(1 - 4 \cdot \min(0; 4 \cdot 1648 \cdot 33) / (\pi \cdot 1648^2))} \\ &= 33.0000 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (33 - 25.4) / 33 \\ &= 0.2303 \end{aligned}$$

$h/p = 7.818181$; $\mu^* = 0.230303$; $E^* = 36704808 \text{ KPa}$.

$E^*/E = 0.191969$; $v^* = 0.418359$

Note: As h/p (7.818) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - v_s^2))^{1/4} / \sqrt{(D_s + t_s)t_s}) \\ &= ((12(1 - 0.3^2))^{1/4} / \sqrt{(1676 + 31)31}) \\ &= 0.0079 \text{ 1/mm} \end{aligned}$$



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ASME TS Calc:

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Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0079 \cdot 192580832 \cdot 31^3 / (6(1 - 0.3^2))$$

$$= 8303546880.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1676 + 31) \cdot 8.30E+09 / 258^3 \cdot (1 + 258 \cdot 0.0079 + (258 \cdot 0.0079)^2 / 2)$$

$$= 25340868.0 \text{ KPa}$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1676^2 / (4 \cdot 192580832 \cdot 31) \cdot (1 - 1676 \cdot 0.3 / (2(1676 + 31)))$$

$$= 0.10030E+00 \text{ mm/N} \cdot \text{mm}^2$$

Parameter [ωs]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 8.30E+09 \cdot 0.0079 \cdot 0.100299 (1 + 258 \cdot 0.0079)$$

$$= 20341.9863 \text{ mm}^2$$

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4} / \sqrt{(D_c + t_c)t_c})$$

$$= ((12(1 - 0.3^2))^{1/4} / \sqrt{(1676 + 11)11})$$

$$= 0.0133 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.013 \cdot 199214080 \cdot 11^3 / (6(1 - 0.3^2))$$

$$= 648048128.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1676 + 11 / \cos(0)) \cdot 648048128 / 258^3 \cdot (1 + 258 \cdot 0.013 \cdot (1 + 1) / 2 + (258 \cdot 0.013)^2 / 2 \cdot 1)$$

$$= 3960732.2 \text{ KPa}$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1676^2 / (4 \cdot 199214080 \cdot 11) \cdot (1 - 1676 \cdot 0.3 / (2(1676 + 11)))$$

$$= 0.27269E+00 \text{ mm/N} \cdot \text{mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ωc]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.02 \cdot 648048128 [0.27268949 (1 + 0.013 \cdot 1 \cdot 258) - 0.000 (1/258 + 1 \cdot 0.013/2)]$$

$$= 10656.4512 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1738 / 1648$$

$$= 1.0547$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$



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ASME TS Calc:

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$$= (1 - 0.42) / 36704808 \cdot (25340868 + 3960732 + 191201920 \cdot \ln(1.05))$$

$$= 0.6256$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$$

$$= 1648^2/16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot 0 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 1000]$$

$$= -5889873.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega c \cdot Pt - \omega s \cdot Ps$$

$$= -5889873 + 10656 \cdot 1000 - 20342 \cdot 0$$

$$= 4766578.5000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M*) - Do^2/32 \cdot F(Ps - Pt) / (1 + F)$$

$$= (4766578) - 1648^2/32 \cdot 0.63(0 - 1000) / (1 + 0.63)$$

$$= 35591240.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu) \cdot (Ps - Pt)$$

$$= 35591240 + 1648^2/64 \cdot (3 + 0.42) \cdot (0 - 1000)$$

$$= -109450904.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|Mp|; |Mo|)$$

$$= \max(|35591240|; |-109450904|)$$

$$= 109450904.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 109450904 / (0.23(258 - 5)^2)$$

$$= 44.5508 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 76683272 / (0.23(90.3 - 5)^2)$$

$$= 274.6823 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 90.3 + 3 + 3$$

$$= 96.2851 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB; HreqS)$$

$$= \max(96.3; 20.5)$$

$$= 96.2851 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [|Ps - Pt|]:

$$= |Ps - Pt|$$



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ASME TS Calc:

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$$= | 0 - 1000 |$$

$$= 1000.000 \text{ KPa.}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)$$

$$= 4 \cdot 0.23 \cdot 258 / 1648 \cdot \min(0.8 \cdot 138; 0.533 \cdot 213)$$

$$= 15.923 \text{ N./mm}^2$$

$|Ps - Pt|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot |Ps - Pt| + Cats + Catc$$

$$= 1 / (4 \cdot 0.23 \cdot 110) \cdot (4 \cdot 18959 / 5147) \cdot |0 - 1000| + 6$$

$$= 20.4884 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [$\sigma_{s,m}$]:

$$= Ds^2 / (4 \cdot ts(Ds + ts)) \cdot Ps$$

$$= 1676^2 / (4 \cdot 31(1676 + 31)) \cdot 0$$

$$= 0.0000 \text{ N./mm}^2$$

Axial Shell Bending Stress [$\sigma_{s,b}$]:

$$= 6 \cdot ks / ts^2 \cdot [\beta S \cdot P's + 6(1 - \nu^*) / E^* \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))]$$

$$= 6 \cdot 1866754 / 31^2 \cdot [0.0079 \cdot 0 + 6(1 - 0.42) / 36704808 \cdot 1648 / 258^3 \cdot (1 + 258 \cdot 0.0079 / 2) \cdot (35591240 + 1648^2 / 32 \cdot (0 - 1000))]$$

$$= -47.062 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= |\sigma_{s,m}| + |\sigma_{s,b}| \text{ (Should be } \leq \text{SPS}_s)$$

$$= 47.062 \text{ N./mm}^2 \text{ (Should be } \leq 450.02)$$

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

$$= Dc^2 / (4 \cdot tc(Dc + tc) \cdot \cos(\theta)) \cdot Pt$$

$$= 1676^2 / (4 \cdot 11(1676 + 11) \cdot \cos(0)) \cdot 1000$$

$$= 37.845 \text{ N./mm}^2$$

Axial Channel Bending Stress [$\sigma_{c,b}$]:

$$= 6 \cdot kc / tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\theta) \cdot Pt - 6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (Cm\theta + h \cdot \beta_c / 2 \cdot Cm\Delta) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))]$$

$$= 6 \cdot 145690 / 11^2 \cdot [(0.27268949 \cdot 1 \cdot 0.013 - 0.000 \cdot 1) \cdot 1000 - 6((1 - 0.42) / 36704808) \cdot 1648 / 258^3 \cdot (1 + 0.013 \cdot 258 / 2 \cdot 1) \cdot (35591240 + 1648^2 / 32 \cdot (0 - 1000))]$$

$$= 156.260 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq \text{SPS}_c)$$

$$= 194.105 \text{ N./mm}^2 \text{ (Should be } \leq 485.22)$$

Note: The Cylinder-to-Tubesheet Junction stress is < SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.7.4 Step 4:



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ASME TS Calc:

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Intermediate Shell Side parameters for the Simply Supported case:

β_s , k_s , λ_s , δ_s are all = 0

Intermediate Channel side parameters for the Simply Supported condition:

β_c , k_c , λ_c , δ_c are all = 0

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.42) / 36704808 \cdot (0 + 0 + 191201920 \cdot \ln(1.05))$$

$$= 0.1613$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s , P_t) [Mts]:

$$= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t]$$

$$= 1648^2 / 16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot 0 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 1000]$$

$$= -5889873.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -5889873 + 0 \cdot 1000 - 0 \cdot 0$$

$$= -5889873.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= 6715576 + 1648^2 / 64 \cdot (3 + 0.42) \cdot (0 - 1000)$$

$$= -138326576.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= |M_o|$$

$$= |-138326576|$$

$$= 138326576.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 138326576 / (0.23 (258 - 5)^2)$$

$$= 56.3043 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 138326576 / (0.23 (119 - 5)^2)$$

$$= 275.9795 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 119 + 3 + 3$$

$$= 125.2754 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(125; 22.2)$$



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ASME TS Calc:

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= 125.2754 mm

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

= | Ps - Pt |

= | 0 - 1000 |

= 1000.000 KPa.

Shear Stress check [τ limit]:

= $4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)$

= $4 \cdot 0.23 \cdot 258 / 1648 \cdot \min(0.8 \cdot 138; 0.533 \cdot 213)$

= 15.923 N./mm²

| Ps - Pt | is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

= $1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot | Ps - Pt | + Cats + Catc$

= $1 / (4 \cdot 0.23 \cdot 110) \cdot (4 \cdot 18959 / 5147) \cdot | 0 - 1000 | + 6$

= 20.4884 mm

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

= $Ds^2 / (4 \cdot ts (Ds + ts)) \cdot Ps$

= $1676^2 / (4 \cdot 31 (1676 + 31)) \cdot 0$

= 0.0000 N./mm²

Axial Shell Bending Stress [σ_s, b]:

= $6 \cdot ks / ts^2 \cdot [\beta S \cdot P's + 6(1 - \nu^*) / E^* \cdot$

$Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))]$

= $6 \cdot 0 / 31^2 \cdot [0 \cdot 0 + 6(1 - 0.42) / 36704808 \cdot$

$1648 / 258^3 \cdot (1 + 258 \cdot 0 / 2) \cdot (6715576 + 1648^2 / 32 \cdot$

$(0 - 1000))]$

= -0.000 N./mm²

Shell Membrane + Bending Stress [σ_s]:

= | σ_s, m | + | σ_s, b | (Should be $\leq$ SPS,s)

= 0.000 N./mm² (Should be $\leq$ 450.02)

Axial Channel Membrane Stress [σ_c, m]:

= $Dc^2 / (4 \cdot tc (Dc + tc) \cdot \cos(\theta)) \cdot Pt$

= $1676^2 / (4 \cdot 11 (1676 + 11) \cdot \cos(0)) \cdot 1000$

= 37.845 N./mm²

Axial Channel Bending Stress [σ_c, b]:

= $6 \cdot kc / tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\Theta) \cdot Pt -$

$6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (Cm\Theta +$

$h \cdot \beta C / 2 \cdot Cm\Delta) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))]$

= $6 \cdot 0 / 11^2 \cdot [(0.000 \cdot 1 \cdot 0 - 0.000 \cdot 1) \cdot 1000 -$

$6((1 - 0.42) / 36704808) \cdot 1648 / 258^3 \cdot (1 +$

$0 \cdot 258 / 2 \cdot 1) \cdot (6715576 + 1648^2 / 32 \cdot (0 - 1000))]$

= 0.000 N./mm²

Channel Membrane + Bending Stress [σ_c]:

= | σ_c, m | + | σ_c, b | (Should be $\leq$ SPS,c)

= 37.845 N./mm² (Should be $\leq$ 485.22)

Analysis Finished.



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ASME TS Calc:

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Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.77) 2.77$$

$$= 1.969 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.77 (25.4 - 2.77) 92.4$$

$$= 1855.452 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 92.4$$

$$= 1855.4518 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 92.4/92.4$$

$$= 1.000$$

Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 92.4$$

$$= 1229.1031 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 3.372 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.6859 mm

Minimum Required Groove Weld Leg [agr]: 1.6859 mm

Tube-Tubesheet Jt allowable, 1855 is >= tube strength 1855 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	56.3 <=	276.0 N./mm ²	Ok
Tubesheet shear stress	6.2 <=	110.4 N./mm ²	Ok
Stress in Shell at Tubesheet	0.0 <=	450.0 N./mm ²	Ok
Stress in Channel at Tubesheet	37.8 <=	485.2 N./mm ²	Ok

Thickness results for loadcase D1 corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	125.275	264.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.686	2.770	Ok
Tube-Tubesheet Groove Weld Leg :	1.686	2.770	Ok

Detailed Results for load Case D3 corr. (Psd,max + Ptd,max)



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ASME TS Calc:

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Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

Note:

The Tubesheet stress reduction option was Simply Supported. Performing the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 811 + 25.4 \\ &= 1647.890 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (33 - 25.4) / 33 \\ &= 0.230 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 1676 / 1648 \\ &= 1.0171 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 1676 / 1648 \\ &= 1.0171 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned} &= l_{tx} / h \\ &= 0 / 258 \\ &= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2t_t(E_t T / E) \cdot (S_t T / S) \cdot (p); dt - 2t_t) \\ &= \max(25.4 - 2 \cdot 2.77 (192580832 / 191201920) \cdot (92.4 / 138) \cdot (0); 25.4 - 2 \cdot 2.77) \\ &= 25.4000 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\ &= 33 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1648 \cdot 33) / (\pi \cdot 1648^2)) } \\ &= 33.0000 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (33 - 25.4) / 33 \\ &= 0.2303 \end{aligned}$$

$h/p = 7.818181$; $\mu^* = 0.230303$; $E^* = 36704808 \text{ KPa}$.

$E^*/E = 0.191969$; $v^* = 0.418359$

Note: As h/p (7.818) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:



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ASME TS Calc:

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Shell coefficient [β_s]:

$$= \left(\left(12(1 - v_s^2) \right)^{1/4} / \sqrt{(D_s + t_s)t_s} \right)$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(1676 + 31)31} \right)$$

$$= 0.0079 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2))$$

$$= 0.0079 \cdot 192580832 \cdot 31^3 / (6(1 - 0.3^2))$$

$$= 8303546880.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1676 + 31) \cdot 8.30E+09 / 258^3 \cdot (1 + 258 \cdot 0.0079 + (258 \cdot 0.0079)^2 / 2)$$

$$= 25340868.0 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 1676^2 / (4 \cdot 192580832 \cdot 31) \cdot (1 - 1676 \cdot 0.3 / (2(1676 + 31)))$$

$$= 0.10030E+00 \text{ mm/N./mm}^2$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 8.30E+09 \cdot 0.0079 \cdot 0.100299 (1 + 258 \cdot 0.0079)$$

$$= 20341.9863 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= \left(\left(12(1 - v_c^2) \right)^{1/4} / \sqrt{(D_c + t_c)t_c} \right)$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(1676 + 11)11} \right)$$

$$= 0.0133 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.013 \cdot 199214080 \cdot 11^3 / (6(1 - 0.3^2))$$

$$= 648048128.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1676 + 11 / \cos(0)) \cdot 648048128 / 258^3 \cdot (1 + 258 \cdot 0.013 \cdot (1 + 1) / 2 + (258 \cdot 0.013)^2 / 2 \cdot 1)$$

$$= 3960732.2 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1676^2 / (4 \cdot 199214080 \cdot 11) \cdot (1 - 1676 \cdot 0.3 / (2(1676 + 11)))$$

$$= 0.27269E+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [w_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.02 \cdot 648048128 [0.27268949 (1 + 0.013 \cdot 1 \cdot 258) - 0.000 (1/258 + 1 \cdot 0.013/2)]$$

$$= 10656.4512 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1738/1648$$



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ASME TS Calc:

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$$= 1.0547$$

Coefficient [F]:

$$\begin{aligned} &= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.42) / 36704808 \cdot (25340868 + 3960732 + \\ &\quad 191201920 \cdot \ln(1.05)) \\ &= 0.6256 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned} &= Do^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (pc - 1) \cdot (pc^2 + 1) \cdot Pt] \\ &= 1648^2 / 16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot 4550 - \\ &\quad (1.02 - 1) \cdot (1.02^2 + 1) \cdot 1000] \\ &= 20909052.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned} &= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps \\ &= 20909052 + 10656 \cdot 1000 - 20342 \cdot 4550 \\ &= -60990536.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned} &= (M^* - Do^2 / 32 \cdot F (Ps - Pt)) / (1 + F) \\ &= (-60990536) - 1648^2 / 32 \cdot 0.63 (4550 - 1000) / (1 + 0.63) \\ &= -153457808.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned} &= Mp + Do^2 / 64 \cdot (3 + v^*) \cdot (Ps - Pt) \\ &= -153457808 + 1648^2 / 64 \cdot (3 + 0.42) \cdot (4550 - 1000) \\ &= 361441824.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(|Mp|; |Mo|) \\ &= \max(|-153457808|; |361441824|) \\ &= 361441824.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - hg')^2) \\ &= 6 \cdot 361441824 / (0.23 (258 - 5)^2) \\ &= 147.1209 \text{ N./mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 138 \\ &= 275.99 \text{ N./mm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - hg')^2) \\ &= 6 \cdot 339908192 / (0.23 (184 - 5)^2) \\ &= 274.8897 \text{ N./mm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned} &= h + Cats + Catc \\ &= 184 + 3 + 3 \\ &= 190.4899 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned} &= \max(HreqB; HreqS) \\ &= \max(190; 57.4) \\ &= 190.4899 \text{ mm} \end{aligned}$$



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ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |4550 - 1000| \\ &= 3550.000 \text{ KPa.} \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\ &= 4 \cdot 0.23 \cdot 258 / 1648 \cdot \min(0.8 \cdot 138; 0.533 \cdot 213) \\ &= 15.923 \text{ N./mm}^2 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$\begin{aligned} &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.23 \cdot 110) \cdot (4 \cdot 18959 / 5147) \cdot |4550 - 1000| + 6 \\ &= 57.4338 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [$\sigma_{s,m}$]:

$$\begin{aligned} &= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s \\ &= 1676^2 / (4 \cdot 31 (1676 + 31)) \cdot 4550 \\ &= 60.3850 \text{ N./mm}^2 \end{aligned}$$

Axial Shell Bending Stress [$\sigma_{s,b}$]:

$$\begin{aligned} &= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot \\ &\quad D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))] \\ &= 6 \cdot 1866754 / 31^2 \cdot [0.0079 \cdot 0.45 + 6(1 - 0.42) / 36704808 \cdot \\ &\quad 1648 / 258^3 \cdot (1 + 258 \cdot 0.0079 / 2) \cdot (-153457808 + 1648^2 / 32 \cdot \\ &\quad (4550 - 1000))] \\ &= 328.161 \text{ N./mm}^2 \end{aligned}$$

Shell Membrane + Bending Stress [σ_s]:

$$\begin{aligned} &= |\sigma_{s,m}| + |\sigma_{s,b}| \text{ (Should be } \leq \text{SPS}_s) \\ &= 388.546 \text{ N./mm}^2 \text{ (Should be } \leq 450.02) \end{aligned}$$

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

$$\begin{aligned} &= D_c^2 / (4 \cdot t_c (D_c + t_c)) \cdot \cos(\theta) \cdot P_t \\ &= 1676^2 / (4 \cdot 11 (1676 + 11)) \cdot \cos(0) \cdot 1000 \\ &= 37.845 \text{ N./mm}^2 \end{aligned}$$

Axial Channel Bending Stress [$\sigma_{c,b}$]:

$$\begin{aligned} &= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - \\ &\quad 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + \\ &\quad h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))] \\ &= 6 \cdot 145690 / 11^2 \cdot [(0.27268949 \cdot 1 \cdot 0.013 - 0.000 \cdot 1) \cdot 1000 - \\ &\quad 6((1 - 0.42) / 36704808) \cdot 1648 / 258^3 \cdot (1 + \\ &\quad 0.013 \cdot 258 / 2 \cdot 1) \cdot (-153457808 + 1648^2 / 32 \cdot (4550 - 1000))] \\ &= -0.980 \text{ N./mm}^2 \end{aligned}$$

Channel Membrane + Bending Stress [σ_c]:

$$\begin{aligned} &= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq \text{SPS}_c) \\ &= 38.824 \text{ N./mm}^2 \text{ (Should be } \leq 485.22) \end{aligned}$$

Note: The Cylinder-to-Tubesheet Junction stress is $<$ SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.



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ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate Shell Side parameters for the Simply Supported case:

$\beta_s, k_s, \lambda_s, \delta_s$ are all = 0

Intermediate Channel side parameters for the Simply Supported condition:

$\beta_c, k_c, \lambda_c, \delta_c$ are all = 0

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Coefficient [F]:

$$\begin{aligned} &= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.42) / 36704808 \cdot (0 + 0 + 191201920 \cdot \ln(1.05)) \\ &= 0.1613 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s, P_t) [Mts]:

$$\begin{aligned} &= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t] \\ &= 1648^2 / 16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot 4550 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 1000] \\ &= 20909052.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned} &= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s \\ &= 20909052 + 0 \cdot 1000 - 0 \cdot 4550 \\ &= 20909052.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned} &= M_p + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (P_s - P_t) \\ &= -23840294 + 1648^2 / 64 \cdot (3 + 0.42) \cdot (4550 - 1000) \\ &= 491059360.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= |M_o| \\ &= |491059360| \\ &= 491059360.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - h_g')^2) \\ &= 6 \cdot 491059360 / (0.23 (258 - 5)^2) \\ &= 199.8802 \text{ N./mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 138 \\ &= 275.99 \text{ N./mm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - h_g')^2) \\ &= 6 \cdot 491059360 / (0.23 (220 - 5)^2) \\ &= 275.9936 \text{ N./mm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 220 + 3 + 3 \\ &= 226.3059 \text{ mm} \end{aligned}$$



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Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB; HreqS)$$

$$= \max(226 ; 63.5)$$

$$= 226.3059 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | Ps - Pt |$$

$$= | 4550 - 1000 |$$

$$= 3550.000 \text{ KPa.}$$

Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h/Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)$$

$$= 4 \cdot 0.23 \cdot 258/1648 \cdot \min(0.8 \cdot 138 ; 0.533 \cdot 213)$$

$$= 15.923 \text{ N./mm}^2$$

| Ps - Pt | is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1/(4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap/Cp) \cdot | Ps - Pt | + Cats + Catc$$

$$= 1/(4 \cdot 0.23 \cdot 110) \cdot (4 \cdot 18959/5147) \cdot | 4550 - 1000 | + 6$$

$$= 57.4338 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$= Ds^2 / (4 \cdot ts(Ds + ts)) \cdot Ps$$

$$= 1676^2 / (4 \cdot 31 (1676 + 31)) \cdot 4550$$

$$= 60.3850 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot ks/ts^2 \cdot [\beta S \cdot P's + 6(1 - \nu^*)/E^* \cdot Do/h^3 \cdot (1 + h \cdot \beta S/2) \cdot (Mp + Do^2/32 \cdot (Ps - Pt))]$$

$$= 6 \cdot 0/31^2 \cdot [0 \cdot 0.45 + 6(1 - 0.42)/36704808 \cdot 1648/258^3 \cdot (1 + 258 \cdot 0/2) \cdot (-23840294 + 1648^2/32 \cdot (4550 - 1000))]$$

$$= 0.000 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_s, m | + | \sigma_s, b | \text{ (Should be } \leq \text{SPS, s)}$$

$$= 60.385 \text{ N./mm}^2 \text{ (Should be } \leq 450.02)$$

Axial Channel Membrane Stress [σ_c, m]:

$$= Dc^2 / (4 \cdot tc(Dc + tc) \cdot \cos(\theta)) \cdot Pt$$

$$= 1676^2 / (4 \cdot 11 (1676 + 11) \cdot \cos(0)) \cdot 1000$$

$$= 37.845 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot kc/tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\Theta) \cdot Pt - 6((1 - \nu^*) / E^*) \cdot Do/h^3 \cdot (Cm\Theta + h \cdot \beta_c/2 \cdot Cm\Delta) \cdot (Mp + Do^2/32 \cdot (Ps - Pt))]$$

$$= 6 \cdot 0/11^2 \cdot [(0.000 \cdot 1 \cdot 0 - 0.000 \cdot 1) \cdot 1000 - 6((1 - 0.42)/36704808) \cdot 1648/258^3 \cdot (1 + 0 \cdot 258/2 \cdot 1) \cdot (-23840294 + 1648^2/32 \cdot (4550 - 1000))]$$

$$= -0.000 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_c, m | + | \sigma_c, b | \text{ (Should be } \leq \text{SPS, c)}$$

$$= 37.845 \text{ N./mm}^2 \text{ (Should be } \leq 485.22)$$



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[Analysis Finished.](#)

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.77) 2.77$$

$$= 1.969 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.77 (25.4 - 2.77) 92.4$$

$$= 1855.452 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 92.4$$

$$= 1855.4518 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 92.4/92.4$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.77 (25.4 + 0.67 \cdot 2.77) 92.4$$

$$= 1229.1031 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) - 0.75 \cdot d_o} \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.77 \cdot (25.4 - 2.77) 1 \cdot 1) - 0.75 \cdot 25.4} \right)$$

$$= 3.372 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.6859 mm

Minimum Required Groove Weld Leg [agr]: 1.6859 mm

Tube-Tubesheet Jt allowable, 1855 is $\geq$ tube strength 1855 Kgf

[Note: This tube-tubesheet joint is a Full Strength joint](#)

Stress/Force summary for loadcase D3 corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	199.9	$\leq 276.0 \text{ N./mm}^2$	Ok
Tubesheet shear stress	22.0	$\leq 110.4 \text{ N./mm}^2$	Ok
Stress in Shell at Tubesheet	60.4	$\leq 450.0 \text{ N./mm}^2$	Ok
Stress in Channel at Tubesheet	37.8	$\leq 485.2 \text{ N./mm}^2$	Ok

Thickness results for loadcase D3 corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	226.306	264.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.686	2.770	Ok
Tube-Tubesheet Groove Weld Leg :	1.686	2.770	Ok



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U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 8 Load Cases:

Case#	--Reqd. Tbsht	Thk. + CA Extnsn	---- Bend	Tubesheet Allwd	Stresses Shear	Allwd	Case Type	Pass/Fail
D1uc	121.832	...	55	276	6	110	Ps+Pt D1	Ok
D2uc	253.528	...	254	276	28	110	Ps+Pt D2	Ok
D3uc	222.452	...	194	276	22	110	Ps+Pt D3	Ok
D4uc	44.519	...	6	276	1	110	Ps+Pt D4	Ok
D1c	125.275	...	56	276	6	110	Ps+Pt-c D1	Ok
D2c	257.506	...	262	276	29	110	Ps+Pt-c D2	Ok
D3c	226.306	...	200	276	22	110	Ps+Pt-c D3	Ok
D4c	47.672	...	6	276	1	110	Ps+Pt-c D4	Ok
Max:	257.5059	... mm		0.949		0.261	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Req'd. Thk.							
Case#	Shell	Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	450	30	485	Ok
D2uc	...	...	55	450	3	485	Ok
D3uc	...	...	55	450	30	485	Ok
D4uc	...	...	...	450	3	485	Ok
D1c	...	...	...	450	38	485	Ok
D2c	...	...	60	450	4	485	Ok
D3c	...	...	60	450	38	485	Ok
D4c	...	...	...	450	4	485	Ok
Max	...	...	0.134		0.078		

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	257.506	264.000	Ok
Tube thickness :	1.118	2.770	Ok
Tube-tubesheet fillet weld leg :	1.686	2.770	Ok
Tube-tubesheet groove weld leg :	1.686	2.770	Ok

Given Shell Thickness: 34.0000 mm

Given Channel Thickness: 14.0000 mm

Note: This is a full strength Tube to Tubesheet Joint.



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Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	4901.84	1.000	4901.84	1.000
Tubesheet Shear Stress	17807.33	1.000	17807.33	1.000
Shell Stress (Axial, Junction)	9562.21	1.000	4700.49	1.000
Tubesheet-Channel Junction	2499.76	1.000	14040.55	1.000
Tube Pressure Stress	22078.02	1.000	13878.91	1.000
Minimum MAWP	2499.76		4700.49	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5058.90	1.000	5058.90	1.000
Tubesheet Shear Stress	18221.45	1.000	18221.45	1.000
Shell Stress (Axial, Junction)	12503.72	1.000	6055.19	1.000
Tubesheet-Channel Junction	3345.09	1.000	17168.77	1.000
Tube Pressure Stress	22078.02	1.000	14664.89	1.000
Minimum MAPnc	3345.09		5058.90	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(264/4, \min(264, 34))$$

$$= 66.000 \text{ mm}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{ps}/\text{Shellside MAPnc}, 1)$$

$$= \min(4550/5059, 1)$$

$$= 0.899$$

Thickness Ratio = 0.9, Temperature Reduction per Fig. UCS 66.1 = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	23 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	1 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness}))$$

$$= \max(264/4, \min(264, 14))$$

$$= 66.000 \text{ mm}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{pt}/\text{Tubeside MAPnc}, 1)$$

$$= \min(1000/3345, 1)$$

$$= 0.299$$

Thickness Ratio = 0.3, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: